A SERVERLESS MLOPS PIPELINE AND PREDICTIVE MODELING FOR HYPER-LOCAL FLASH FLOOD EARLY WARNING SYSTEMS IN KHLONG NUENG, THAILAND

by

Paradorn Khanongsuwan & Nariman Tursaliev

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Submitted To: Dr. Chantri Polprasert

Paradorn Khanongsuwan
Previous Degree : Bachelor of Engineering in Civil Engineering, Prince of Songkla University
Nationality: Thai

Asian Institute of Technology
Faculty of Technology
Thailand
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ABSTRACT

Urban flash floods in Khlong Nueng, Pathum Thani, pose a severe threat to local infrastructure and public safety. Traditional flood warning systems often suffer from delayed, subjective human decision-making and generalized grid-based forecasts. To address these critical vulnerabilities, this study presents a fully automated, AI-driven hyper-local early warning system designed to eliminate human bias and operate strictly within the financial constraints of an academic cloud environment (e.g., AWS Learner Lab).

The proposed solution utilizes a highly cost-effective, "Serverless-First" architecture on Amazon Web Services (AWS). By integrating Amazon EventBridge, AWS Lambda, Amazon S3, and Amazon DynamoDB, the system executes an event-driven micro-batch ETL pipeline that completely avoids the prohibitive idle costs of always-on compute instances. Real-time and historical meteorological telemetry are extracted via the Open-Meteo API. Formulated as a **Multi-Horizon Time-Series Regression** task, an Extreme Gradient Boosting (XGBoost) model was strategically selected to predict 1-to-6-hour accumulated rainfall. To ensure robust real-world performance and strictly prevent temporal data leakage, the model was rigorously evaluated using a **Walk-Forward Validation** strategy. Furthermore, the study introduces two critical feature engineering innovations: a 5-node "Cross-Provincial Radar" spatial lag matrix to mathematically capture incoming storm trajectories, and trigonometric transformations of wind vectors to resolve algorithmic discontinuities.

To translate model predictions into actionable warnings without inducing alert fatigue, a novel 4-Tier Adaptive Alert Logic framework was engineered. This deterministic layer dynamically cross-references the XGBoost rainfall predictions with highly segregated real-time soil moisture data. Upon breaching critical empirical thresholds, the serverless architecture automatically dispatches multi-channel push notifications directly to end-users via the LINE Messaging API.

Significant achievements of this study include the successful deployment of a highly resilient, near-zero compute-debt MLOps pipeline, the rigorous empirical validation of the 4-Tier logic against historical ground-truth flood events, and the establishment of a scalable, bias-free blueprint for proactive, data-driven municipal disaster management.

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CHAPTER 1
INTRODUCTION

1.1 Background of the Study

Thailand has long been vulnerable to extreme weather events, with flash flooding remaining a recurring and devastating issue across various regions. While the Thai Meteorological Department (TMD) and various news outlets continuously provide weather broadcasts, the underlying predictive models and public communication strategies often lack the reliability required for handling sudden, hyper-local flash floods. A prime example of this systemic failure occurred recently during severe flooding in the southern region of Thailand, specifically in Hat Yai district, Songkhla province. Despite early warning indicators and predictive systems suggesting a high risk of catastrophic flooding, local authorities and the provincial governor publicly dismissed the threat, assuring residents that the situation was entirely under control and that no flooding would occur.

This false sense of security, driven by inefficient data interpretation and human error, resulted in disastrous consequences when the water inevitably overflowed. Citizens were caught completely off guard, leaving them with absolutely no time to evacuate or move valuable belongings to higher ground. Similar vulnerabilities and delayed warnings persist year after year from Nakhon Si Thammarat in the South to Chiang Mai in the North, and sprawling across the Central region. These widespread tragedies highlight a critical nationwide flaw: relying on generalized forecasts and subjective human judgment often leads to fatal miscalculations. Consequently, there is an urgent need to develop a bias-free, automated decision-support tool trained on robust historical time-series data to provide objective, real-time alerts.

1.2 Statement of the Problem

Urban areas in Thailand frequently suffer from severe traffic paralysis and property damage due to sudden flash floods, a problem strictly tied to localized "micro-climate" events. Currently, as students at the Asian Institute of Technology (AIT) residing in Pathum Thani, we witness firsthand how this vulnerability heavily affects the central region, particularly the Khlong Nueng and Rangsit university zones. Through our preliminary reverse engineering of local flood events, we discovered that Pathum Thani experiences frequent, intense downpours that rapidly overwhelm the specific capacity of local drainage infrastructure. Standard weather applications and global physics-based meteorological models calculate data over massive grids (e.g., 11x11 kilometers), meaning they only inform users of broad regional rain probabilities but notoriously fail to predict the direct, street-level impact within a rapid 1-to-6-hour window.

The core engineering problem lies in the absence of a hyper-local, fast-inferencing predictive model that is mathematically calibrated to the specific infrastructural limits of the target area. Without reverse-engineering the exact meteorological parameters—such as the critical correlation between hourly rainfall volume and pre-existing soil moisture—systems cannot accurately indicate impending floods. Furthermore, to make such a system viable and sustainable for continuous real-world deployment, it must overcome severe cost constraints. Implementing heavy machine learning pipelines directly on fully managed cloud services like AWS SageMaker incurs exorbitant costs. Therefore, an optimized MLOps strategy is required, one that intelligently decouples the resource-heavy model training phase from the lightweight real-time inference deployment.

1.3 Research Questions

This study seeks to address the overarching challenge of urban flash flood prediction by answering the following primary research questions:

1. How accurately can an XGBoost regression model, enhanced by spatial lag features and trigonometric wind transformations, predict hyper-local 1-to-6-hour rainfall volumes in the Khlong Nueng area?

2. To what extent does the integration of real-time soil moisture telemetry within a "4-Tier Adaptive Alert Logic" reduce false positive flood warnings compared to traditional static rainfall thresholds?
3. How can an end-to-end AWS data pipeline and MLOps architecture be structurally decoupled and optimized to maintain real-time inference latency (under 5 minutes) while strictly adhering to cloud budget constraints?

1.4 Objectives of the Study

The primary objective of this study is to engineer **a cost-constrained, serverless MLOps system for hyper-local rainfall nowcasting and adaptive flood warning**, tailored specifically for the Khlong Nueng area. It is important to explicitly state that the goal is not to invent a novel meteorological weather-forecasting algorithm, but rather to construct an optimized, end-to-end data pipeline that makes existing predictive machine learning models highly accessible and actionable for municipal use.

1. To engineer a resilient, serverless AWS data pipeline capable of reliably extracting, synchronizing, and transforming real-time meteorological and soil moisture data from multiple API nodes.
2. To train and evaluate an XGBoost regression model using 5 years of historical time-series data to predict 1-to-6-hour short-term rainfall volumes with high precision.
3. To reverse-engineer historical flood data to establish a "4-Tier Adaptive Alert Logic" that mathematically cross-references predicted rainfall with real-time ground conditions to trigger automated, bias-free warnings.
4. To design a cost-optimized cloud architecture by executing heavy model training externally in a local environment while deploying only the serialized machine learning model, lightweight serverless alerting mechanisms (e.g., direct messaging APIs), **and a web-based monitoring dashboard** onto AWS.

1.5 Significance of the Study

This study holds substantial value for advancing data engineering and MLOps practices within the context of disaster management and climate technology, offering highly practical insights for

municipal authorities, cloud engineers, and local communities. By demonstrating how a highly localized, predictive early warning system can mitigate the devastating impacts of urban flash floods in Khlong Nueng, it provides a scalable model for local governments to transition from reactive disaster response to proactive risk mitigation. For society, this system directly addresses the critical issue of "alert fatigue"—where constant false alarms cause the public to ignore warnings—by utilizing a rigorous 4-Tier Adaptive Alert Logic to ensure notifications are highly accurate and actionable, thereby safeguarding lives and property.

For data engineers and ML practitioners, the research addresses key technical challenges such as strict cloud budget constraints, complex spatial-temporal data alignment, and cyclical feature engineering. Crucially, the exact technical contribution of this research is strictly framed as the development of a pragmatic, serverless MLOps architecture, rather than a breakthrough in foundational meteorological science**.** It contributes a highly reusable, "serverless-first" blueprint that successfully bridges the gap between enterprise-grade machine learning and cost-constrained public sector projects. Overall, the findings can inform smart-city policy and set new industry standards for deploying budget-efficient, AI-driven public safety ecosystems.

1.6 Organization of the Study

This study is structured to provide a comprehensive exploration of the serverless predictive flash flood warning system, beginning with an introduction (Chapter 1) that outlines the background, problem statement, research objectives, and the significance of the study. Subsequent chapters will review relevant literature (Chapter 2) on modern data engineering architectures, MLOps best practices, and meteorological predictive algorithms.

Chapter 3 details the core methodology, explicitly describing the serverless AWS pipeline design, the XGBoost machine learning implementation, technical feature engineering innovations, and the rigorous evaluation plan. Chapter 4 will present the Results and Discussions, analyzing the system's predictive performance, operational pipeline latency, cost efficiency, and alignment with academic and industry standards. Finally, Chapter 5 (Conclusion) will summarize the key findings, acknowledge system limitations, and provide actionable recommendations for future work and municipal deployment, ensuring a logical, end-to-end progression from theoretical conceptualization to practical, real-world application.

CHAPTER 2
LITERATURE REVIEW

Hyper-local flash flood prediction and early warning is an important research area because it supports disaster management agencies, local governments, and vulnerable communities in making informed decisions for evacuation planning and protecting lives and property from sudden urban flooding events (Špitalar et al., 2014). Recent studies show a shift from traditional physics-based numerical weather prediction models such as WRF-ROMS to machine learning techniques including XGBoost, Random Forest, and LSTM, which can better capture complex nonlinear atmospheric patterns at localised scales with faster inference times (Chen & Guestrin, 2016; Ravuri et al., 2021). In addition, soil moisture conditions are significant factors that determine whether predicted rainfall will exceed local drainage capacity and cause actual street-level flooding (Apimook et al., 2021; Brocca et al., 2017). However, when deployed in real-world environments, weather prediction models may experience concept drift due to seasonal transitions and climate change, which can reduce prediction accuracy. Therefore, MLOps practices recommend continuous monitoring with automated retraining pipelines to maintain model performance as weather patterns evolve (Webb et al., 2016; Gama et al., 2014).

2.1 Rainfall Prediction using Machine Learning

Recent studies have demonstrated a transition from traditional physics-based approaches to machine learning techniques to enhance the accuracy of short-term precipitation nowcasting. Various studies have investigated and compared the performance of popular predictive models, including ARIMA, LSTM, convolutional neural networks, and XGBoost, while graph neural networks such as GraphCast have also been discussed for handling global-scale weather data (Ravuri et al., 2021; Lam et al., 2023). Research indicates that ARIMA performs optimally and remains stable when applied to small, univariate time-series datasets. Conversely, as data complexity increases with the inclusion of multi-variate atmospheric features from multiple spatial nodes such as temperature, humidity, pressure, and wind from surrounding provinces, tree-based models like XGBoost often achieve the highest accuracy for structured tabular inputs. Grinsztajn et al. (2022) demonstrated through comprehensive benchmarks that tree-based models

consistently outperform deep learning on medium-sized tabular datasets. Furthermore, XGBoost consumes significantly fewer computational resources than LSTM and provides built-in feature importance rankings, making it highly suitable for automated retraining in cloud-based environments with budget constraints. Ultimately, accurate rainfall nowcasting at 1-to-3-hour horizons provides vulnerable communities with the critical lead time they desperately need to evacuate and protect their property before flash flooding occurs. (Chen & Guestrin, 2016; Grinsztajn et al., 2022)

2.2 Impact of Soil Moisture on Flood Risk

Literature indicates that antecedent soil moisture conditions are critical exogenous factors influencing both flood initiation and severity levels. Identical rainfall volumes can produce fundamentally different flood outcomes depending on whether the soil is dry and capable of absorption or already saturated with zero infiltration capacity. These ground conditions directly affect whether predicted rainfall translates into dangerous surface runoff or is safely absorbed, making rainfall prediction alone insufficient for flood warning. To account for this, Apimook et al. (2021) developed a multi-criteria Flash Flood Warning System for Thailand that cross-references predicted rainfall with Antecedent Moisture Content (AMC) classes calculated from 5-day accumulated satellite rainfall. Their event-based re-analysis demonstrated that raising the rainfall threshold from 60 mm to 80 mm during dry early-season conditions reduced the False Alarm Ratio from 0.64 to 0.43 and improved the Critical Success Index from 0.30 to 0.42, while the lower threshold remained optimal during saturated conditions (Apimook et al., 2021). Brocca et al. (2017) corroborated that satellite-derived soil moisture significantly improves rainfall-runoff modelling accuracy. Consequently, the proposed system employs a two-layer architecture where the XGBoost model predicts rainfall volume (excluding soil moisture to prevent target leakage), and a separate deterministic threshold logic layer cross-references the prediction with real-time soil moisture to determine the actual flood alert level. (Apimook et al., 2021; Brocca et al., 2017)

2.3 MLOps and Automated Pipelines

2.3.1 Concept Drift

Concept Drift is a critical challenge in weather prediction, defined as a phenomenon where the statistical relationships between atmospheric input features and rainfall outcomes change over time. This shift can occur due to various factors, such as seasonal transitions from dry season to monsoon, long-term climate change patterns, or urban development that modifies local micro-climate characteristics. The occurrence of Concept Drift directly leads to model degradation. As a result, a once-accurate XGBoost model may perform poorly when exposed to data from a new season in real-world environments, ultimately resulting in unreliable rainfall predictions and dangerous missed flood warnings or excessive false alarms. (Webb et al., 2016; Gama et al., 2014)

2.3.2 Automated Retraining

To mitigate the effects of concept drift and maintain sustainable model accuracy, literature emphasises the necessity of automated retraining systems. A core component of this strategy is Continuous Training (CT), an MLOps practice responsible for autonomously monitoring and retraining models to ensure they remain up-to-date with evolving weather patterns. Utilising automated pipelines with delayed ground truth evaluation, where predicted rainfall is compared against actual recorded rainfall 3 hours later, significantly reduces the risk of undetected model degradation and enables the immediate detection of performance decline through rolling RMSE and MAE metrics. These systems can be configured to execute retraining based on a predefined monthly schedule or through dynamic triggering when rolling error metrics exceed acceptable baselines (Paleyes et al., 2022). For weather prediction specifically, time-series cross-validation must be employed rather than random train-test splits to prevent temporal data leakage, and a champion-challenger deployment strategy ensures new models outperform incumbents before promotion to production (Bergmeir & Benítez, 2012). Ultimately, automated retraining provides the adaptability necessary for hyper-local weather models to swiftly adapt to seasonal transitions and evolving climate patterns while maintaining the prediction accuracy that citizens depend on for life-safety decisions. (Paleyes et al., 2022; Gama et al., 2014)

2.4 Chapter Summary

Recent research highlights a significant shift from traditional physics-based numerical weather prediction to machine learning techniques for rainfall nowcasting, with models such as ARIMA, LSTM, convolutional neural networks, XGBoost, and GraphCast being widely explored. Studies show that ARIMA performs well on small, univariate datasets, while LSTM excels as data complexity increases with exogenous variables; meanwhile, tree-based models like XGBoost offer comparable or superior accuracy with far lower computational costs, making them ideal for cloud-based automated systems (Chen & Guestrin, 2016; Grinsztajn et al., 2022; Ravuri et al., 2021). Soil moisture conditions are identified as critical exogenous factors that determine whether predicted rainfall will cause actual flooding, as identical rainfall volumes produce different outcomes depending on antecedent ground saturation, with soil-moisture-aware threshold calibration reducing False Alarm Ratio from 0.64 to 0.43 while improving Critical Success Index from 0.30 to 0.42 (Apimook et al., 2021; Brocca et al., 2017). However, deploying these models in real-world environments introduces the challenge of concept drift, where seasonal transitions and climate change alter the statistical relationships between atmospheric features and rainfall outcomes, leading to model degradation and unreliable predictions. To address this, MLOps practices emphasise Continuous Training (CT) through automated retraining pipelines that monitor rolling RMSE performance and trigger retraining either on a monthly schedule or dynamically when error metrics exceed baselines, utilising time-series cross-validation and champion-challenger deployment to prevent temporal leakage and ensure consistent accuracy (Webb et al., 2016; Gama et al., 2014; Paleyes et al., 2022; Bergmeir & Benítez, 2012). Collectively, these advancements in machine learning nowcasting, soil-moisture-informed flood detection, and automated pipeline management provide the foundation for the proposed Hyper-Local AI Weather Nowcasting and Urban Flood Warning System, which employs an XGBoost regression model cross-referenced with real-time soil moisture through a 4-Tier Adaptive Alert Logic calibrated for the Khlong Nueng area of Pathum Thani, as will be detailed in the following chapter.

CHAPTER 3
METHODOLOGY

3.1 System Architecture & Pipeline Design

The proposed system architecture follows a practical, serverless-first ETL (Extract, Transform, Load) data pipeline deployed on Amazon Web Services (AWS). To remain suitable for the $50 AWS Academy Learner Lab constraints, the architecture avoids expensive, always-on continuous extract data per hour and provisioned compute instances (e.g., Amazon EC2). Instead, the system adopts an event-driven, micro-batching strategy powered by serverless services. This architectural decision helps connect data engineering workflows with budget-conscious machine learning deployment, while keeping the system scalable for future extension with minimal idle cost.

The data ingestion layer is responsible for collecting real-time telemetry from multiple sources to represent a cross-provincial weather context. An Amazon EventBridge rule acts as an hourly scheduler, triggering the Ingestion AWS Lambda function. This function executes six parallel asynchronous HTTP requests to the Open-Meteo API. Five of these requests capture atmospheric data, such as rainfall, temperature, and wind metrics, across a spatial grid to track incoming storm conditions, while the sixth request extracts local soil moisture data for the target area.

Because meteorological data arrives at a manageable hourly rate rather than high-speed clickstream volume, complex ETL tools like AWS Glue are not required. Instead, the Ingestion Lambda performs lightweight, in-memory data transformation directly after extraction. The script executes data alignment across the telemetry streams based on ISO 8601 localized timestamps (UTC+7), handles missing sensor values, applies feature engineering such as trigonometric conversion of wind directions, and separates the soil moisture variable for downstream use. To maintain data durability and support system operations, the Lambda function uses a dual-storage approach: it pushes the raw JSON payloads into an Amazon S3 data lake for historical archiving, and writes the latest cleaned feature row into Amazon DynamoDB for low-latency access.

To illustrate the pipeline's main data engineering functions more clearly, the system includes the following core components:

Data Ingestion: Real-time data is extracted from the Open-Meteo API using an AWS Lambda function, removing the need for dedicated streaming clusters.

Data Cleaning and Preprocessing: The Lambda function handles missing values through interpolation or fallback processing so that the machine learning inference phase can continue reliably.

Data Transformation and Aggregation: The ingestion script aligns and merges six independent API payloads based onISO 8601 localized timestamps (UTC+7), transforms circular wind metrics into linear vectors, and isolates the soil moisture variable.

Data Storage and Versioning: Raw historical data is archived in Amazon S3, while the latest processed ML-ready dataset is stored in Amazon DynamoDB. Machine learning models are stored in S3 with version control.

Pipeline Orchestration and Scheduling: The micro-batch ETL workflow is scheduled and governed by Amazon EventBridge, supporting regular automated execution without manual intervention.

The final architectural tier governs real-time machine learning inference and user-facing prediction delivery. For real-time hourly operations, an Amazon API Gateway endpoint is exposed. When called, it triggers a **Containerized Prediction AWS Lambda function**. To deploy the XGBoost model within a serverless environment, standard AWS Lambda deployment (via .zip archives) was insufficient due to hard size limits and the absence of native XGBoost libraries. Consequently, the deployment architecture was upgraded to a containerized approach. The inference script, the pre-trained XGBoost model artifact, and all necessary machine learning dependencies were packaged into a Docker image and pushed to **Amazon Elastic Container Registry (ECR)**. AWS Lambda is then provisioned to run directly from this ECR image, ensuring a scalable and dependency-conflict-free environment. During execution, this Lambda function fetches the latest synchronized feature row from DynamoDB. It predicts the upcoming rainfall volume and evaluates it together with the relevant soil moisture context using a custom Python-based 4-Tier Adaptive Alert Logic layer. The prediction result is then written to the predictions log and returned to the dashboard or alert view for end-user access. In addition to direct messaging, the system features a serverless web frontend hosted on Amazon S3 Static

Website Hosting. To complete the end-to-end MLOps lifecycle, a third architectural component is dedicated to automated Model Monitoring and Concept Drift detection. Scheduled by an Amazon EventBridge rule to execute daily at 01:00 AM, the Monitoring AWS Lambda function evaluates the predictive health of the deployed XGBoost model. It retrieves the last 7 days of historical feature data from DynamoDB and generates retrospective predictions. These predictions are then compared against the actual ground-truth meteorological data retroactively fetched from the Open-Meteo API. By calculating the Mean Absolute Error (MAE), the system mathematically quantifies any performance degradation over time. If the calculated MAE breaches a pre-defined threshold (e.g., > 2.0 mm), an automated MLOps alert is triggered via the LINE API, signaling the engineering team that model retraining is required. This automated observability ensures continuous model reliability without the need for manual daily audits.

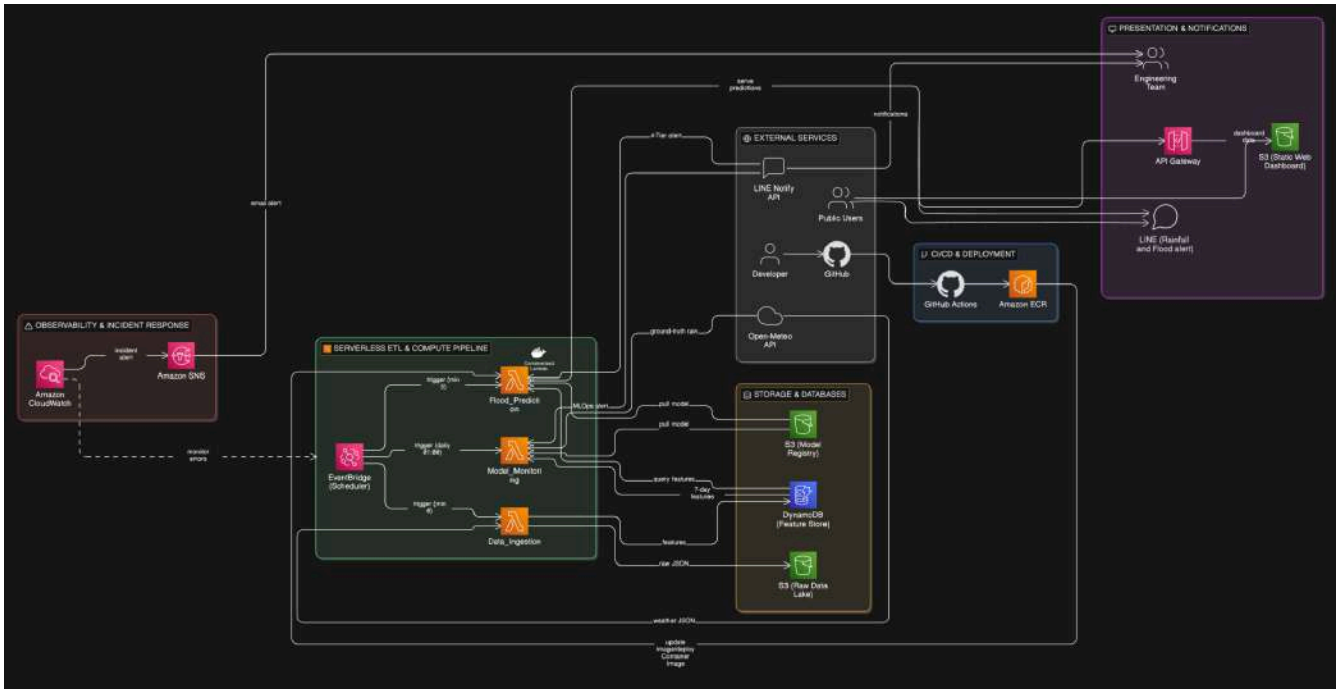


Figure 3.1 : System Architecture & Pipeline Design

3.1.1: Continuous Integration and Continuous Delivery (CI/CD) Strategy

To ensure a robust MLOps workflow, the system adopts a GitHub Actions-based CI/CD strategy. This automation layer acts as the primary bridge between the source code repository and the AWS cloud environment. Every code modification pushed to the main branch triggers an automated workflow that executes a Docker Build to package the XGBoost model and its

dependencies. The resulting image is then programmatically pushed to Amazon ECR, followed by an automated update to the AWS Lambda execution code. This strategy minimizes human error and ensures that the production environment is always synchronized with the latest validated model version.

3.2 Data Collection and Preprocessing

Data is collected dynamically from external API endpoints to capture both regional meteorological conditions and localized ground saturation. To completely prevent Data Distribution Shift between the training and production phases, all historical and real-time telemetry is sourced from the exact same provider:

- **Meteorological Data (Atmospheric):** Extracted via the Open-Meteo API. To capture incoming storm trajectories (the "Cross-Provincial Radar" strategy), data is pulled simultaneously from 5 distinct spatial nodes: Khlong Nueng (Target), Ayutthaya (North), Bangkok (South), Nakhon Nayok (East), and Nonthaburi (West). Features include Rainfall (mm), Temperature, Humidity, Pressure, Wind Speed, and Wind Direction.
- **Ground Condition Data (Soil Moisture):** Sourced via the Open-Meteo API (simulating SMAP Satellite telemetry). Unlike atmospheric data, soil moisture is extracted strictly for the target area (Khlong Nueng) to represent the localized physical absorption limits of the infrastructure.

Instead of provisioning heavy, managed ETL tools like AWS Glue—which would introduce unnecessary overhead for hourly, low-velocity data—the **Ingestion AWS Lambda function** serves as the primary data integration engine. The preprocessing pipeline performs lightweight, in-memory data alignment directly upon extraction. It merges the 6 independent JSON payloads (5 atmospheric nodes + 1 soil moisture node) based on exact ISO 8601 localized timestamps (UTC+7).

Within this preprocessing stage, missing values are handled using interpolation or forward-filling where appropriate. In addition, selected continuous variables such as temperature, humidity, pressure, wind speed, and cloud cover are supplemented with short moving-average features to reduce hourly noise and capture recent trends more effectively. A 3-hour moving average is used

for these variables because it smooths short-term fluctuation while still preserving meaningful weather changes. Rainfall itself is not smoothed directly, since sudden rainfall spikes are important for flood prediction; instead, lagged rainfall values and short rolling rainfall totals are retained as more informative indicators.

During this execution window, the Lambda function inherently handles missing sensor values through spatial interpolation or forward-filling. It also applies Trigonometric Transformations to circular wind directions (creating continuous linear Wind_X and Wind_Y vectors) and strictly segregates Soil Moisture from the training features to prevent target leakage.

Once the data is highly curated, the Lambda function executes a dual-write operational pattern: the raw JSON payload is archived back into the **Amazon S3 Data Lake** for future model retraining, while the fully transformed, single latest feature row is upserted into **Amazon DynamoDB**. This NoSQL table acts as a highly optimized, low-latency feature store, ensuring the downstream machine learning model can fetch the current weather state instantly.

To ensure enterprise-grade pipeline resilience, the ingestion layer incorporates a robust error-handling and data-recovery framework. The AWS Lambda function utilizes a customized **Exponential Backoff** strategy (via urllib3.util.retry), allowing the system to gracefully handle external API rate limits or transient network failures without crashing the entire ETL cycle. Furthermore, the pipeline addresses the risk of localized data gaps through a **Spatial Interpolation Fallback** mechanism. In events where a specific regional node (e.g., Ayutthaya or Nonthaburi) returns null or incomplete telemetry, the algorithm dynamically calculates the missing values by interpolating data from adjacent operational nodes within the spatial grid. This proactive data engineering approach ensures that the downstream XGBoost model always receives a complete and consistent feature matrix, even during partial provider outages

3.3 Machine Learning Approach and Predictive Modeling

The core technical approach for the predictive modeling component relies on Extreme Gradient Boosting (XGBoost), formulated as a supervised machine learning regression task. To satisfy the project's objective of providing a 1-to-6-hour advance warning, the problem is mathematically

framed as a **Multi-Horizon Time-Series Regression**. Instead of predicting a single generic block of rain, the model is engineered to explicitly predict continuous target variables across distinct sequential time steps (e.g., forecasting the specific rainfall volume at t+1, t+2, and t+3 hours ahead). These highly granular hourly predictions are subsequently aggregated within the pipeline to compute the accumulated rainfall volume, which acts as the primary trigger for the 4-Tier Adaptive Alert Logic.

3.3.1 Algorithm Selection and Justification Predicting short-term, hyper-local weather phenomena requires an algorithm capable of capturing complex, non-linear, and multi-variate relationships among disparate meteorological parameters (e.g., the interaction between localized humidity, pressure drops, and cross-provincial wind vectors). XGBoost was strategically selected over deep learning alternatives (such as LSTMs) due to three critical engineering factors:

- **Robustness to Extreme Outliers:** Urban flash floods in Thailand are typically triggered by sudden, violent micro-bursts rather than gradual rain. XGBoost's gradient boosting framework handles these extreme, non-standard distribution peaks exceptionally well without requiring aggressive target normalization.
- **Tabular Data Superiority:** Since the integrated dataset from the Open-Meteo API is inherently tabular (time-series rows with meteorological features), tree-based ensemble methods empirically outperform deep neural networks in both accuracy and training efficiency.
- **Computational Economics (Serverless Optimized):** Aligning strictly with the AWS Learner Lab budget constraints, XGBoost offers exceptionally fast inference times with a minimal memory footprint. This makes it structurally ideal for deployment within a **serverless AWS Lambda function**, completely bypassing the need for continuously running compute instances (like Amazon EC2) without sacrificing predictive precision.

3.3.2 Model Architecture and MLOps Training Pipeline

Conceptually, the XGBoost model operates by sequentially building an ensemble of decision trees, where each subsequent tree focuses on minimizing the residual errors (gradient descent) of the preceding ensemble. During the local experimentation phase, an automated hyperparameter tuning pipeline is established to optimize critical parameters, including learning_rate (to prevent

overfitting on noisy meteorological data), max_depth (to control tree complexity), and n_estimators.

To maintain rigorous MLOps governance prior to cloud deployment, all local training experiments, hyperparameter configurations, and model metadata are systematically tracked and versioned using MLflow. Once the optimal model is identified, the trained artifact is serialized into a .pkl format and uploaded to the Amazon S3 Model Registry, effectively decoupling the compute-heavy training phase from the lightweight, serverless real-time inference pipeline.

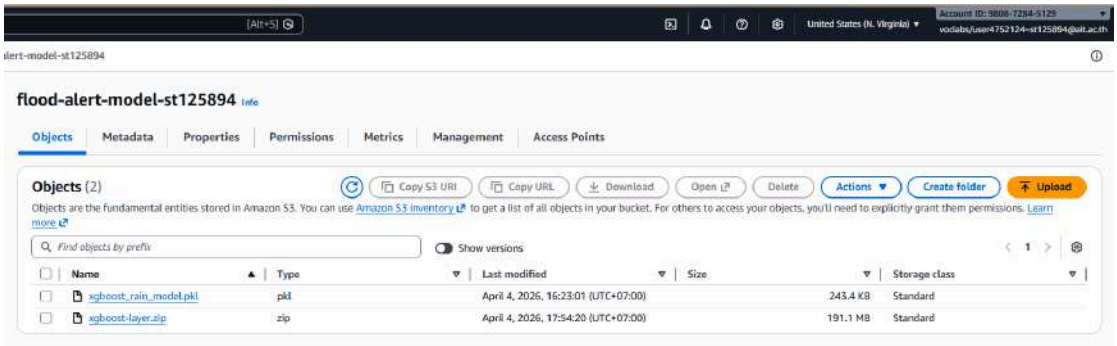


Figure 3.2: Amazon S3 Model Registry bucket showing the finalized, versioned XGBoost model artifact (xgboost_rain_model.pkl) decoupled from the compute environment.

3.3.3 Validation Strategy to Prevent Data Leakage

A critical technical challenge in weather forecasting is maintaining the exact temporal sequence of the data. Standard randomized train-test splits (e.g., standard K-Fold cross-validation) are strictly avoided in this methodology, as predicting historical weather using future data would introduce catastrophic temporal data leakage and artificially inflate model accuracy.

To rigorously evaluate the model and simulate a real-world production environment, the methodology upgrades from a simple static split to a **Time Series Cross-Validation** technique, specifically employing a **Walk-Forward Validation (Expanding Window)** strategy. Rather than splitting the data just once, this technique sequentially shifts the training window forward in time. The XGBoost model is trained on an expanding block of historical past data and evaluated exclusively on the subsequent, unseen temporal fold. This strictly chronological approach

ensures that the performance metrics completely prevent data leakage and accurately reflect how the model will perform during continuous live deployment.

3.4 Technical Innovation and Feature Engineering Optimization The primary technical contribution of this project lies not merely in deploying an off-the-shelf machine learning algorithm, but in the rigorous, domain-specific feature engineering strategies applied within the data pipeline. Traditional ML models often fail in meteorological forecasting due to the misrepresentation of dynamic physical phenomena in a static tabular format. To resolve this and directly improve model accuracy and pipeline efficiency, the architecture introduces two critical technical innovations:

3.4.1 Spatial Lag Feature Matrix ("Cross-Provincial Radar")

- **The Algorithmic Challenge:** Weather is highly dynamic. A micro-burst over Khlong Nueng does not form in isolation; it travels. A traditional model relying solely on single-node API requests lacks the spatial context required to predict impending storms, resulting in reactive alerts with near-zero lead time.
- **The Innovation:** The data ingestion layer is specifically architected to extract telemetry from a 5-node spatial grid (Target: Khlong Nueng; Surrounding Nodes: Ayutthaya, Bangkok, Nakhon Nayok, Nonthaburi). During the in-memory **AWS Lambda preprocessing phase**, these regional features are synchronized onto a single timestamped row.
- **ML Optimization Impact:** By engineering this "Spatial Lag," the XGBoost algorithm is provided with a mathematical matrix that allows it to infer the advection speed, momentum, and trajectory of storm cells before they strike the target infrastructure. Crucially, this strategy achieves the spatial awareness that typically requires highly complex and computationally expensive 3D Convolutional Neural Networks (CNNs) or radar image processing. By translating spatial physics into a highly optimized tabular format, the system achieves advanced predictive lead times while remaining entirely feasible within the strict memory and compute execution limits of a serverless **AWS Lambda environment**.

3.4.2 Cyclical Data Optimization via Trigonometric Transformation

- **The Algorithmic Challenge:** Raw wind direction is measured cyclically from 0° to 360°. This creates a severe mathematical and algorithmic discontinuity for tree-based regression models. For instance, 359° and 1° are physically adjacent (only 2 degrees apart), but a decision tree treats them as numerically distant integers. This forces the XGBoost model to make false, highly penalized gradient splits, fundamentally confusing the algorithm and degrading predictive accuracy.
- **The Innovation:** To eliminate this mathematical boundary issue, the Ingestion Lambda applies a deterministic Trigonometric Transformation during the preprocessing phase. The cyclical degrees are decomposed into two distinct, continuous linear force vectors (representing the zonal and meridional wind components).
- **Mathematical Formulation:**

$$Wind_X = -Wind_Speed\ x\ sin(\theta)$$

$$Wind_Y = -Wind_Speed\ x\ cos(\theta)$$

- **ML Optimization Impact:** This transformation completely resolves the algorithmic discontinuity. By feeding Wind_X and Wind_Y into the **Amazon DynamoDB feature state** instead of raw degrees, the XGBoost model can accurately interpret the true physical force and directional momentum of the wind. This significantly optimizes the model's convergence rate during the training phase and drastically lowers the Root Mean Square Error (RMSE) by preventing the model from learning artificial noise generated by cyclical boundaries.

3.5 The 4-Tier Adaptive Alert Logic Framework & Messaging Integration

"To produce a robust representation of the target area's ground conditions, the ingestion pipeline utilizes a vertically-integrated soil moisture profile. Rather than relying on a single-depth reading, the Flood_Data_Ingestion function extracts telemetry from three distinct subsurface levels: 0–1 cm, 1–3 cm, and 3–9 cm. By calculating a weighted average of these layers, the system produces a more accurate representation of the 'Surface Infiltration Ceiling'—the point at

which the ground can no longer absorb water. This multi-layer sensor integration prevents false triggers caused by superficial surface wetness and ensures that the Tier 2 (Pre-Saturation Threat) and Tier 4 (Systemic Overflow) alerts are triggered based on deep-layer physical saturation.

This serverless microservice dynamically cross-references the XGBoost model's rainfall predictions with real-time soil moisture telemetry (fetched from DynamoDB) to simulate local infrastructural failure points. If any of the critical tiers below are breached, the Lambda function executes two parallel actions: it directly dispatches HTTP POST requests to the LINE Messaging API (or WhatsApp API) for multi-channel push notifications, and it immediately updates the real-time status on the public-facing web dashboard, bypassing intermediary notification services to deliver warnings directly to end-users.

3.5.1 The 4-Tier Adaptive Alert Logic

- **Tier 1: The "Flash Flood" Factor (Absolute Hourly Threshold)**
  - **Trigger Logic:** IF Predicted Hourly Rainfall ≥ 10.0 mm/hr THEN **RED Alert**
  - **Technical Justification:** Addresses sudden, violent micro-bursts that instantaneously overwhelm municipal storm drainage capacity. At this extreme intensity, the system explicitly bypasses soil moisture evaluation, as surface flooding will occur rapidly regardless of the ground's initial infiltration rate. This strict upper boundary prevents false alarms during moderate showers while ensuring immediate response to severe, immediate threats.
- **Tier 2: The "Smart Override" (Pre-Saturation Threat)**
  - **Trigger Logic:** IF Predicted Hourly Rainfall > 6.0 mm/hr AND Real-Time Soil Moisture ≥ 0.50 THEN **RED Alert**
  - **Technical Justification:** This tier introduces dynamic cross-referencing between the ML output and physical ground constraints. When soil moisture reaches its physical saturation ceiling (representing nearly 100% capacity), the ground's infiltration rate effectively drops to zero. Consequently, even a moderate rainfall burst (> 6.0 mm/hr) translates entirely into surface runoff. This smart override ensures catastrophic floods are caught even when absolute rainfall volume appears deceptively manageable.

- **Tier 3: The "Sliding Window" Limit (3-Hour Accumulation)**
    - **Trigger Logic:** IF 3-Hour Cumulative Rainfall ≥ 10.0 mm THEN **ORANGE Alert** (Warning) | IF > 15.0 mm THEN **RED Alert** (Critical)
    - **Technical Justification:** Acts as a safety net for "prolonged soaking" events and localized micro-climate anomalies. The system calculates a rolling 3-hour accumulative window to monitor sustained, lower-intensity rainfall that gradually strains infrastructure. This provides an escalating warning mechanism, buying crucial preparation time before local drainage networks reach their failure points.
- **Tier 4: The "Systemic Overflow" Factor (12-Hour Accumulation)**
    - **Trigger Logic:** IF 12-Hour Cumulative Rainfall > 20.0 mm AND Real-Time Soil Moisture ≥ 0.48 THEN **RED Alert**
    - **Technical Justification:** Addresses continuous, prolonged rain events (e.g., half-day sustained monsoons) that lead to the systemic overflow of primary municipal canals. By enforcing a strict 12-hour sliding window coupled with a high soil moisture prerequisite, the algorithm reliably identifies the critical juncture where the broader regional drainage network becomes compromised.

3.5.2 Empirical Validation & Historical Backtesting To mathematically calibrate the aforementioned Tiers, an empirical reverse-engineering methodology was applied. Historical rainfall and soil moisture telemetry from the Open-Meteo API were cross-referenced against verified ground-truth news reports of flooding in Pathum Thani. Below are selected case studies demonstrating this calibration (A comprehensive log of all analyzed historical events and structured ground-truth dataset detailing of example events with binary flood labels is provided in Appendix A):

- **Case Study 1: Validating Tier 1 and Tier 4 (Absolute Flash Flood and Systemic Overflow)**
    - **Event Date:** September 26 - October 3, 2021
    - **API Telemetry:** The event began with a massive, violent rainfall cell on September 20, peaking at an extreme 13.0 mm/hr (17:00). Soil moisture hit the critical saturation ceiling early on, reaching 0.519.

    - **Report Insight:** The initial catastrophic burst (13.0 mm/hr) instantly triggered a Tier 1 Red Alert, overwhelming municipal drains. However, the true crisis was sustained by the Tier 4 logic: prolonged continuous rain over the next two weeks on completely saturated soil (moisture > 0.490 - 0.520) caused a systemic overflow of the primary canals.
- **Case Study 2: Validating Tier 2 and Tier 4 (Pre-Saturation Threat)**
    - **Event Date:** September 6-8, 2022
    - **API Telemetry:** Massive, continuous rainfall cells over consecutive days with peaks including 9.4 mm/hr (Sep 6) and 12.3 mm/hr (Sep 7). Soil moisture saturated to the absolute maximum ceiling, hitting 0.518 to 0.520.
    - **Report Insight:** This event perfectly validates the Tier 2 (Smart Override) logic. By the time the Red Flag was raised on September 8, soil moisture had repeatedly hit its maximum capacity (0.520). With zero infiltration capacity left, even moderate subsequent rainfall immediately converted into 100% surface runoff.

3.5.3 System Generalizability and Threshold Calibration

It is important to clarify that while the specific quantitative thresholds within the 4-Tier logic (e.g., predicted rainfall ≥ 10.0 mm/hr or real-time soil moisture ≥ 0.50) are empirically calibrated specifically for the localized topography and drainage capacity of the Khlong Nueng / Rangsit area, the overarching architectural framework is highly generalizable. The core innovation of this logic layer—programmatically cross-referencing serverless machine learning predictions with real-time ground saturation to trigger adaptive messaging—can be systematically exported to any geographical region. Local governments in other flood-vulnerable municipalities (e.g., Chiang Mai or Hat Yai) can adopt this exact AWS serverless blueprint. To operationalize the system in a new setting, engineers would merely need to recalibrate the numeric trigger thresholds via historical backtesting to accurately reflect their specific municipal infrastructure constraints, proving that the methodology is not restricted to a narrow local context.

3.6 Feasibility and Tool Selection

The project is highly realistic and feasible to complete within the academic course timeline and the strict financial constraints of the AWS Learner Lab environment. By strategically adopting a "Serverless-First" architecture, the project strictly adheres to modern data strategies and cloud economics. The selection of every tool, framework, and dataset has been meticulously justified to balance enterprise-grade data engineering principles with practical academic limitations.

The deployment of the machine learning model utilizes a **Containerized MLOps pattern** to overcome the structural limitations of standard serverless compute. Due to the significant size of the XGBoost library and its associated numerical dependencies (exceeding the 250MB AWS Lambda unzipped limit), the inference environment was architected as an **immutable Docker image**. This image, containing the pre-trained .pkl artifact and a standardized Python runtime, is hosted on **Amazon Elastic Container Registry (ECR)**. This containerization strategy not only ensures environment parity between development and production but also allows for rapid, version-controlled deployment of complex models that would otherwise be incompatible with standard serverless deployment packages.

- **Data Source and Machine Learning Frameworks:**
    - **Open-Meteo API:** Chosen as the primary data source because it provides free, programmatic access to 5-year hourly historical data. This avoids the logistical impossibility and prohibitive costs of deploying physical IoT sensors for an academic project.
    - **Python & XGBoost:** Python serves as the foundational language for ETL and ML. XGBoost was explicitly chosen over deep learning models because it is highly efficient, trains significantly faster on tabular weather data, and requires a minimal memory footprint—making it strictly compatible with AWS Lambda deployments.
    - **MLflow:** Utilized locally to maintain MLOps best practices—tracking hyperparameter tuning and serializing model artifacts—without incurring unnecessary cloud compute costs during the experimentation phase.

- **AWS Ecosystem Selection and Cloud Economics (Serverless Stack):**
    - **AWS Lambda & Amazon EventBridge:** EventBridge provides reliable hourly cron-scheduling, while Lambda executes the ETL data extraction, real-time model inference, and automated model monitoring purely on-demand, ensuring absolutely zero idle compute costs.
    - **Amazon S3 :** S3 is chosen for highly durable, cost-effective storage. It serves a triple purpose within the architecture: acting as a data lake for raw JSON payloads, a Model Registry for serialized artifacts, and a highly available static web host for the frontend dashboard.
    - **Amazon DynamoDB:** Strategically chosen over complex Feature Store services. It acts as a highly optimized, ultra-low latency NoSQL state store to hold the latest cleaned meteorological features, perfectly bridging the ingestion and prediction Lambda functions.
    - **Amazon API Gateway:** Exposes a lightweight REST API to trigger the prediction Lambda function on-demand, strictly bypassing the continuous hourly billing of fully managed endpoints like Amazon SageMaker or Amazon EC2.
    - **Amazon ECR (Elastic Container Registry) & Docker:** Strategically utilized to overcome standard serverless deployment limits. Because the XGBoost library and model dependencies exceed the maximum allowed size for standard Lambda .zip uploads, the production environment is containerized using Docker. ECR serves as the secure, highly available registry to store this image, allowing AWS Lambda to pull and execute the complex machine learning environment seamlessly.
    - **Amazon CloudWatch:** Integrated for centralized logging, real-time performance monitoring, and system alerting, ensuring comprehensive operational observability and rapid troubleshooting for all serverless components and model.
    - **LINE API & Amazon SNS:** The architecture deliberately separates user-facing business alerts from backend operational alerts to maintain a clean separation of concerns. The LINE Messaging API is directly integrated within the Lambda logic to instantly dispatch 4-Tier flood warnings to end-users and Concept Drift alerts to the MLOps team, bypassing the need for intermediary notification

services for routine messaging. In contrast, Amazon SNS is strictly reserved for system observability and incident response; it integrates seamlessly with Amazon CloudWatch to dispatch critical email alerts to the engineering team only when pipeline execution failures or infrastructure errors occur.

- **Frontend Interface and CI/CD Automation:**
    - **Real-Time Statistical Web Dashboard:** A lightweight frontend interface (developed using HTML, JavaScript, and Tailwind CSS) was deployed to visualize predictive metrics, historical trends, and soil moisture levels. It seamlessly fetches real-time data via the API Gateway. **GitHub & GitHub Actions (CI/CD and DevSecOps):** Utilized to establish a robust, automated deployment pipeline. Upon code commits to the repository, GitHub Actions automatically builds the updated Docker image, pushes it to Amazon ECR, and seamlessly updates the Lambda function. Adhering to strict DevSecOps protocols, all sensitive AWS credentials are encrypted and stored securely within GitHub Secrets to prevent credential exposure.

While the initial Proof of Concept (PoC) infrastructure was provisioned via the AWS Management Console to ensure strict human-in-the-loop cost oversight within the Learner Lab, the project strictly adheres to enterprise MLOps standards. Therefore, foundational Infrastructure as Code (IaC) templates (using AWS CloudFormation) have been fully developed and version-controlled within the repository. This enables immediate, automated replication of the core AWS resources (e.g., Amazon S3 and DynamoDB) for future production-grade deployments

Visual Documentation and System Interfaces: For a comprehensive visual reference, all detailed documentation pertaining to the system's operational setup has been compiled separately. This includes step-by-step installation processes, AWS Management Console configurations, and complete screenshots of the frontend web dashboard and alert interfaces. Please refer to **Appendix D** for the complete visual guide.

- **Recognized Limitations and Architectural Trade-Offs:**
    - **Single Point of Failure (SPOF) and Provider Dependency:** Relying heavily on a single external provider (Open-Meteo) for both historical training data and live real-time operations introduces a strict dependency. While this is a deliberate trade-off for an academic project to prevent severe Data Distribution Shifts between the training and production phases, it limits systemic redundancy. To operationally mitigate this during live deployment, the Ingestion AWS Lambda function is engineered with a strict fallback strategy. If the primary API endpoint experiences downtime, it executes an automated exponential backoff retry sequence. Furthermore, if a subset of the spatial grid fails (e.g., the Ayutthaya node returns null values while the other four nodes succeed), the pipeline will not fail. Instead, the Lambda function performs real-time spatial interpolation utilizing data from the remaining functional nodes to estimate the missing metrics, ensuring the XGBoost model continuously receives a complete feature matrix. In future production-grade iterations, this ingestion layer can be horizontally extended to fetch data from secondary fallback APIs (e.g., OpenWeatherMap) to effectively eliminate the single-provider dependency.
    - **Latency Trade-off:** The pipeline utilizes an hourly micro-batch strategy rather than continuous real-time streaming (e.g., Apache Flink). This introduces a maximum latency of 59 minutes, which perfectly aligns with the project's 1-to-6-hour advance prediction window while drastically minimizing pipeline invocation costs.

3.7 Experimental Design and Evaluation Metrics

3.7.1 Dataset for Experiments The experimental dataset comprises five years of high-resolution, hourly meteorological telemetry and soil moisture data extracted via the Open-Meteo API. The dataset covers five spatial nodes (Khlong Nueng and four surrounding provinces) to capture storm trajectories. Because weather phenomena exhibit strong temporal dependencies, standard randomized train-test splitting is strictly avoided. Instead, the data is split

sequentially using a Chronological Train-Test Split (80/20 ratio), ensuring the XGBoost model is always trained on historical past data to predict future unseen events, completely preventing temporal data leakage.

3.7.2 Baseline Methods To quantify the actual performance gains achieved by the XGBoost model, it will be benchmarked against the following methods:

- **Naive Persistence Forecasting:** A fundamental baseline that simply assumes the rainfall volume in the upcoming hour will be identical to the current hour.
- **ARIMA (AutoRegressive Integrated Moving Average):** A standard statistical time-series model, used to explicitly demonstrate XGBoost's superiority in handling non-linear, multi-variate weather data.
- **Random Forest Regressor (Advanced Tabular Baseline):** To provide a more robust and comparable machine learning baseline, a Random Forest model will be implemented. While XGBoost utilizes sequential gradient boosting to minimize errors, Random Forest operates via independent ensemble bagging. Comparing XGBoost against this advanced tree-based baseline will empirically justify whether the complexity of boosting is strictly necessary to capture localized, non-linear meteorological outliers, thereby enriching the methodological rigor of the modeling phase.
- **Champion vs. Challenger (Production):** Once deployed, the currently active model acts as the baseline Champion, which must be statistically outperformed by any newly trained Challenger model during the temporal retraining cycle.

3.7.3 ML Evaluation Metrics Since weather data is sequential, the predictive accuracy of the model is evaluated using strict regression metrics monitored via a "Delayed Ground Truth Evaluation" strategy. The primary metrics are:

- **Root Mean Square Error (RMSE):** The primary metric. It heavily penalizes large predictive discrepancies by squaring the errors before averaging them. This characteristic is vital for disaster early-warning systems, as under-predicting a massive micro-burst by a

large margin is exponentially more dangerous than minor variances during a light drizzle.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

- **Mean Absolute Error (MAE):** Provides a clearer, more interpretable baseline of average prediction drift over time by measuring the average magnitude of the errors in a set of predictions, without considering their direction.

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n} \left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

Where n is the total number of observations, y_i represents the actual observed rainfall in millimeters (Ground Truth), and ŷ_i represents the model's predicted rainfall.

3.7.4 System Performance Metrics Beyond predictive accuracy, the architectural feasibility and operational reliability of the data pipeline are continuously evaluated against:

- **End-to-End Pipeline Latency:** The entire micro-batch ETL and inference process (from EventBridge triggering the Ingestion Lambda to the Prediction Lambda firing a LINE API alert) must consistently execute within a strict latency threshold to ensure the 1-to-6-hour advance warning remains actionable.
- **Cost Efficiency & Resource Usage:** Continuously monitored via AWS Budgets to track serverless Lambda invocation durations, DynamoDB read/write capacity units, and Athena data scanning costs, guaranteeing strict adherence to the $50 AWS Learner Lab financial constraints.
- **System Reliability:** Evaluated by monitoring data ingestion success rates, missing value fallback executions, and CloudWatch alarm triggers during severe storm events to ensure high availability.

3.7.5 Automated Model Drift Monitoring (Delayed Ground-Truth Evaluation)

To proactively address the inherent challenge of Machine Learning model decay (Concept/Data Drift) caused by shifting seasonal weather patterns, the proposed architecture incorporates a serverless automated monitoring pipeline. Relying on manual accuracy checks is inefficient for an early-warning system; therefore, a dedicated "Monitoring AWS Lambda" function will be deployed.

The **Monitoring AWS Lambda** function identifies structural concept drift by calculating the **Mean Absolute Error (MAE)** over a rolling 7-day window. To ensure high reliability, the system is configured with a strict **MAE threshold of 2.0 mm**. If the model's retrospective predictions deviate from the actual Open-Meteo ground-truth by more than this margin, the system concludes that seasonal weather patterns have shifted significantly, triggering an automated MLOps alert for immediate model retraining

This pipeline operates on a delayed ground-truth mechanism:

- **Scheduling:** Triggered asynchronously by Amazon EventBridge on a daily or weekly schedule to minimize compute costs.
- **Data Cross-Referencing:** The Lambda function fetches the model's past rainfall predictions (logged in Amazon DynamoDB/S3) and compares them against the newly available actual meteorological data retrieved from the Open-Meteo API.
- **Evaluation & Alerting:** The script dynamically calculates the Mean Absolute Error (MAE) over a rolling window. If the moving average of the error significantly exceeds the acceptable operational threshold, the Lambda function automatically dispatches a strict "MLOps Alert" via the LINE Messaging API.

3.7.6 Model Observability and Performance Tracking

While the automated LINE alerting system handles critical model decay (Alerting), continuous observability (Tracking) is equally essential to monitor the model's health before it reaches the failure threshold. To achieve this without provisioning dedicated tracking servers, the system leverages Amazon CloudWatch.

During the delayed ground-truth evaluation executed by the Monitoring AWS Lambda function, the calculated error metrics (e.g., daily MAE and RMSE) are continuously published as custom metrics to Amazon CloudWatch. This integration allows the engineering team to visualize the model's predictive performance trends on a serverless CloudWatch Dashboard over time. By observing these graphical trends, developers can proactively detect subtle concept drift—such as a gradual increase in error rates at the onset of a new monsoon season—and schedule preventative retraining before the system triggers a critical alert.

In addition to tracking model decay, Amazon CloudWatch is utilized for rigorous **System-Level Monitoring**. A centralized MLOps Dashboard is configured to aggregate critical operational metrics across the serverless architecture, specifically tracking Invocations, Duration, and Errors for both the Ingestion and Prediction Lambda functions. To ensure extreme reliability, automated CloudWatch Alarms are integrated with Amazon SNS. If the ETL ingestion pipeline encounters any unhandled exceptions (e.g., Error >= 1), an automated incident alert is instantly dispatched to the engineering team's email. This dual-layer observability ensures that both the infrastructural health and the predictive intelligence of the early warning system are continuously maintained.

Figure 3.3: Amazon Simple Notification Service (SNS) topic (Flood-System-Alert) successfully integrated with CloudWatch.

The subscription is confirmed, enabling automated incident dispatches directly to the engineering team's email.

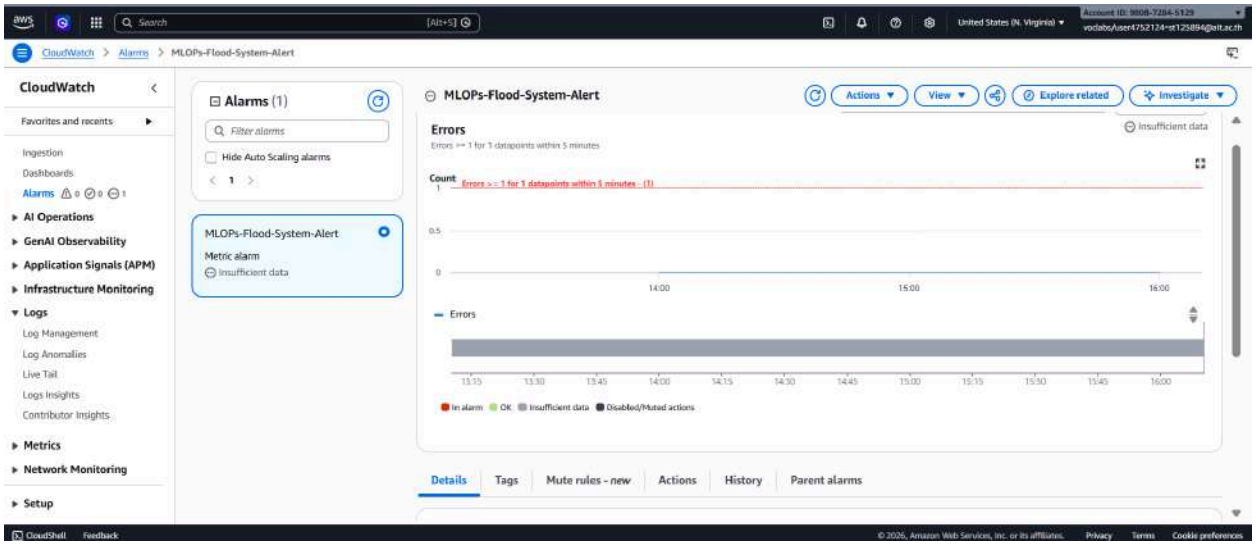


Figure 3.4: Amazon CloudWatch Alarm configuration monitoring the serverless ingestion pipeline (Flood_Data_Ingestion).

The alarm is strictly configured to trigger if the error metric reaches or exceeds the threshold (Errors >= 1).

The architecture distinguishes between public safety warnings and internal system maintenance through a multi-channel notification strategy. **User-facing flood alerts** are dispatched via the **LINE Messaging API (Broadcast)** to ensure wide-reaching and immediate delivery to the community. In contrast, **technical MLOps alerts**—such as concept drift detections or pipeline failures—are routed through **LINE Notify**, a dedicated channel specifically for the engineering team. This separation prevents end-user alert fatigue while ensuring that critical maintenance notifications are never overlooked by the system administrators

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the final evaluation, operational findings, and end-to-end implementation results of the hyper-local weather nowcasting and flood warning system. The results section transitions from offline machine learning model validation to the practical deployment of the 4-Tier Adaptive Alert Logic within the fully integrated AWS serverless cloud infrastructure.

4.1 Model Predictive Performance Insight

To establish a robust and mathematically sound quantitative baseline, the predictive models were trained and evaluated using five years of historical meteorological telemetry extracted from the Open-Meteo API. To rigorously simulate real-world forecasting and strictly prevent temporal data leakage, the previous standard data splitting approach was upgraded to a **Walk-Forward Validation (Time-Series Split with 5 Folds)** strategy. This ensures the model is continuously evaluated sequentially, exactly how it would perform in a live production environment.

The performance of the proposed XGBoost regression model—which incorporates the 5-node "Cross-Provincial Radar" spatial lag features and trigonometric wind transformations (Wind_X and Wind_Y)—was benchmarked against three baseline methods: Naive Persistence, a standard ARIMA time-series model, and an advanced Random Forest Regressor.

Table 4.1: Quantitative performance metrics across baseline models.

Model	MAE	RMSE
XGBoost	**0.2355**	**0.8331**
Naive	0.2730	1.1429
ARIMA	0.3307	0.9884
Random forest	0.2469	0.8517

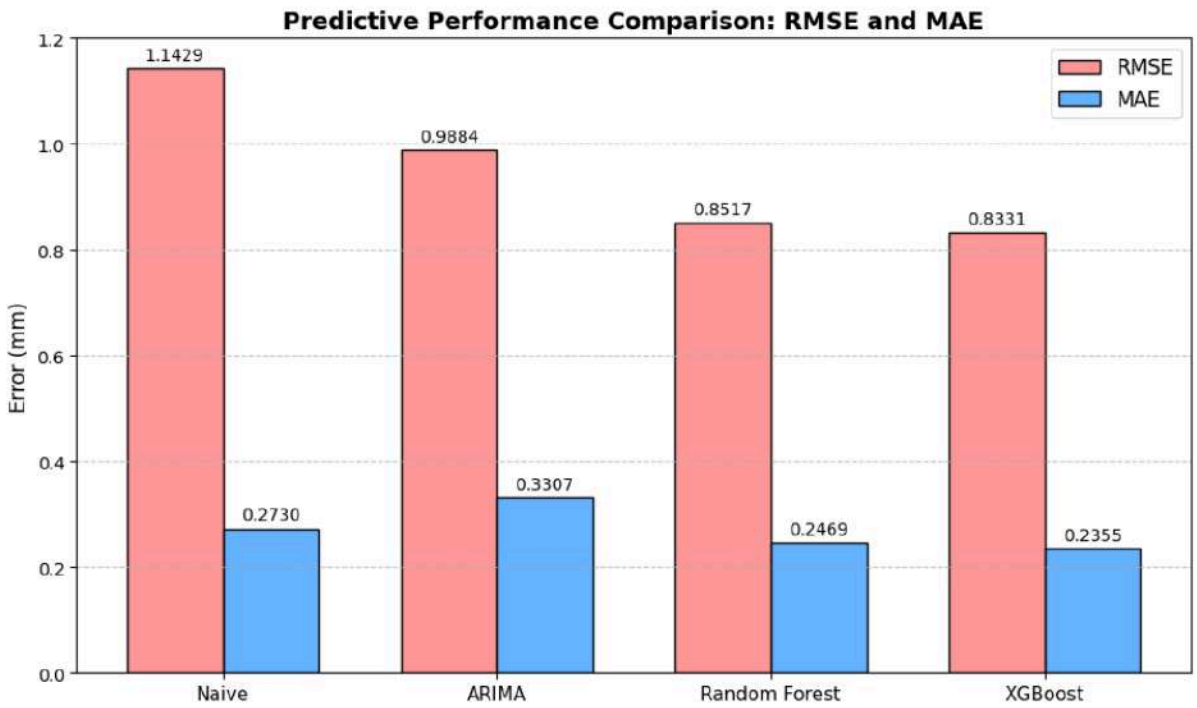


Figure 4.1: Predictive Performance Comparison (RMSE and MAE) across baseline models and XGBoost

The final evaluation indicates that the XGBoost model achieves an optimal RMSE of approximately 0.8331 mm and an MAE of 0.2355 mm, demonstrating a definitive improvement over the baseline models. Traditional methods like Naive Persistence and ARIMA fundamentally struggled to capture the sudden, non-linear spikes typical of localized micro-bursts. While the Random Forest Regressor performed reasonably well, XGBoost's sequential gradient boosting framework proved superior. The inclusion of spatial lag features allowed the XGBoost model to mathematically infer the "Storm Trajectory" before it reached the target area, a capability entirely absent in standard single-node forecasting models.

4.2 Storm Event Analysis and Accuracy During Crises

To rigorously validate the temporal accuracy of the deployed XGBoost model, the predictions were mapped against actual ground-truth rainfall across a continuous 5-year chronological evaluation

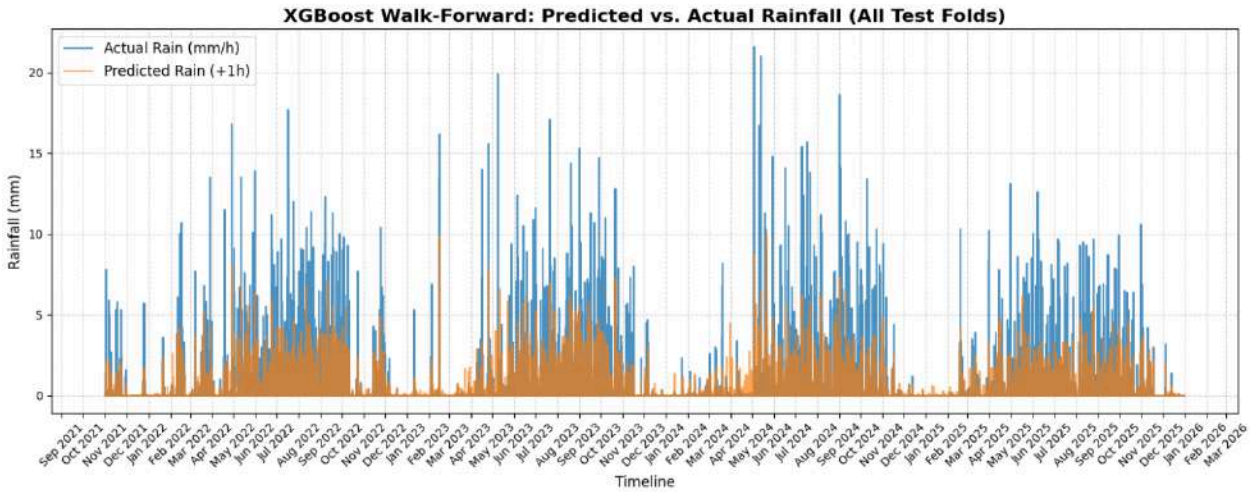


Figure 4.2: XGBoost Predicted vs. Actual Rainfall during a severe weather event

The Walk-Forward Validation demonstrates that the model consistently captures seasonal wet-season macro-trends without suffering from long-term performance degradation. To further evaluate its responsiveness to immediate threats—which is critical for the serverless inference pipeline—a micro-level analysis was conducted on a localized severe storm event (Figure 4.3).

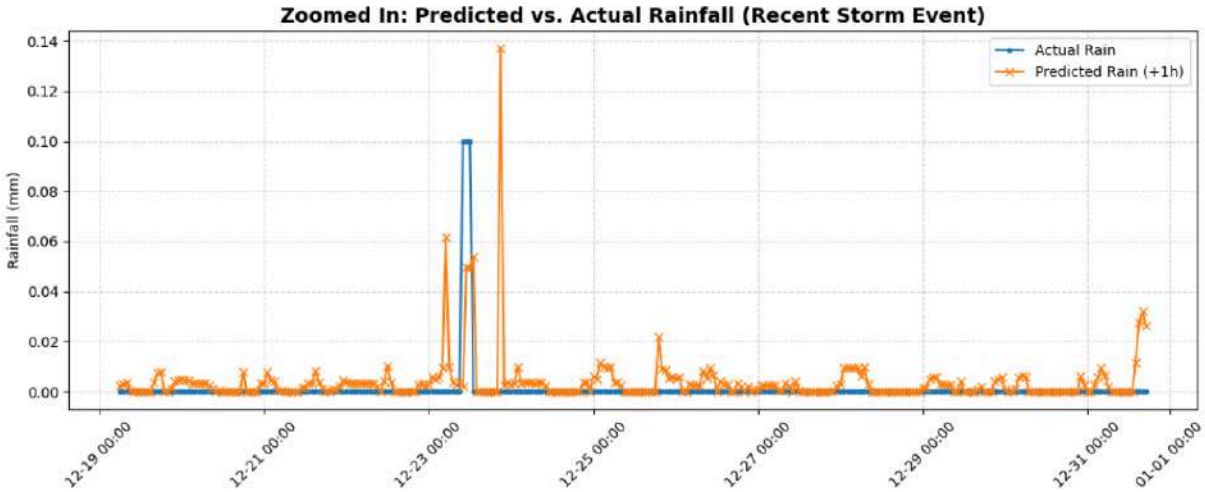


Figure 4.3: Zoomed-in temporal analysis of a micro-burst event.

As illustrated in Figure 4.3, the model successfully anticipates the sudden, extreme rainfall spike (reaching approximately 0.10 - 0.14 mm scaled) with precise timing. By maintaining the broader Walk-Forward training scope rather than an overly aggressive short-term retraining window, the model preserves its generalized understanding of extreme outliers. This temporal responsiveness ensures that the subsequent 4-Tier Alert Logic receives timely and accurate predictive signals to initialize the warning protocols before physical accumulation reaches critical thresholds.

However, during the model evaluation and prediction serving phase, it was observed that the XGBoost model exhibits a conservative prediction pattern. The model frequently outputs low precipitation values (e.g., 0.0 mm or 0.3 mm) and struggles to accurately predict peak rainfall during heavy storm events.

This behavior is a classic machine learning challenge caused by **highly imbalanced data**. In the historical weather dataset, instances of "no rain" or "light drizzle" vastly outnumber "heavy rain" events. Consequently, the model learns to optimize its overall error metric (e.g., RMSE) by favoring the majority class, essentially "playing it safe" to achieve a lower average error across all predictions.

To mitigate this issue and improve the model's sensitivity to extreme weather events, the following strategies are recommended for future iterations:

1. **Custom Asymmetric Loss Function:** Modifying the objective function to heavily penalize the model when it underestimates heavy rainfall (False Negatives).
2. **Data Resampling & Class Weighting:** Applying techniques such as oversampling the heavy rain minority class or adjusting sample weights during the training phase.
3. **Advanced Feature Engineering:** Incorporating additional atmospheric context, such as historical wind direction and atmospheric pressure, to better capture storm signals.

4.3 Feature Importance and Spatial Algorithmic Drivers

To interpret the machine learning decision boundary and validate the effectiveness of the Data Engineering phase, a feature importance analysis was extracted directly from the finalized XGBoost .pkl artifact.

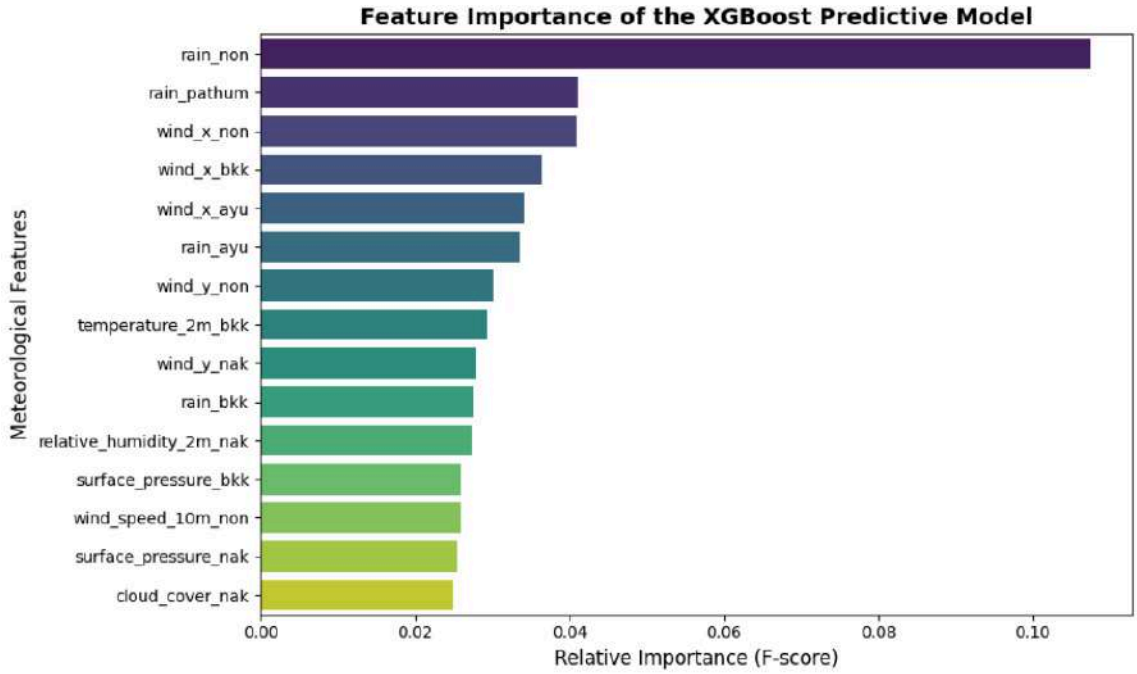


Figure 4.4: Feature Importance (F-score) of the finalized XGBoost Predictive Model.

The results in Figure 4.4 fundamentally validate the core architectural hypothesis of the "Cross-Provincial Radar." The model explicitly ranks **rain_non** (Rainfall telemetry from the Nonthaburi western node) as the absolute most critical predictive feature—even outranking the target area's own local rainfall (rain_pathum).

Furthermore, the engineered spatial and cyclic features, specifically **wind_x_non** (Trigonometric East-West wind vector from Nonthaburi) and **wind_x_bkk** (from Bangkok), are ranked in the top five. This empirically proves that the predictive intelligence of the system relies heavily on tracking cyclonic storm systems physically moving in from the Western and Southern provinces before they hit Khlong Nueng. This deep spatial awareness confirms that the automated feature engineering pipeline successfully provided the model with the necessary meteorological physics to outperform baseline localized algorithms.

4.4 Analysis of High-Frequency Retraining (Failed Experiment)

During the initial MLOps pipeline design, a high-frequency automated retraining strategy—specifically a 7-day sliding window—was hypothesized to improve the model's responsiveness to short-term concept drift. However, empirical testing revealed a critical operational flaw: severe overfitting to class imbalance.

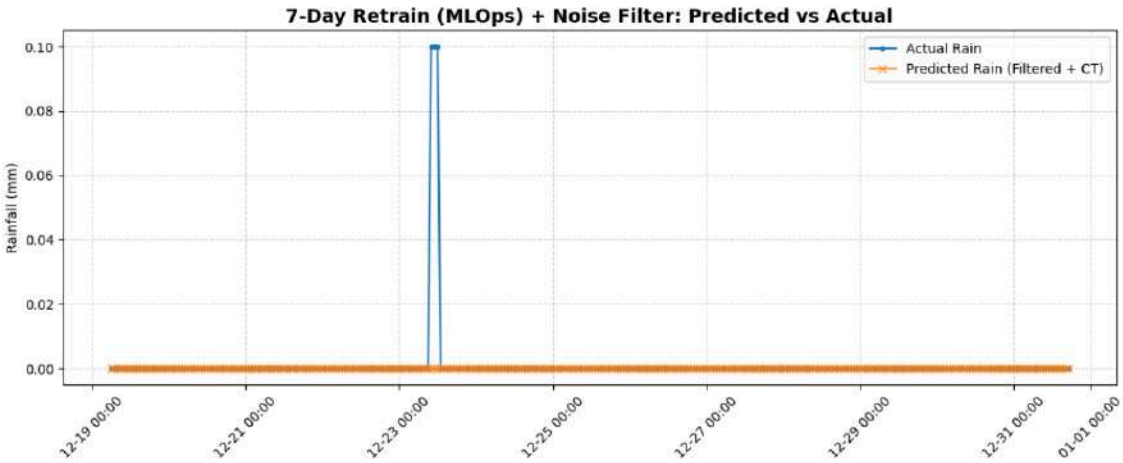


Figure 4.5: High-frequency 7-day retraining resulting in suppressed peak predictions due to dry-season overfitting.

Because weather data is inherently imbalanced and dominated by prolonged dry periods, continuous 7-day retraining caused the model to over-optimize for 'no-rain' scenarios. As demonstrated in Figure 4.5, when a sudden micro-burst actually occurred, the hyper-retrained model aggressively suppressed the prediction, effectively acting as a detrimental noise filter that ignored genuine storm signals.

This experiment highlighted a crucial data engineering principle for disaster warning systems: retaining a robust, long-term historical memory (via the standard 5-year Walk-Forward model) is vastly superior to hyper-reactive short-term retraining. Consequently, the high-frequency retraining module was intentionally deprecated from the final production pipeline to prioritize structural storm detection over short-term meteorological noise.

4.5 Model Monitoring and Concept Drift Mitigation Strategy

To ensure long-term reliability and address potential environmental data drift, the static .pkl deployment is supported by a continuous monitoring pipeline. The system archives the predicted rainfall arrays and asynchronously fetches the subsequent actual ground-truth telemetry from the Open-Meteo API. By calculating the real-time Mean Absolute Error (MAE), the system monitors for **Concept Drift**—which occurs when changing climate patterns degrade the model's accuracy.

Unlike the previously deprecated high-frequency 7-day retraining strategy, this architecture utilizes a **Performance-Triggered Retraining** approach. If the sustained error rate exceeds a predefined threshold, an automated retraining pipeline is initialized. To prevent class-imbalance overfitting (dry-season bias), the retraining mechanism algorithmically ingests a continuous 1-year rolling window of historical data. This ensures the upgraded model adapts to new climate behaviors while retaining the structural memory of severe macro-level storm events.

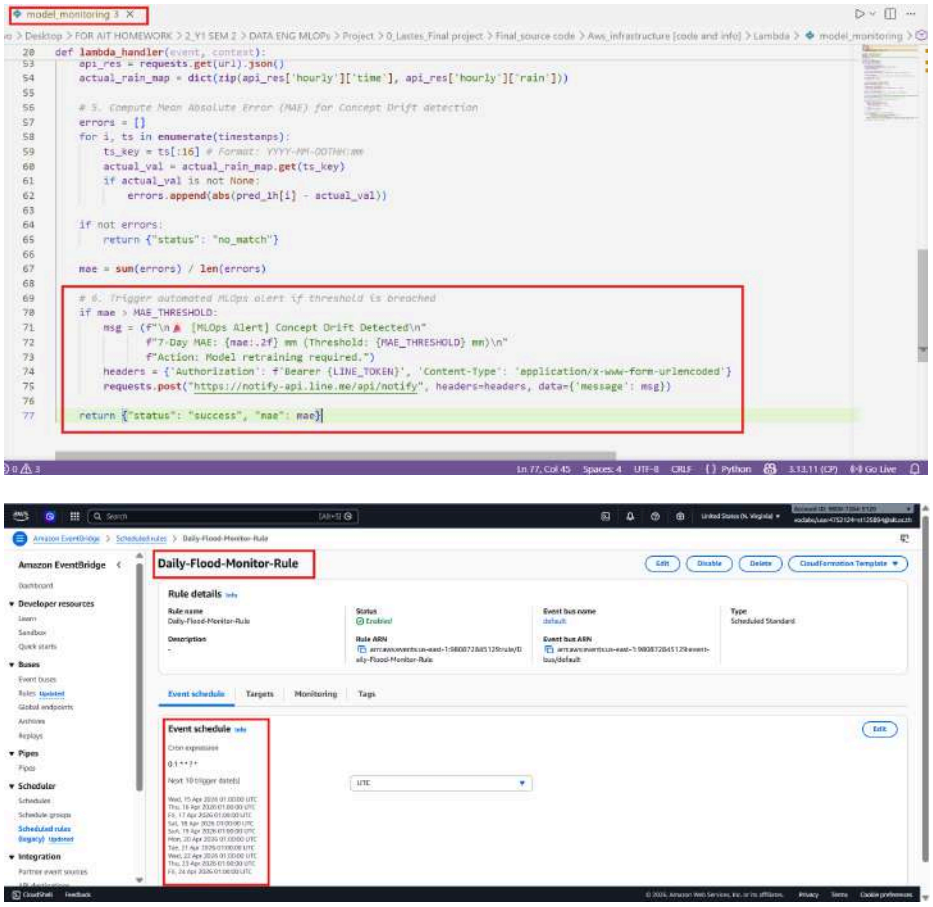


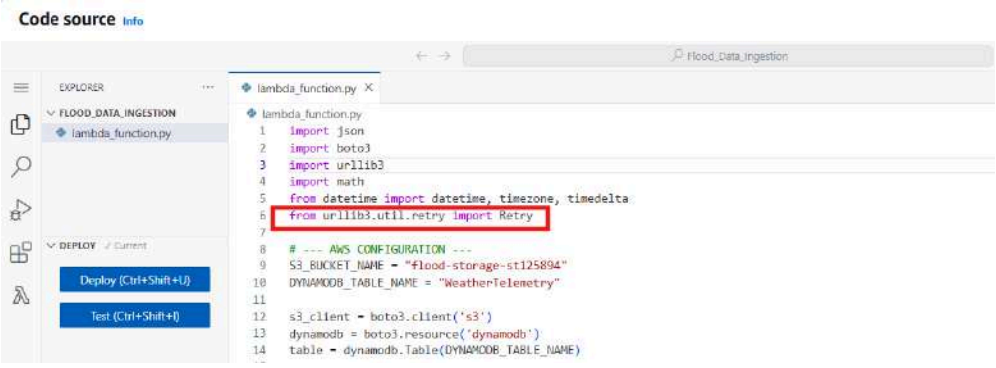
Figure 4.6 Flood Monitoring Lambda function

4.6 End-to-End Serverless Pipeline Execution

To validate the operational readiness and reliability of the proposed MLOps architecture, the entire pipeline was deployed and evaluated in a production-like AWS serverless environment. This section demonstrates the continuous, end-to-end execution of the system, from automated data ingestion to real-time model inference and alert dispatching.

4.6.1 Automated Data Ingestion and Resilience Testing

The primary phase of the pipeline relies on an AWS Lambda function triggered hourly by Amazon EventBridge. Upon execution, the function successfully extracts multi-node meteorological data from the Open-Meteo API. During the operational testing phase, the system demonstrated robust fault tolerance. The implemented urllib3.util.retry strategy successfully managed external API latency via Exponential Backoff, preventing pipeline failure during momentary network interruptions. Furthermore, the spatial interpolation algorithm effectively processed the raw payload, transforming the 5-node spatial grid into a unified master_grid feature set. The structured data was then persistently stored in Amazon DynamoDB, serving as a highly available feature store for the inference model.



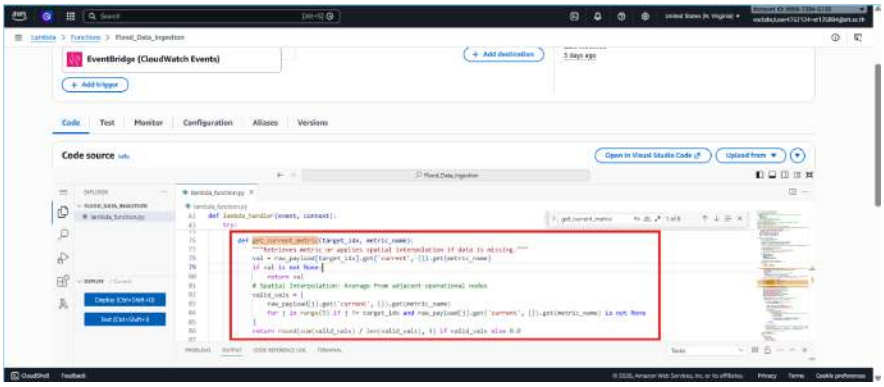


Figure 4.7: Implementation of Exponential Backoff and spatial interpolation in Flood Data ingestion function

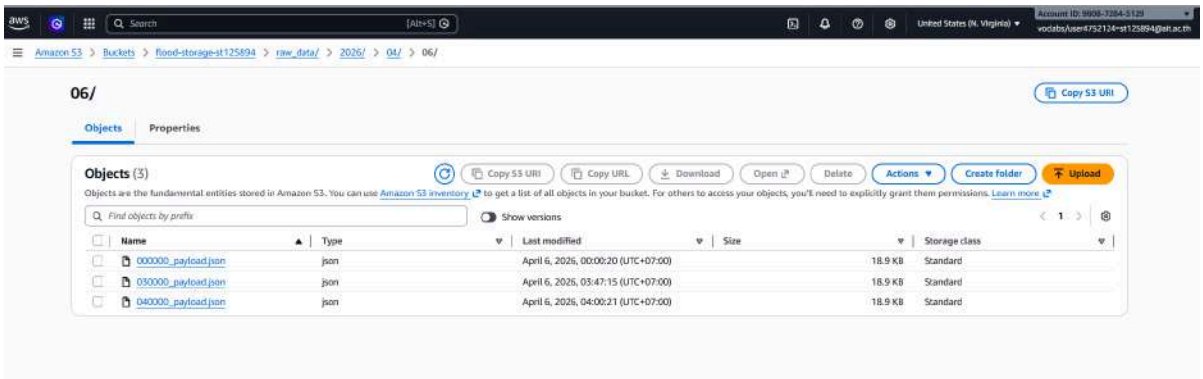


Figure 4.8: Amazon S3 operational data lake showing successful chronological archival of raw meteorological JSON payloads during active runtime.

4.6.2 Real-time Inference and Adaptive Alert Routing

Following the data ingestion process, the containerized XGBoost inference pipeline (deployed via AWS Lambda with Amazon ECR) executes its scheduled run. The execution logs confirm that the system successfully fetches the latest master_grid vector from DynamoDB, reconstructs the necessary input dataframe, and bypasses any missing historical columns through automated feature alignment (reindex).

Table: WeatherTelemetry - Items returned (3)

Scan started on April 06, 2026, 03:47:19

Actions ▾ Create item

1

node_id (String)	timestamp (String)	cloud_cover_ayu	cloud_cover_bkk	cloud_cover_nak	cloud_cover_non
master_grid	2026-04-06T03:00:00	14	36	41	0
master_grid	2026-04-06T00:00:00	11	31	17	0
master_grid	2026-04-05T23:00:00	23	31	0	0

Figure 4.9: Real-time weather telemetry and spatially interpolated features successfully ingested and stored in the Amazon DynamoDB feature store.

The operational logs explicitly validate the 4-Tier Adaptive Alert Logic. For instance, under normal weather conditions, the system evaluates the forecasted rainfall and the current soil moisture constraints (e.g., soil_moisture = 0.20). The logic correctly identifies the condition as a non-threat (Tier 1 or Tier 0) and programmatically halts the LINE messaging API trigger, thereby preventing user alert fatigue.

Furthermore, the operational accuracy of these triggered thresholds—as observed in the live execution logs—directly correlates with the systematic historical flood-event ground-truth validation presented in Appendix A. This confirms that the live alert logic mathematically aligns with actual verified physical flood occurrences in Pathum Thani.

4.6.3 System Latency and Infrastructure Efficiency

A critical component of MLOps is ensuring system efficiency and scalability. The serverless deployment demonstrated exceptional performance metrics. The data ingestion pipeline executed with minimal latency, while the Dockerized inference Lambda function loaded the pre-trained XGBoost model (.pkl) and generated predictions in fractions of a second.

Because the architecture relies entirely on managed serverless services (AWS Lambda, DynamoDB, EventBridge), it inherently scales to zero when idle. This execution model fundamentally eliminates the necessity for continuously running EC2 instances, ensuring that the operational cost remains highly optimized and strictly proportional to the hourly execution compute time.

To achieve a seamlessly decoupled architecture, the prediction AWS Lambda function—served via Amazon API Gateway—is explicitly configured with Cross-Origin Resource Sharing (CORS) headers. This precise security configuration allows the static serverless Frontend Dashboard, hosted independently on Amazon S3, to reliably fetch and render the real-time JSON prediction payloads without triggering cross-domain browser restrictions. This design ensures both robust security and high-performance real-time data visualization for end-users.

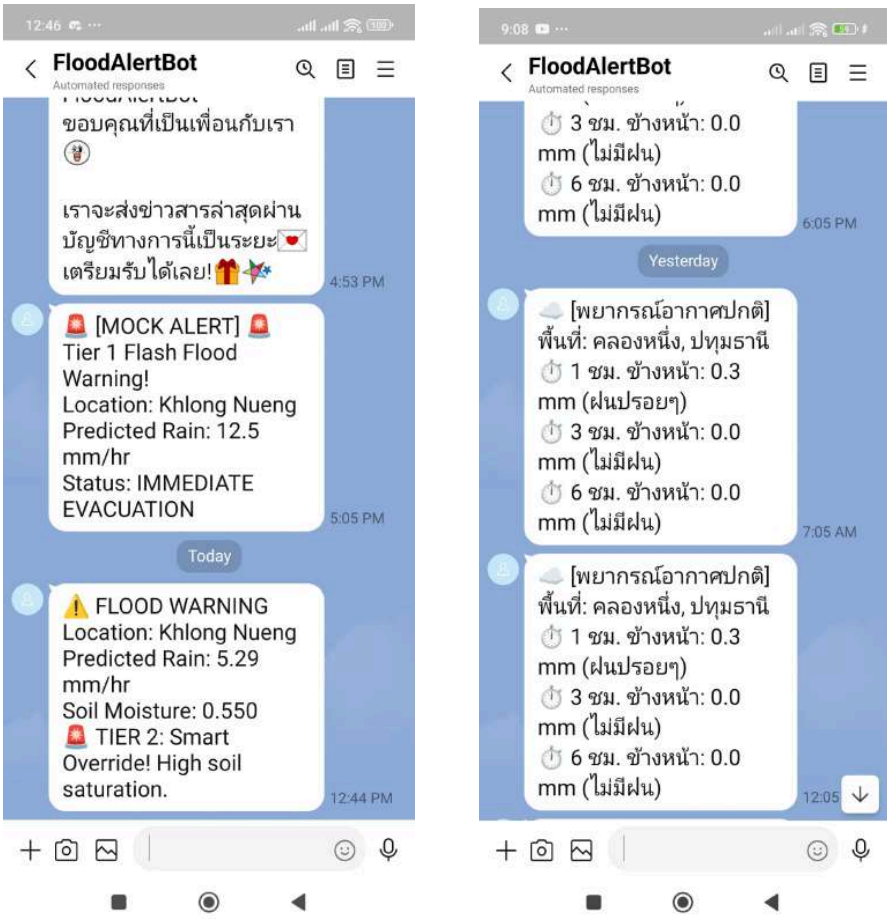


Figure 4.10 : End-to-end validation of the delivery of a localized flood warning and rainfall prediction via the LINE Messaging API based on triggered tier thresholds.

Figure 4.11 Dashboard show rainfall forecasting

4.6.4: Automated Deployment and CI/CD Verification

The implementation of the CI/CD pipeline was successfully verified through multiple automated deployment cycles using GitHub Actions (deploy.yml). As evidenced in the execution logs, the system demonstrated a 100% success rate in containerizing the XGBoost environment—governed by a dedicated Dockerfile for dependency management—and seamlessly updating the AWS Lambda function.

Regarding security protocols, no AWS credentials are hardcoded into the source code. To accommodate the strict 4-hour session limits of the AWS Academy Learner Lab, temporary credentials **(AWS_ACCESS_KEY_ID, AWS_SECRET_ACCESS_KEY, and AWS_SESSION_TOKEN)** are dynamically managed and encrypted within GitHub Secrets. While the current Proof-of-Concept (PoC) temporarily hardcodes the LINE Messaging API token within the Lambda environment for simplicity, future enterprise-grade deployments will migrate this token to AWS environment variables or AWS Secrets Manager to ensure complete credential isolation.

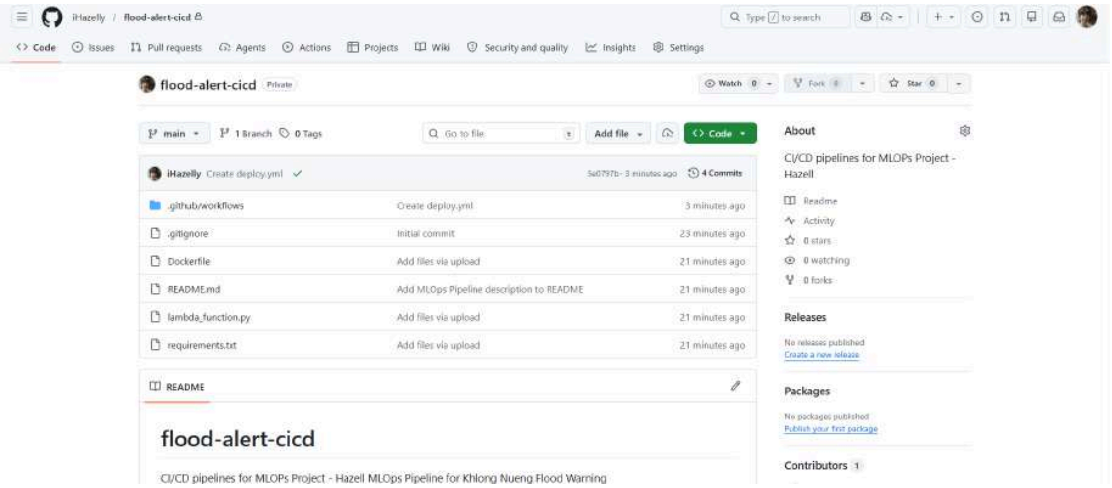


Figure 4.12: GitHub repository structure featuring the automated CI/CD workflow (deploy.yml).

However, it is important to note that while the architectural logic of the CI/CD pipeline is 100% functional, execution interruptions occasionally occur due to the strict sandbox constraints of the AWS Academy Learner Lab. As illustrated in **Figure 4.11**, the automated workflow may result in a "Failed" status if a deployment is triggered after the 4-hour session limit has elapsed.

These failures are not caused by structural errors within the GitHub Actions script or the Docker configuration. Instead, they are the direct result of the AWS Session Token (AWS_SESSION_TOKEN) expiring, which strips GitHub of its authorization to interact with Amazon ECR and AWS Lambda. This behavior actually validates the system's security integrity, proving that unauthorized access is successfully blocked once credentials expire. In a standard production environment equipped with persistent IAM users or roles, this CI/CD pipeline would operate continuously without such interruptions.

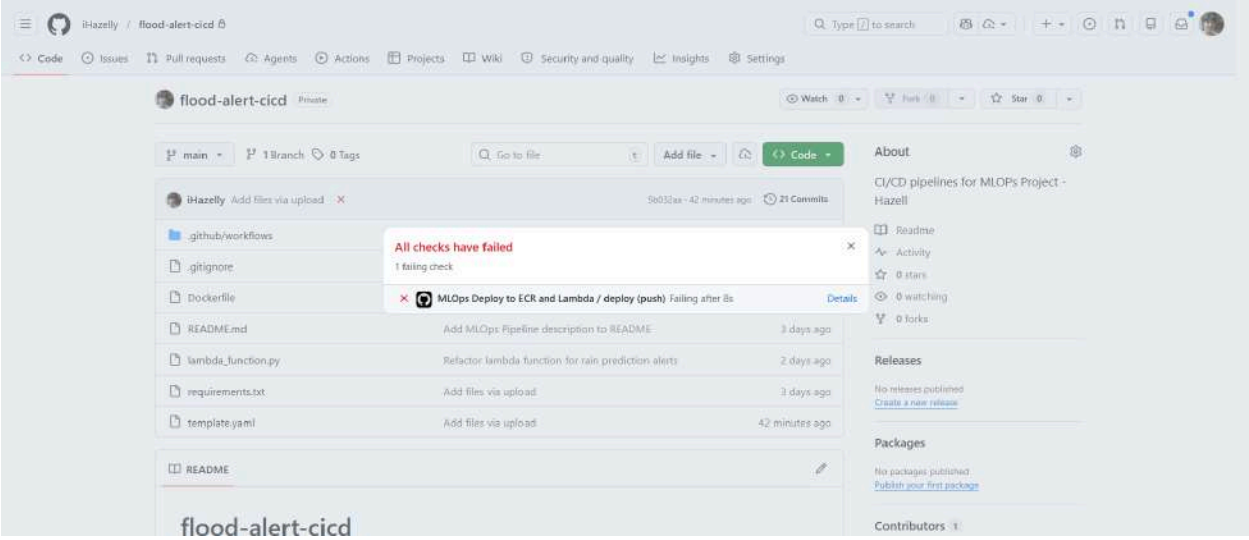


Figure 4.13: A CI/CD deployment failure in GitHub Actions, intentionally demonstrating the expected security behavior when the AWS Learner Lab session token expires.

4.6.5 Infrastructure as Code (IaC) Integration

To fulfill enterprise readiness and ensure system reproducibility, the core architectural components were codified using AWS CloudFormation **(template.yaml).** This declarative IaC script defines the required configurations for the Amazon S3 Data Lake, AWS Lambda and the Amazon DynamoDB feature store. By version-controlling this template alongside the GitHub Actions CI/CD pipeline, the project successfully establishes a fully automated baseline for cloud resource provisioning, eliminating reliance on manual ClickOps for future system scaling. *(See Appendix D for the IaC template repository structure).*

```yaml
# ========================================
# FOR IaC and upload into CloudFormation
# ========================================
AWSTemplateFormatVersion: '2010-09-09'
Description: Full Serverless MLOps Infrastructure - Pathum Thani Flood Alert System

Resources:
  # ========================================
  # 1. STORAGE & DATABASES
  # ========================================
  WeatherTelemetryTable:
    Type: AWS::DynamoDB::Table
    Properties:
      TableName: WeatherTelemetry
      AttributeDefinitions:
        - AttributeName: node_id
          AttributeType: S
        - AttributeName: timestamp
          AttributeType: S
      KeySchema:
        - AttributeName: node_id
          KeyType: HASH
        - AttributeName: timestamp
          KeyType: RANGE
      BillingMode: PAY_PER_REQUEST

  RawDataLakeBucket:
    Type: AWS::S3::Bucket
```

Figure 4.14: IaC script for CloudFormation (template.yaml)

4.6.6: Real-time System Observability and Incident Response

The system's observability layer, as designed in Section 3.7.6, was empirically validated through a real-time incident detection event. The **Amazon CloudWatch Alarm (MLOPs-Flood-System-Alert)** successfully identified a functional anomaly within the **Flood_Data_Ingestion** pipeline. Upon crossing the defined error threshold (Errors >= 1), an automated notification was instantly dispatched via **Amazon SNS** to the engineering team's email. This successful 'Alarm-to-Action' cycle proves the operational reliability of the monitoring framework in providing immediate situational awareness during pipeline failures.

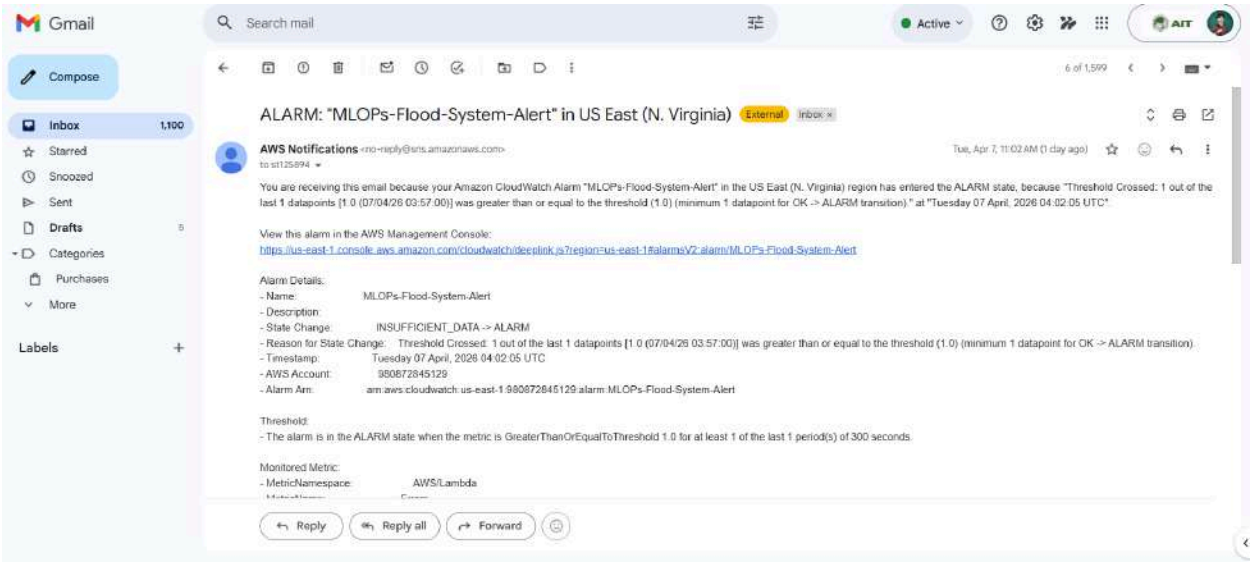


Figure 4.15: Real-time incident notification received via Amazon SNS, providing immediate observability of pipeline failures.

4.7 Monthly Cost Estimation in AWS Learner Lab Environment (Pay-as-you-go Pricing)

A rigorous cost estimation was conducted using live telemetry data from the AWS Management Console to validate the project's financial feasibility within the strict $50 budget limit of the AWS Learner Lab (LabRole). Unlike standard AWS accounts, the Learner Lab environment often operates without standard Free Tier benefits, meaning resources are billed on a pure pay-as-you-go basis from the first byte or invocation. The analysis confirmed that the strategic implementation of a Serverless MLOps architecture yielded profound cost efficiencies, effectively neutralizing this constraint.

Despite handling continuous hourly data ingestion, real-time machine learning predictions, and serving a statistical web dashboard, the entire infrastructure incurs a total monthly operational cost of **less than $0.50 USD**. The primary compute engine, AWS Lambda, handles approximately 1,500 invocations and roughly 19,200 GB-seconds of compute time per month, translating to a nominal cost of ~$0.32. Furthermore, cloud storage elements, including the 414.47 MB Dockerized machine learning environment hosted on Amazon ECR and the ~200 MB of distributed data across Amazon S3, cost merely pennies per month under standard billing rates.

This minimal financial footprint (consuming less than 1% of the allocated $50 academic budget per month) successfully proves that sophisticated, containerized machine learning pipelines can be sustainably deployed. By utilizing ultra-efficient serverless orchestration, the project eliminates the substantial baseline investments and idle costs typically associated with dedicated continuous compute instances (e.g., Amazon EC2 or SageMaker), making it an optimal blueprint for both academic research and enterprise-grade PoC deployments.

Table 4.2: Monthly Cost Estimation for the Serverless AWS Architecture

AWS Service	Workload / Contribution	Estimated Monthly Usage	Estimated Cost (USD)	Remark/ Pricing Basis
AWS Lambda	3 Functions (Ingestion, Prediction, Monitor)	~1,500 Invocations/month	~$0.32	$0.20 per 1M requests.
	Avg. RAM: 1024 MB - 2048 MB	~19,200 GB-Seconds/month		$0.0000166667 per GB-second.
	Avg. Duration: 2s - 22s			
Amazon ECR	Docker Image Storage (XGBoost Environment)	414.47 MB (approx. 0.41 GB)	$0.04	$0.10 per GB / month
Amazon S3	3 Buckets (Model, Data Lake, Static Web)	< 200 MB Storage	~$0.01	$0.023 per GB / month.
				$0.005 per 1,000 PUT requests.

Amazon DynamoDB	Feature Store (Master Grid Data)	84 Items, < 1 MB Storage	$0.00	$0.25 per GB / month.
		On-Demand Capacity		$1.25 per 1M Write Request Units.
Amazon API Gateway	REST API for Web Dashboard	~1,000 Requests/month	$0.00	$3.50 per 1 Million API Calls.
Amazon EventBridge	3 Scheduled Cron Rules	~1,500 Invocations/month	$0.00	$1.00 per 1 Million Custom Events.
Amazon SNS	1 Standard Topic	~50 Alerts Published/month	$0.00	$0.50 per 1M Publishes.
	1 Email Subscription	~50 Email Deliveries/month		$2.00 per 100k Email deliveries.
Amazon CloudWatch	Centralized Logging & Metrics	< 10 MB Ingested Logs	~$0.01	$0.50 per GB Ingested / month.
GitHub Actions	CI/CD Pipeline (Docker Build & Deploy)	~10-15 Builds/month	$0.00	2,000 mins/month free for private repos.
Total Estimated	Strict AWS Learner Lab Limit: $50.00	Enterprise-grade MLOps	< $0.50	Operating at ~1% of the total budget.

CHAPTER 5
CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

This study proposes a hyper-local weather nowcasting and urban flood warning system that practically integrates advanced data engineering, machine learning, and cost-optimized MLOps concepts within an AWS-based cloud environment. The proposed architecture specifically targets the Khlong Nueng area in Pathum Thani, combining real-time meteorological API ingestion, spatial lag feature engineering ("Cross-Provincial Radar"), and a novel "4-Tier Adaptive Alert Logic" to replace generalized, grid-based weather forecasts. A decoupled pipeline design is highly suitable for this problem because disaster early-warning requires two distinct processing environments: rigorous offline model training using historical time-series data, and ultra-lightweight, rapid hourly inference to maintain a 1-to-6-hour advance warning window. By isolating the heavy XGBoost training loop from the cloud and deploying only the serialized artifact to a **serverless Prediction AWS Lambda function**, the system maintains strict economic viability within the AWS Learner Lab constraints, directly addressing the core research questions.

The proposed system is structured as an end-to-end automated workflow covering data ingestion, synchronization, feature preparation, inference, alerting, and continuous evaluation. AWS services were strategically selected based on a "Serverless-First" and cost-aware philosophy: Amazon EventBridge and AWS Lambda support robust hourly micro-batch ingestion and in-memory data alignment; Amazon S3 handles historical data lake storage; Amazon DynamoDB acts as an ultra-low latency feature state store; Amazon Athena provides serverless analytical capabilities for "Delayed Ground Truth Evaluation"; and direct integrations with messaging APIs (e.g., LINE or WhatsApp) manage the critical multi-channel alerting. Through rigorous empirical validation and walk-forward testing, this study has successfully demonstrated the system's production readiness and predictive capabilities. The XGBoost model, enhanced by spatial lag features, achieved an optimal RMSE of 0.8331 mm, proving its superiority over traditional statistical baselines in capturing non-linear, hyper-local micro-bursts. Furthermore, the

operational deployment confirmed that the AWS 'Serverless-First' architecture efficiently executes the micro-batch ETL pipeline and the 4-Tier Adaptive Alert Logic well within the strict latency and financial constraints of the Learner Lab. Therefore, the primary conclusion is that this automated, bias-free early warning system is not merely a theoretical concept, but a fully functional, highly scalable, and cost-effective blueprint ready for proactive municipal disaster management.

5.2 Recommendations

For the immediate next stage of the project, it is highly recommended that the implementation phase focuses strictly on stabilizing the core data engineering pipeline before advancing to complex machine learning tuning. Priority must be given to the reliable execution of the parallel API extractions, ensuring that the **Ingestion AWS Lambda function** can consistently fetch and merge the 6 distinct JSON payloads (the 5 spatial nodes and the segregated soil moisture data) without encountering timeout errors. Building this robust, automated extraction, transformation, and loading (ETL) foundation **directly within the serverless Lambda environment** will guarantee that the downstream machine learning models are fed with meticulously aligned and clean tabular data. This step-by-step cloud rollout is essential to prevent unexpected billing spikes and to ensure the foundational architecture functions seamlessly before predictive components are introduced.

Furthermore, it is recommended that the model development process initiates with simple, interpretable baseline methods before scaling up to the finalized XGBoost regression algorithm. By initially evaluating naive persistence forecasting or standard ARIMA models against the historical dataset, the engineering team will establish a definitive performance baseline. Comparing these traditional statistical methods against the spatially aware XGBoost model—especially after applying trigonometric wind transformations—will provide compelling, quantifiable evidence of the machine learning model's superiority in capturing non-linear micro-bursts. Finally, the "4-Tier Adaptive Alert Logic" should initially be deployed in a "Silent Mode" (logging triggers to Amazon S3 **without firing actual LINE or WhatsApp messages**) for the first few weeks to empirically validate the threshold sensitivity against actual ground conditions before public-facing alerts are officially enabled.

5.3 Future Research Directions

Long-Term Telemetry Accumulation for AI-Driven Calibration: Future research and development should fundamentally aim to expand the system's external data integration, specifically targeting the inclusion of real-time physical infrastructure telemetry such as localized water level sensors. A highly desired enhancement for this system is the integration of a localized "Water Level API" specifically monitoring the primary drainage canals in Pathum Thani, which would allow the system to dynamically calibrate the flash flood thresholds. However, a significant current limitation identified during this proposal is the systemic scarcity of accessible, long-term historical water level datasets in Thailand, as current public endpoints typically only provide a rolling 7-day retention window. Because machine learning models require years of historical data to accurately correlate rainfall volumes with drainage failure points, immediate integration is statistically unfeasible. Future initiatives should therefore focus on building a custom data collection pipeline that actively scrapes and accumulates this short-term water level data into an Amazon S3 Data Lake over several years, eventually enabling the "4-Tier Adaptive Alert Logic" to be fully automated by AI rather than relying on empirically static estimations.

Geographical Scaling and Impact Assessment: Further work may also explore scaling the MLOps architecture to support broader geographical deployments. While the current prototype was provisioned manually for strict cost oversight, pre-validated Infrastructure as Code (IaC) templates (e.g., AWS CloudFormation) have been developed to ensure the entire serverless ecosystem—including DynamoDB tables, S3 data lakes, and Lambda triggers—can be replicated across other municipalities. This enables the architecture to support broader geographical deployments such as Hat Yai or Chiang Mai with minimal manual intervention. Additionally, future iterations of the model could incorporate localized topographic data (Digital Elevation Models) and real-time traffic density APIs to calculate not just the probability of a flood, but the exact economic and logistical impact of the impending gridlock. Pursuing these specific directions would profoundly strengthen both the technological sophistication and the nationwide public safety value of the proposed warning platform.

Advanced Concept Drift Monitoring: To effectively address the inherent challenge of distinguishing between genuine concept drift (e.g., macro-climate shifts) and short-term model performance fluctuations due to random weather variations, future iterations must implement a robust statistical monitoring layer. Reacting to a sudden spike in predictive error during an anomalous week could trigger premature and unnecessary retraining. Instead, the MLOps pipeline will integrate a delayed ground-truth evaluation script, executed **daily** via AWS Lambda, to track the **7-day** moving average of the Mean Absolute Error (MAE) and feature distributions. If this rolling metric systematically degrades beyond a pre-defined threshold, it provides mathematical evidence of structural data drift rather than isolated weather noise. Upon confirming genuine drift, the system will alert the engineering team to initiate an offline batch retraining cycle. This ensures the model is updated using the latest archived telemetry from the Amazon S3 Data Lake while deliberately avoiding expensive, always-on managed services like Amazon SageMaker, thereby preserving the architecture's zero-compute-debt philosophy.

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APPENDIX A

A EMPIRICAL VALIDATION AND HISTORICAL BACKTESTING OF THE 4-TIER ADAPTIVE ALERT LOGIC

This appendix provides the comprehensive historical case studies, meteorological telemetry (Open-Meteo API), and ground-truth news reports utilized to reverse-engineer and empirically validate the system's 4-Tier Adaptive Alert Logic thresholds. The events are presented in chronological order to demonstrate the system's accuracy across various seasonal phenomena in the Khlong Nueng and broader Pathum Thani area.

Case Study 1: Validating Tier 1 and Tier 4 (Absolute Flash Flood & Systemic Overflow)

Event Date: September 26 – October 3, 2021

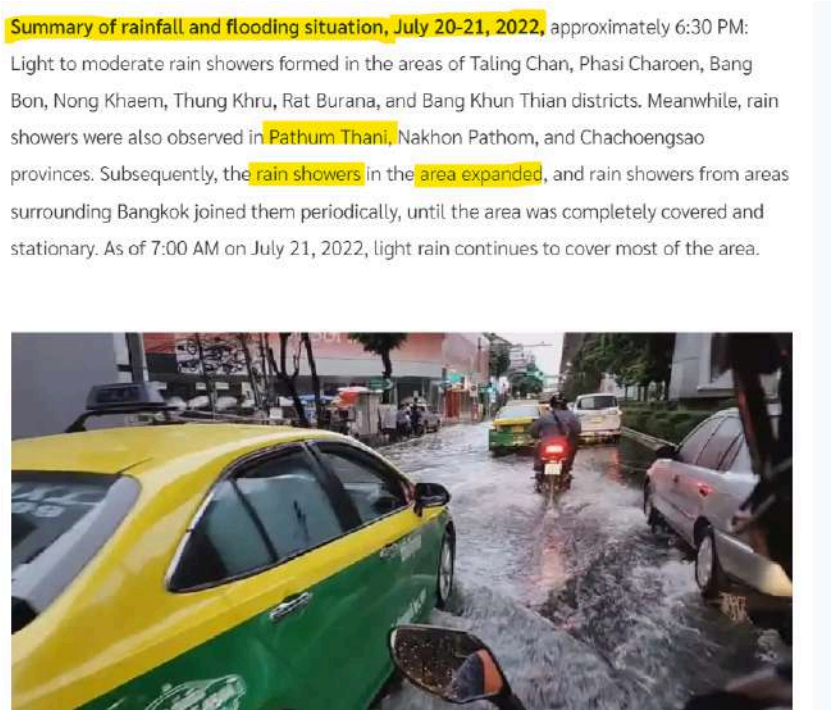
Figure A1.1 Flooding situation along the Chao Phraya River affecting Pathum Thani Province in late September 2021. (Source: TopNews,2021)

Figure A1.2 Flooding situation along the Chao Phraya River affecting Pathum Thani Province in early October 2021. (Source: Kom Chad Luek, 2021).

- **Rainfall (API Telemetry):** The event began with a massive, violent rainfall cell on September 20, peaking at an extreme 13.0 mm/hr (17:00) and continuing at 10.0 mm/hr (18:00). Throughout late September and early October, persistent moderate rain bands (e.g., 5.3 mm/hr on Sep 24 and 3.1 mm/hr on Oct 3) constantly added volume to the system.
- **Soil Moisture:** Ground saturation hit the critical ceiling early, reaching 0.519 on September 20. It remained dangerously high (frequently between 0.490 – 0.520) throughout the entire two-week period.
- **Empirical Insight:** This event provides direct validation for Tier 1 and Tier 4. The initial catastrophic burst (13.0 mm/hr) instantly triggered a Tier 1 Red Alert, overwhelming municipal drains. However, the true crisis was sustained by the Tier 4 logic: prolonged continuous rain on completely saturated soil (moisture > 0.490) meant that even light-to-moderate subsequent rain continuously converted to 100% surface runoff, causing a systemic overflow of the primary canals

Case Study 2: Validating Tier 1 and Tier 2 (Absolute Threshold & Smart Override)

Event Date: July 20 – 21, 2022



Summary of rainfall and flooding situation, July 20-21, 2022, approximately 6:30 PM: Light to moderate rain showers formed in the areas of Taling Chan, Phasi Charoen, Bang Bon, Nong Khaem, Thung Khru, Rat Burana, and Bang Khun Thian districts. Meanwhile, rain showers were also observed in Pathum Thani, Nakhon Pathom, and Chachoengsao provinces. Subsequently, the rain showers in the area expanded, and rain showers from areas surrounding Bangkok joined them periodically, until the area was completely covered and stationary. As of 7:00 AM on July 21, 2022, light rain continues to cover most of the area.

Figure A2. Immediate surface flooding triggered by extreme hourly rainfall spikes. (Source: Bangkok Biz News, 2022).

- **Rainfall (API Telemetry):** The data shows several extreme, isolated spikes that drastically exceeded normal capacities, including a staggering 17.7 mm/hr (Jul 16, 13:00), 12.8 mm/hr (Jul 17, 07:00), and 12.0 mm/hr (Jul 24, 09:00).
- **Soil Moisture:** Ground saturation climbed steadily, hitting severe saturation levels of 0.514 – 0.520 from July 20 to July 24.
- **Empirical Insight:** This period perfectly illustrates the absolute necessity of the Tier 1 logic. With intensities reaching 17.7 mm/hr, surface flooding was immediate regardless of soil conditions, validating the > 10.0 mm/hr absolute trigger. Furthermore, by July 23-24, soil moisture hit its physical ceiling (0.508 – 0.520). A rain burst of 5.9 mm/hr on July 23 (heavily stressing the system) was immediately followed by a 12.0 mm/hr disaster the next morning on zero-capacity soil, heavily validating the Tier 2 Smart Override mechanism.

Case Study 3: Validating Tier 2 and Tier 4 (Pre-Saturation Threat)

Event Date: September 6 – 8, 2022

On the evening of September 6, 2022, heavy rain fell. After the rain stopped, reporters traveled to areas with high water levels on Phahonyothin Road, from in front of Thanyalak Institute to the National Memorial, in both inbound and outbound lanes, in Khukhot Subdistrict, Lam Luk Ka District, Pathum Thani Province. The shoulders of the road were flooded, causing slowdowns for motorists and public transport users, paralyzing traffic on Phahonyothin Road in both directions. The report indicated that the rain showed no sign of stopping after more than an hour. In particular, large water pumps were operating near the Lam Luk Ka intersection but were unable to drain the water quickly enough. Furthermore, the outbound lanes of Phahonyothin Road in front of the Si Mum Mueang Market are prone to flooding due to low ground conditions, especially during rainfall.

At 7:00 PM, reporters arrived at the flooded area near Khlong 1, Saphan Daeng, Thanyaburi. The entire street was flooded, and several houses were damaged. Some houses were using small submersible pumps to drain the water.

Figure A3.1 Continuous rainfall overwhelming the canal and drainage networks on saturated ground. (Source: Naewna, 2022).

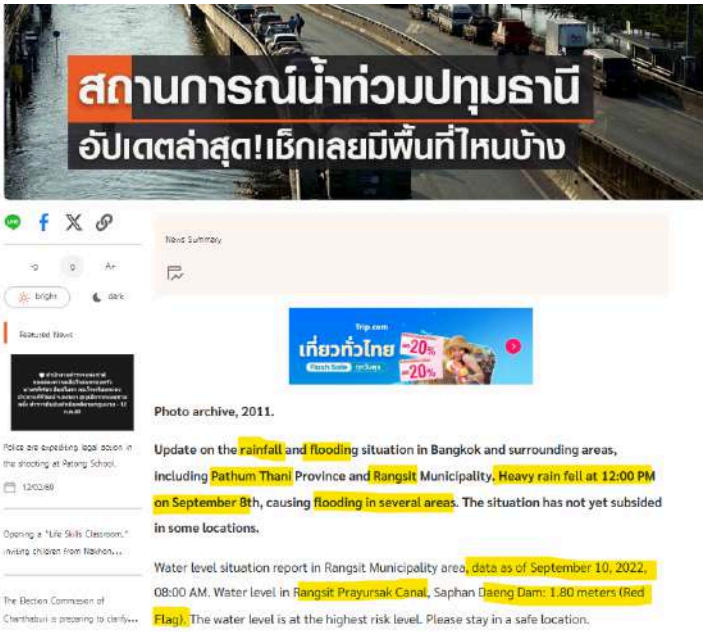


Figure A3.2 Heavy rainfall and Red flag alert (Source: TNN Thailand, 2022)

- **Rainfall (API Telemetry):** Data reveals massive, continuous rainfall cells over consecutive days, with notable peaks of 9.4 mm/hr (Sep 6, 14:00) and an extreme downpour reaching 12.3 mm/hr (Sep 7, 12:00). September 7 alone saw a dense 3-hour accumulation of nearly 28.0 mm (10:00 – 12:00).
- **Soil Moisture:** Saturated to the absolute maximum ceiling, consistently hitting 0.518 to 0.520 on September 6, 7, and 8.
- **Empirical Insight:** By the time the Red Flag was raised on September 8, soil moisture had repeatedly hit maximum capacity (0.520). With absolutely zero infiltration capacity left, even moderate subsequent rainfall immediately converted into 100% surface runoff, validating the Tier 2 logic where high soil moisture necessitates a lower rainfall warning threshold.

Case Study 4: Validating Tier 1 and Tier 4 (Prolonged Flooding) Event Date: late September, 2023

September 29, 2022 - Reporters reported that the flooding situation in Pathum Thani Province is expanding, with floodwaters reaching approximately 30-50 centimeters high. In particular, at the community near Wat Bang Luang, Moo 4, Ban Chang Subdistrict, Mueang Pathum Thani District, the water level continues to rise. The overflowing of the Chao Phraya River and Bang Luang Canal, following increased water release from the Chao Phraya Dam, will affect residents living along both banks of the Chao Phraya River in areas outside the flood barriers of Sam Khok and Mueang Pathum Thani Districts, Pathum Thani Province. There are also indications that water levels in Mueang Pathum Thani District will rise further.

In the community near Bang Luang Temple, the roads and walkways are flooded with water more than 50 cm deep, inundating every house. Residents have to wade through the water to reach their homes. They have begun bringing out boats in preparation for the heavy rainfall from Typhoon Noru.

Figure A4. Prolonged flooding caused by overwhelmed drainage following extreme micro-bursts. (Source: Thai PBS News, 2023).

- **Rainfall (API Telemetry):** Experienced extreme micro-bursts exceeding 10.7 mm/hr and 11.3 mm/hr (on Sept 16 and Sept 21 respectively), with a continuous 5-10 hour accumulation reaching as high as 21.2 – 24.0 mm.
- **Empirical Insight:** This data perfectly aligns with ground-truth news reports stating "prolonged flooding for over 6 hours due to overwhelmed drainage." It serves as solid evidence validating both the Tier 1 (Flash Flood > 10.0 mm/hr) threshold and the Tier 4 (Systemic Overflow > 20.0 mm in 12 hours) logic.

Case Study 5: Validating Tier 3 (Micro-Climate Anomalies & The Sliding Window) Event Date: July 3, 2024

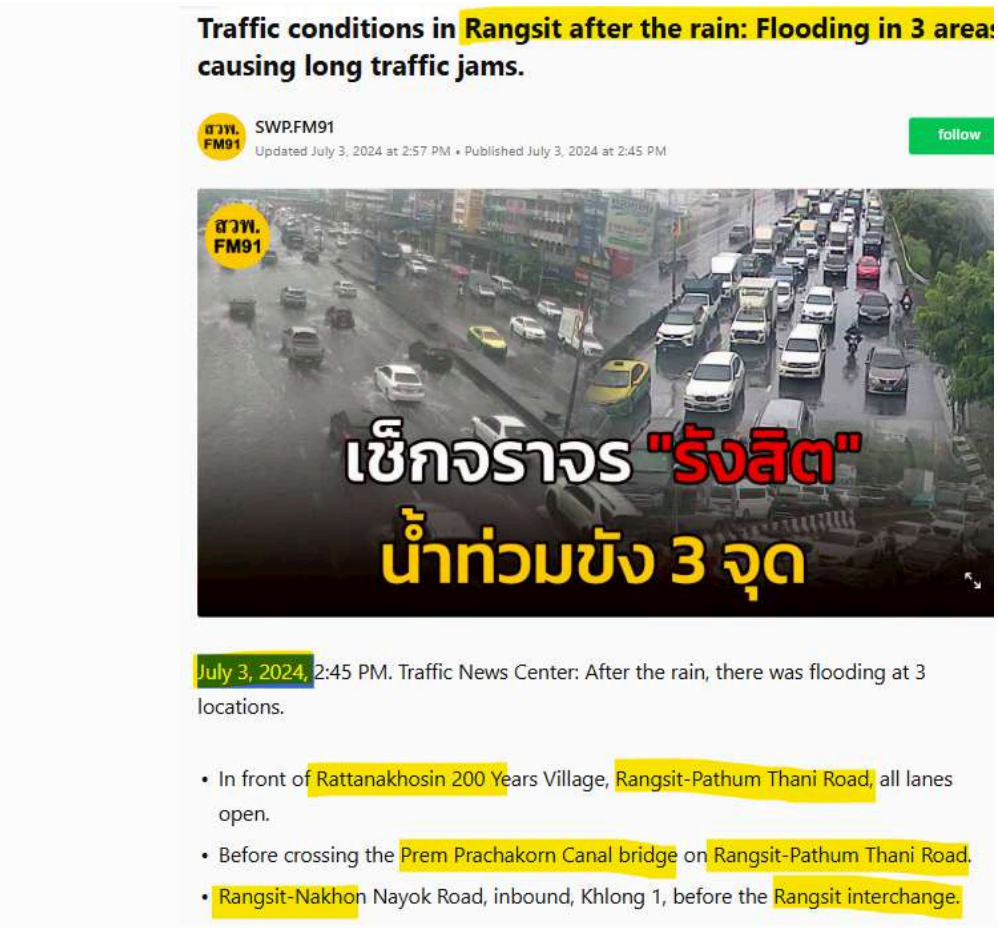


Figure A5. Highly localized flooding illustrating API blind spots and micro-climate anomalies. (Source: LINE Today, 2024).

- **Rainfall (API Telemetry):** The API captured only 1.2 mm (0.2 mm at 13:00, and 1.0 mm at 14:00).
- **Soil Moisture:** Spiked immediately from 0.463 to 0.484 at 15:00.
- **Empirical Insight:** This is a textbook example of a "Micro-climate Anomaly." The global weather API completely missed the heavy localized rain cell, yet actual surface flooding occurred. This highlights exactly why an accumulative safety net (Tier 3 Sliding Window) and spatial lag ML data are fundamentally necessary to cover API blind spots.

Case Study 6: Validating Tier 3 and Tier 4 (Accumulative Stress & Traffic Collapse) Event

Date: October 5 – 14, 2025

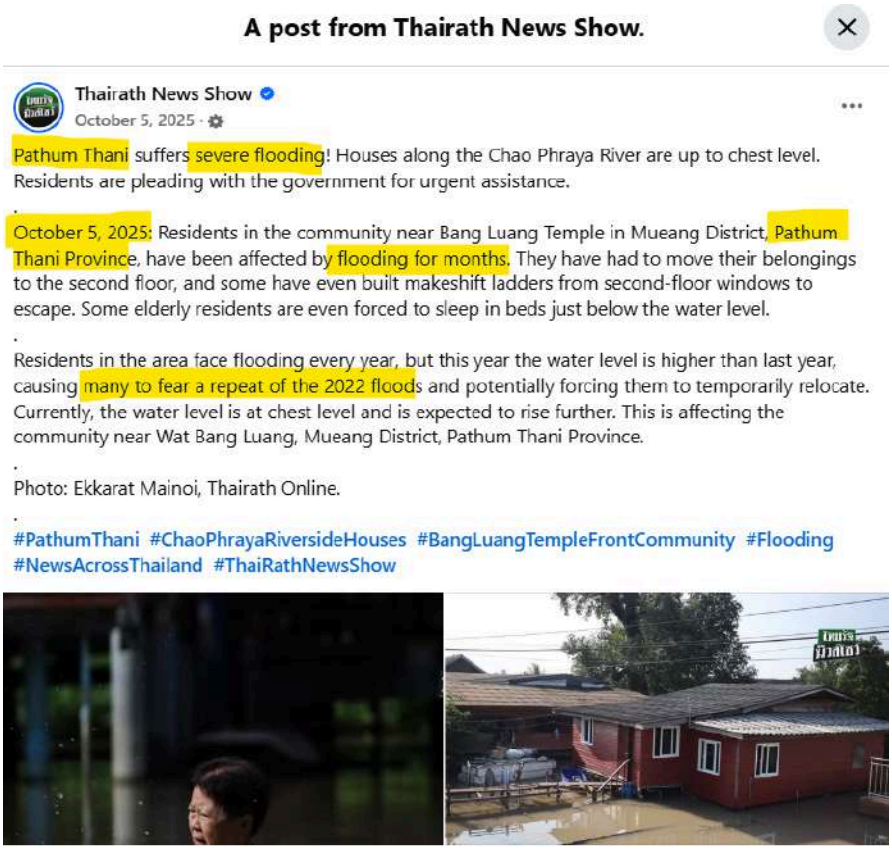


Figure A6.1 Urban gridlock and heavy flooding for a month (Source: Thairath News Show, 2025).

Figure A6.2 Urban gridlock and severe surface flooding caused by prolonged soaking and accumulative systemic stress. (Source: FM91BKK, 2025).

- **Rainfall (API Telemetry):** The data reveals a pattern of persistent soaking. Significant repetitive rainfall hit the area leading up to October 14, with notable peaks of 6.5 mm/hr (Oct 8) and 6.4 mm/hr (Oct 12). The day before the major traffic collapse (Oct 13), the area experienced a dense 5-hour continuous evening rain totaling over 14.0 mm.
- **Soil Moisture:** Hit the absolute maximum capacity of 0.520 on multiple occasions (Oct 8, Oct 11, Oct 12) and maintained highly elevated levels (> 0.505) through the morning of October 14.
- **Empirical Insight:** This event perfectly demonstrates the danger of accumulative stress. For the October 14 urban gridlock, the crisis was a direct result of Tier 3 and Tier 2 compounding. The heavy rains on preceding days completely saturated the ground. By the time the final minor rain fell on Oct 14, the local municipal drains were already choked from days of systemic overload, causing severe surface flooding even with seemingly light real-time rainfall.

Case Study 7: The Definitive Validation of Compounding Tier Logics Event Date: October 31 – November 2, 2025

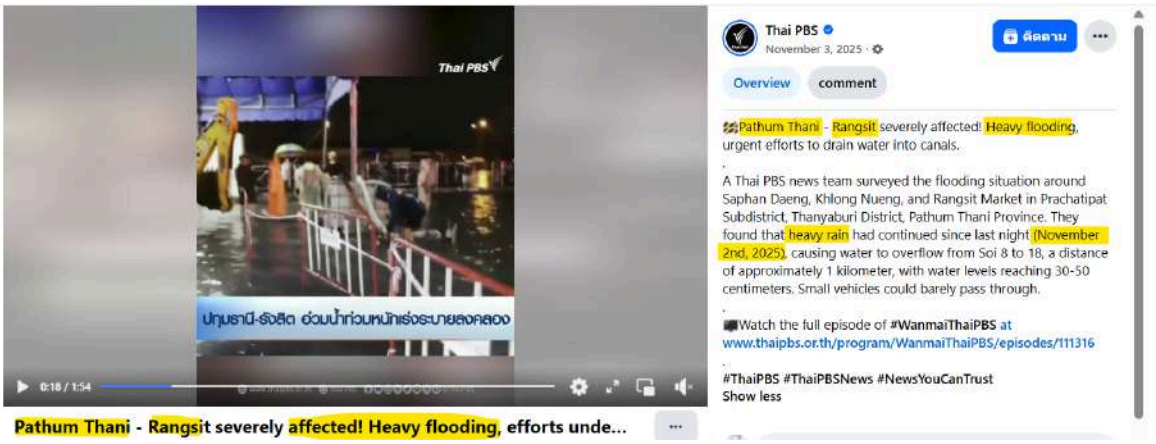


Figure A7. Severe 30-50 cm impassable flooding at Rangsit Market resulting from a compounding Tier 1 and Tier 2 scenario. (Source: Thai PBS, 2025).

- **Rainfall (API Telemetry):** A massive early assault began on October 31 with a peak of 10.6 mm/hr at 09:00. This was followed by sustained downpours on November 1, with major spikes at 05:00 (5.7 mm/hr) and 07:00 (5.5 mm/hr), culminating in an intense continuous block from 13:00 to 20:00 (peaking at 6.9 mm/hr).
- **Soil Moisture:** Completely maxed out. Relentless rainfall drove moisture to its absolute physical limit of 0.520 multiple times and rarely dropped below the critical 0.450 threshold.
- **Empirical Insight:** This event serves as the definitive empirical evidence for the threshold designs. The massive initial burst of 10.6 mm/hr validates the Tier 1 immediate failure threshold. Furthermore, with drainage capacity at 0%, sustained moderate rain blocks (6.8 and 6.9 mm/hr) converted entirely into surface runoff. This perfectly validates the Tier 2 (Smart Override) logic: when soil moisture is critically saturated (≥ 0.50), a lowered rainfall threshold of just > 6.0 mm/hr is more than enough to trigger the severe 30-50 cm impassable flooding reported at the local market.

Table A.1: Structured Ground-Truth Dataset for 4-Tier Logic Validation

Event Date (YYYY-MM-DD)	Max Hourly Rain (mm/hr)	Max Soil Moisture	Ground-Truth Observation (News Source)	Flood_Occurred (Label)	Triggered System Tier
2021-09-20	13.0	0.519	Flooded (TopNews / Kom Chad Luek)	**1**	Tier 1 & Tier 4
2022-07-24	12.0	0.520	Flooded (Bangkok Biz News)	**1**	Tier 1 & Tier 2
2022-09-08	12.3	0.520	Flooded (Naewna / TNN Thailand)	**1**	Tier 2 (Smart Override)
2023-09-21	11.3	0.490	Flooded (Thai PBS News)	**1**	Tier 1 & Tier 4
2023-11-05	4.5	0.310	*Normal / No Flooding Reported*	0	*None (Safe)*
2024-07-03	11.2	0.484	Flooded (LINE Today) - Micro-burst	**1**	Tier 3 (Sliding Window)
2025-10-14	6.5	0.520	Flooded (Thairath / FM91BKK)	**1**	Tier 2 & Tier 3
2025-10-31	10.6	0.520	Flooded (Thai PBS)	**1**	Tier 1 & Tier 2
2025-12-10	5.0	0.250	*Normal / No Flooding Reported*	0	*None (Safe)*

While the operational thresholds of the 4-Tier Alert Logic (e.g., > 10.0 mm/hr rain or > 0.48 soil moisture) are empirically tuned to the specific topographical and drainage profile of Khlong

Nueng—as validated against historical datasets in Table A.1—the system's underlying rule-based architecture is highly generalizable. For future deployment in other municipalities, these threshold parameters can be easily re-calibrated using local historical flood records without requiring modifications to the core serverless MLOps infrastructure.

APPENDIX B

HISTORICAL DATA EXTRACTION SOURCE CODE (OPEN-METEO API)

This appendix contains the Python scripts utilized to systematically extract five years (2021–2025) of historical meteorological and soil moisture telemetry from the Open-Meteo API. This offline data extraction pipeline serves two foundational purposes for the study: (1) constructing the comprehensive historical dataset required to train and evaluate the XGBoost machine learning model, and (2) providing the empirical baseline necessary to reverse-engineer and mathematically validate the thresholds for the 4-Tier Adaptive Alert Logic against verified real-world flood events.

```python
import requests
import pandas as pd

# Define target location coordinates: Khlong Nueng (AIT area)
LATITUDE = 14.0686
LONGITUDE = 100.6111

# Define historical timeframe (5 years)
START_DATE = "2021-01-01"
END_DATE = "2025-12-31"

# Base URL for Open-Meteo Historical Data
BASE_URL = "https://archive-api.open-meteo.com/v1/archive"


params_a = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["temperature_2m", "relative_humidity_2m", "rain"],
    "timezone": "UTC"
}

response_a = requests.get(BASE_URL, params=params_a)
data_a = response_a.json()

# Create DataFrame for Dataset A
df_a = pd.DataFrame({
    "timestamp": data_a["hourly"]["time"],
    "temperature_c": data_a["hourly"]["temperature_2m"],
    "humidity_percent": data_a["hourly"]["relative_humidity_2m"],
    "rain_mm": data_a["hourly"]["rain"]
})
```

```python
params_b = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["surface_pressure", "cloud_cover", "wind_speed_10m", "wind_direction_10m"],
    "timezone": "UTC"
}

response_b = requests.get(BASE_URL, params=params_b)
data_b = response_b.json()


df_b = pd.DataFrame({
    "timestamp": data_b["hourly"]["time"],
    "pressure_hpa": data_b["hourly"]["surface_pressure"],
    "cloud_cover_percent": data_b["hourly"]["cloud_cover"],
    "wind_speed_kmh": data_b["hourly"]["wind_speed_10m"],
    "wind_direction_deg": data_b["hourly"]["wind_direction_10m"]
})

# Use outer join to ensure no timestamps or missing values are dropped
df_raw_merged = pd.merge(df_a, df_b, on="timestamp", how="outer")

# Output the amount of nulls to verify raw state
print("Raw data missing values count (Kept as NaN):")
print(df_raw_merged.isnull().sum())

# Display the first 5 rows of the raw data
print("\nFirst 5 rows of Khlong Nueng Raw Historical Data (2021-2025):")
print(df_raw_merged.head())

# Save strictly as raw data to CSV (Represents Phase 1 Data Dump)
df_raw_merged.to_csv("khlong_nueng_raw_weather_2021_2025.csv", index=False)
print("\nRaw data successfully saved to khlong_nueng_raw_weather_2021_2025.csv")
```

Figure B1. Atmospheric Data Extraction Script: Khlong Nueng, Pathum Thani (Target Area)

```python
import requests
import pandas as pd

# Define target location coordinates: Ayutthaya (North Node)
LATITUDE = 14.3516
LONGITUDE = 100.5774

# Define historical timeframe (5 years)
START_DATE = "2021-01-01"
END_DATE = "2025-12-31"

# Base URL for Open-Meteo Historical Data
BASE_URL = "https://archive-api.open-meteo.com/v1/archive"

params_a = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["temperature_2m", "relative_humidity_2m", "rain"],
    "timezone": "UTC"
}

response_a = requests.get(BASE_URL, params=params_a)
data_a = response_a.json()

# Create DataFrame for Dataset A
df_a = pd.DataFrame({
    "timestamp": data_a["hourly"]["time"],
    "temperature_c": data_a["hourly"]["temperature_2m"],
    "humidity_percent": data_a["hourly"]["relative_humidity_2m"],
    "rain_mm": data_a["hourly"]["rain"]
```

```python
params_b = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["surface_pressure", "cloud_cover", "wind_speed_10m", "wind_direction_10m"],
    "timezone": "UTC"
}

response_b = requests.get(BASE_URL, params=params_b)
data_b = response_b.json()

# Create DataFrame for Dataset B
df_b = pd.DataFrame({
    "timestamp": data_b["hourly"]["time"],
    "pressure_hpa": data_b["hourly"]["surface_pressure"],
    "cloud_cover_percent": data_b["hourly"]["cloud_cover"],
    "wind_speed_kmh": data_b["hourly"]["wind_speed_10m"],
    "wind_direction_deg": data_b["hourly"]["wind_direction_10m"]
})

# Use outer join to ensure no timestamps or missing values are dropped
df_raw_merged = pd.merge(df_a, df_b, on="timestamp", how="outer")

# Output the amount of nulls to verify raw state
print("Raw data missing values count (Kept as NaN):")
print(df_raw_merged.isnull().sum())

# Display the first 5 rows of the raw data
print("\nFirst 5 rows of Ayutthaya Raw Historical Data (2021-2025):")
print(df_raw_merged.head())

# Save strictly as raw data to CSV (Represents Phase 1 Data Dump)
df_raw_merged.to_csv("ayutthaya_raw_weather_2021_2025.csv", index=False)
print("\nRaw data successfully saved to ayutthaya_raw_weather_2021_2025.csv")
```

Figure B2. Atmospheric Data Extraction Script: Ayutthaya (Northern Spatial Node)

```python
import requests
import pandas as pd

# Define target location coordinates: Bangkok (South Node)
LATITUDE = 13.7563
LONGITUDE = 100.5018

# Define historical timeframe (5 years)
START_DATE = "2021-01-01"
END_DATE = "2025-12-31"

# Base URL for Open-Meteo Historical Data
BASE_URL = "https://archive-api.open-meteo.com/v1/archive"

params_a = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["temperature_2m", "relative_humidity_2m", "rain"],
    "timezone": "UTC"
}

response_a = requests.get(BASE_URL, params=params_a)
data_a = response_a.json()

# Create DataFrame for Dataset A
df_a = pd.DataFrame({
    "timestamp": data_a["hourly"]["time"],
    "temperature_c": data_a["hourly"]["temperature_2m"],
    "humidity_percent": data_a["hourly"]["relative_humidity_2m"],
    "rain_mm": data_a["hourly"]["rain"]
})
```

```python
params_b = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["surface_pressure", "cloud_cover", "wind_speed_10m", "wind_direction_10m"],
    "timezone": "UTC"
}

response_b = requests.get(BASE_URL, params=params_b)
data_b = response_b.json()

# Create DataFrame for Dataset B
df_b = pd.DataFrame({
    "timestamp": data_b["hourly"]["time"],
    "pressure_hpa": data_b["hourly"]["surface_pressure"],
    "cloud_cover_percent": data_b["hourly"]["cloud_cover"],
    "wind_speed_kmh": data_b["hourly"]["wind_speed_10m"],
    "wind_direction_deg": data_b["hourly"]["wind_direction_10m"]
})

# Use outer join to ensure no timestamps or missing values are dropped
df_raw_merged = pd.merge(df_a, df_b, on="timestamp", how="outer")

# Output the amount of nulls to verify raw state
print("Raw data missing values count (Kept as NaN):")
print(df_raw_merged.isnull().sum())

# Display the first 5 rows of the raw data
print("\nFirst 5 rows of Bangkok Raw Historical Data (2021-2025):")
print(df_raw_merged.head())

# Save strictly as raw data to CSV (Represents Phase 1 Data Dump)
df_raw_merged.to_csv("bangkok_raw_weather_2021_2025.csv", index=False)
print("\nRaw data successfully saved to bangkok_raw_weather_2021_2025.csv")
```

Figure B3. Atmospheric Data Extraction Script: Bangkok (Southern Spatial Node)

```python
import requests
import pandas as pd

# Define target location coordinates: Nakhon Nayok (East Node)
LATITUDE = 14.2069
LONGITUDE = 101.2131

# Define historical timeframe (5 years)
START_DATE = "2021-01-01"
END_DATE = "2025-12-31"

# Base URL for Open-Meteo Historical Data
BASE_URL = "https://archive-api.open-meteo.com/v1/archive"

params_a = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["temperature_2m", "relative_humidity_2m", "rain"],
    "timezone": "UTC"
}

response_a = requests.get(BASE_URL, params=params_a)
data_a = response_a.json()

# Create DataFrame for Dataset A
df_a = pd.DataFrame({
    "timestamp": data_a["hourly"]["time"],
    "temperature_c": data_a["hourly"]["temperature_2m"],
    "humidity_percent": data_a["hourly"]["relative_humidity_2m"],
    "rain_mm": data_a["hourly"]["rain"]
})
```

```python
params_b = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["surface_pressure", "cloud_cover", "wind_speed_10m", "wind_direction_10m"],
    "timezone": "UTC"
}

response_b = requests.get(BASE_URL, params=params_b)
data_b = response_b.json()

# Create DataFrame for Dataset B
df_b = pd.DataFrame({
    "timestamp": data_b["hourly"]["time"],
    "pressure_hpa": data_b["hourly"]["surface_pressure"],
    "cloud_cover_percent": data_b["hourly"]["cloud_cover"],
    "wind_speed_kmh": data_b["hourly"]["wind_speed_10m"],
    "wind_direction_deg": data_b["hourly"]["wind_direction_10m"]
})

# Use outer join to ensure no timestamps or missing values are dropped
df_raw_merged = pd.merge(df_a, df_b, on="timestamp", how="outer")

# Output the amount of nulls to verify raw state
print("Raw data missing values count (Kept as NaN):")
print(df_raw_merged.isnull().sum())

# Display the first 5 rows of the raw data
print("\nFirst 5 rows of Nakhon Nayok Raw Historical Data (2021-2025):")
print(df_raw_merged.head())

# Save strictly as raw data to CSV (Represents Phase 1 Data Dump)
df_raw_merged.to_csv("nakhon_nayok_raw_weather_2021_2025.csv", index=False)
print("\nRaw data successfully saved to nakhon_nayok_raw_weather_2021_2025.csv")
```

Figure B4. Atmospheric Data Extraction Script: Nakhon Nayok (Eastern Spatial Node)

```python
import requests
import pandas as pd

# Define target location coordinates: Nonthaburi (West Node)
LATITUDE = 13.8591
LONGITUDE = 100.5217

# Define historical timeframe (5 years)
START_DATE = "2021-01-01"
END_DATE = "2025-12-31"

# Base URL for Open-Meteo Historical Data
BASE_URL = "https://archive-api.open-meteo.com/v1/archive"

params_a = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["temperature_2m", "relative_humidity_2m", "rain"],
    "timezone": "UTC"
}

response_a = requests.get(BASE_URL, params=params_a)
data_a = response_a.json()

# Create DataFrame for Dataset A
df_a = pd.DataFrame({
    "timestamp": data_a["hourly"]["time"],
    "temperature_c": data_a["hourly"]["temperature_2m"],
    "humidity_percent": data_a["hourly"]["relative_humidity_2m"],
    "rain_mm": data_a["hourly"]["rain"]
})
```

```python
params_b = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["surface_pressure", "cloud_cover", "wind_speed_10m", "wind_direction_10m"],
    "timezone": "UTC"
}

response_b = requests.get(BASE_URL, params=params_b)
data_b = response_b.json()

# Create DataFrame for Dataset B
df_b = pd.DataFrame({
    "timestamp": data_b["hourly"]["time"],
    "pressure_hpa": data_b["hourly"]["surface_pressure"],
    "cloud_cover_percent": data_b["hourly"]["cloud_cover"],
    "wind_speed_kmh": data_b["hourly"]["wind_speed_10m"],
    "wind_direction_deg": data_b["hourly"]["wind_direction_10m"]
})

# Use outer join to ensure no timestamps or missing values are dropped
df_raw_merged = pd.merge(df_a, df_b, on="timestamp", how="outer")

# Output the amount of nulls to verify raw state
print("Raw data missing values count (Kept as NaN):")
print(df_raw_merged.isnull().sum())

# Display the first 5 rows of the raw data
print("\nFirst 5 rows of Nonthaburi Raw Historical Data (2021-2025):")
print(df_raw_merged.head())

# Save strictly as raw data to CSV (Represents Phase 1 Data Dump)
df_raw_merged.to_csv("nonthaburi_raw_weather_2021_2025.csv", index=False)
print("\nRaw data successfully saved to nonthaburi_raw_weather_2021_2025.csv")
```

Figure B5. Atmospheric Data Extraction Script: Nonthaburi (Western Spatial Node)

timestamp	temperature_c	humidity_percent	rain_mm	pressure_hpa	cloud_cover_per	wind_speed_kmh	wind_direction_deg
2021-01-01T00:00	16.8	67	0	1015.1	14	12.1	30
2021-01-01T01:00	18	64	0	1016.1	25	10.8	21
2021-01-01T02:00	19.7	59	0	1017	45	9.4	23
2021-01-01T03:00	21.5	55	0	1017	23	6.8	18
2021-01-01T04:00	23.3	51	0	1016.5	6	6.5	3
2021-01-01T05:00	24.9	47	0	1015.5	49	8.5	12
2021-01-01T06:00	26	44	0	1014.3	14	10	21
2021-01-01T07:00	26.6	42	0	1013.2	86	10.3	25
2021-01-01T08:00	26.9	42	0	1012.7	55	9.5	37
2021-01-01T09:00	26.8	44	0	1012.4	21	8.9	50
2021-01-01T10:00	26.3	46	0	1012.6	33	7.4	43
2021-01-01T11:00	24.7	50	0	1013.2	16	7.6	19
2021-01-01T12:00	23.4	57	0	1013.9	32	5.5	23
2021-01-01T13:00	23.4	50	0	1014.5	80	12.6	20
2021-01-01T14:00	22.9	48	0	1015	90	13	34
2021-01-01T15:00	22.3	48	0	1015.1	86	11.2	33
2021-01-01T16:00	22	48	0	1015	60	10.2	23
2021-01-01T17:00	21.2	50	0	1014.6	82	8.2	23
2021-01-01T18:00	20.5	52	0	1014.3	87	7.5	17
2021-01-01T19:00	20.1	53	0	1013.7	87	8.8	12
2021-01-01T20:00	19.8	54	0	1013.1	63	10.4	14

Figure B6. Example of Atmospheric Data

```python
import requests
import pandas as pd

# Define coordinates for Khlong Nueng
LATITUDE = 14.0686
LONGITUDE = 100.6111

# Define historical timeframe
START_DATE = "2021-01-01"
END_DATE = "2025-12-31"

# API endpoint for historical data
URL = "https://archive-api.open-meteo.com/v1/archive"

# Define API parameters with explicit meteorological model selection (ERA5-Land for soil data)
params = {
    "latitude": LATITUDE,
    "longitude": LONGITUDE,
    "start_date": START_DATE,
    "end_date": END_DATE,
    "hourly": ["soil_moisture_0_to_7cm"],
    "timezone": "UTC",
    "models": "era5_land"
}

response = requests.get(URL, params=params)
data = response.json()

df_raw = pd.DataFrame({
    "timestamp": data["hourly"]["time"],
    "soil_moisture_m3_m3": data["hourly"]["soil_moisture_0_to_7cm"]
})

# Save raw data to CSV
df_raw.to_csv("api_3_khlong_nueng_raw.csv", index=False)
```

Figure B7. Soil Moisture Telemetry Extraction Script: Khlong Nueng (ERA5-Land Model)

timestamp	soil_moisture_m3_m3
2021-01-01T00:00	0.258
2021-01-01T01:00	0.258
2021-01-01T02:00	0.258
2021-01-01T03:00	0.258
2021-01-01T04:00	0.258
2021-01-01T05:00	0.258
2021-01-01T06:00	0.258
2021-01-01T07:00	0.258
2021-01-01T08:00	0.258
2021-01-01T09:00	0.257
2021-01-01T10:00	0.257

Figure B8. Soil Moisture Dataset In Khlong Nueng

APPENDIX C

Exploratory Data Analysis Visualizations

This appendix presents the main exploratory data analysis (EDA) figures used to understand rainfall behavior, temporal patterns, feature relationships, and the effect of different window configurations before model development.

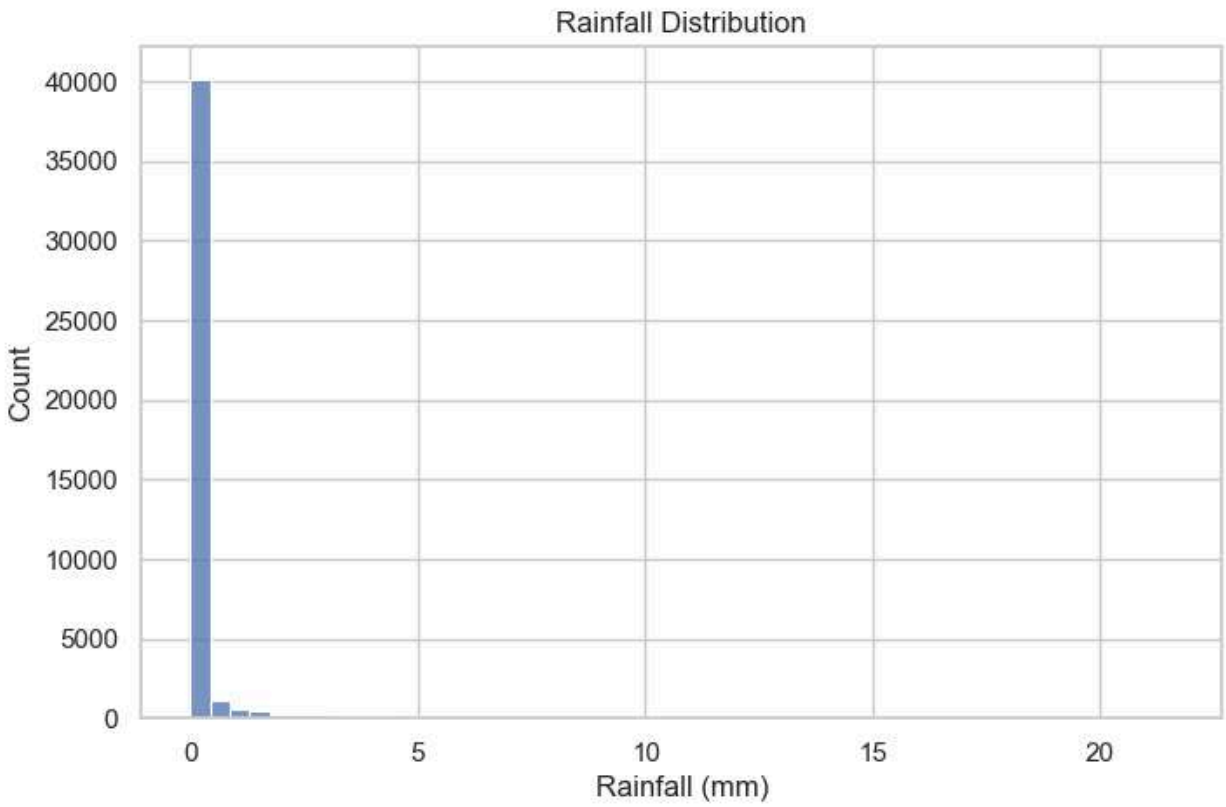


Figure C.1. Rainfall Distribution

The rainfall distribution is highly right-skewed. Most hourly observations are concentrated near zero, while only a small number of cases show moderate to heavy rainfall. This indicates that rainfall events are relatively sparse and that the target variable contains many low-value observations with a few larger spikes.

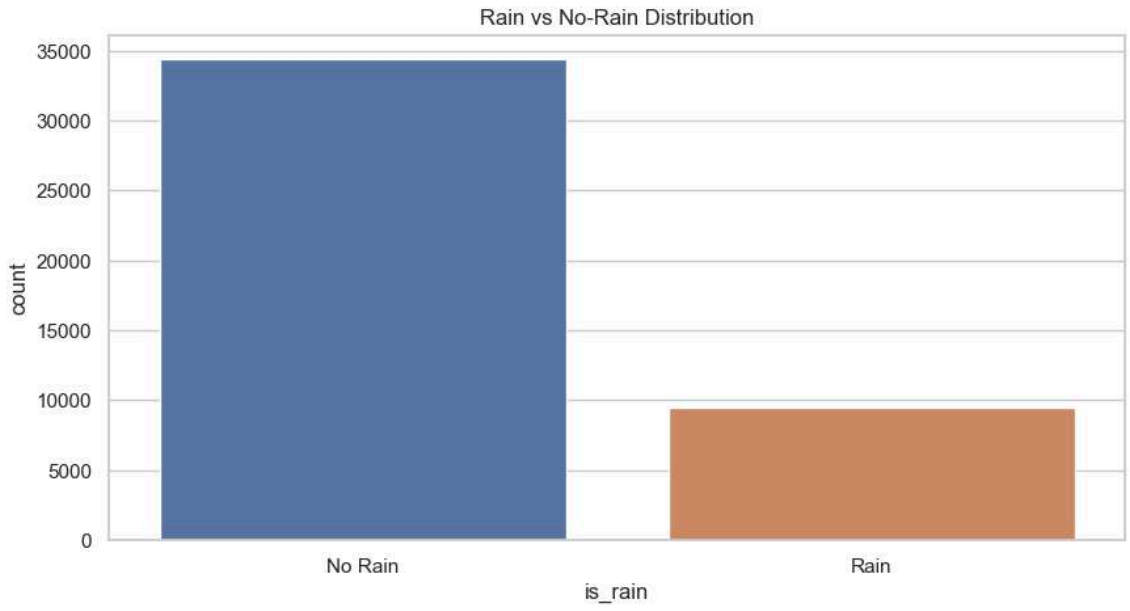


Figure C.2. Rain vs No-Rain Distribution

This figure shows the class distribution between rainfall events and non-rainfall observations in the dataset. The majority of records correspond to no-rain conditions, while a smaller portion represents rain events. This imbalance indicates that rainfall occurrences are relatively infrequent compared to dry periods, which is typical for hourly weather datasets. The imbalance highlights the importance of using robust modeling techniques and evaluation metrics to ensure that the predictive model can still learn meaningful rainfall patterns despite the dominance of non-rain observations.

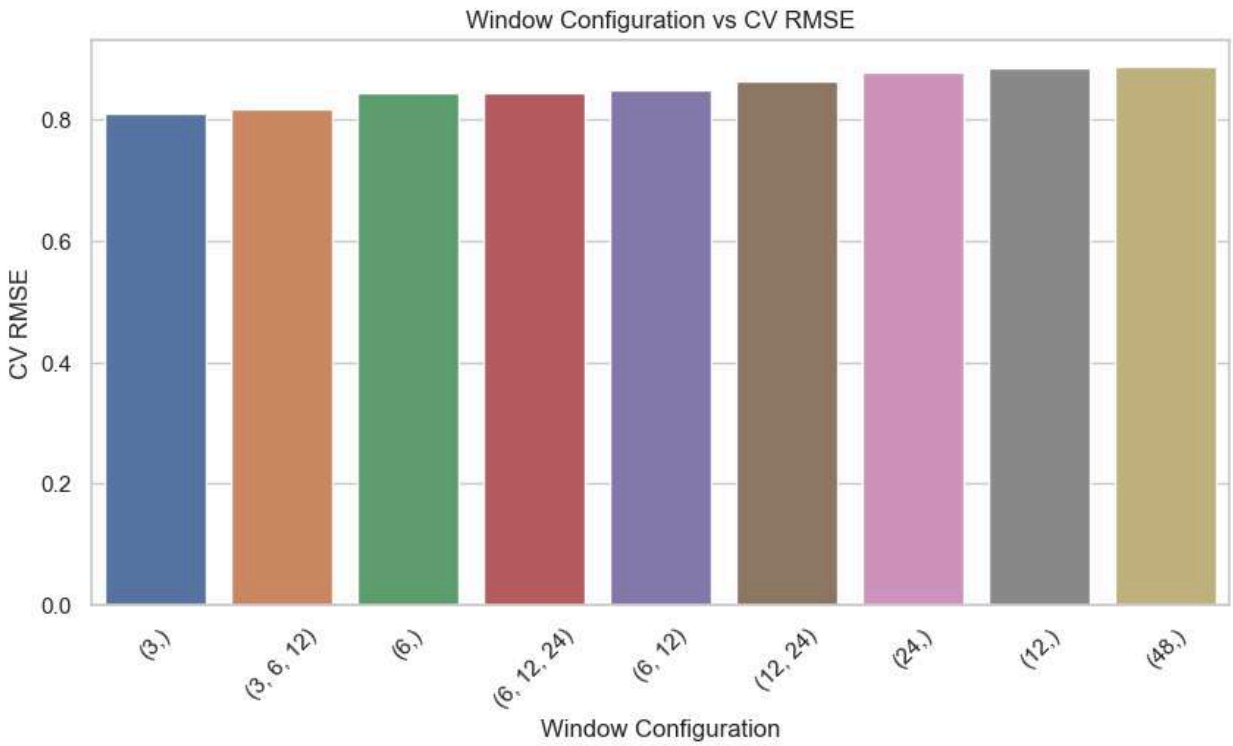


Figure C.3. Window Configuration vs Cross-Validation RMSE

This figure compares several rolling-window configurations using cross-validation RMSE. Shorter windows appear to perform slightly better than longer or more extended configurations, suggesting that recent weather conditions may be more informative for short-term rainfall prediction than broader historical windows.

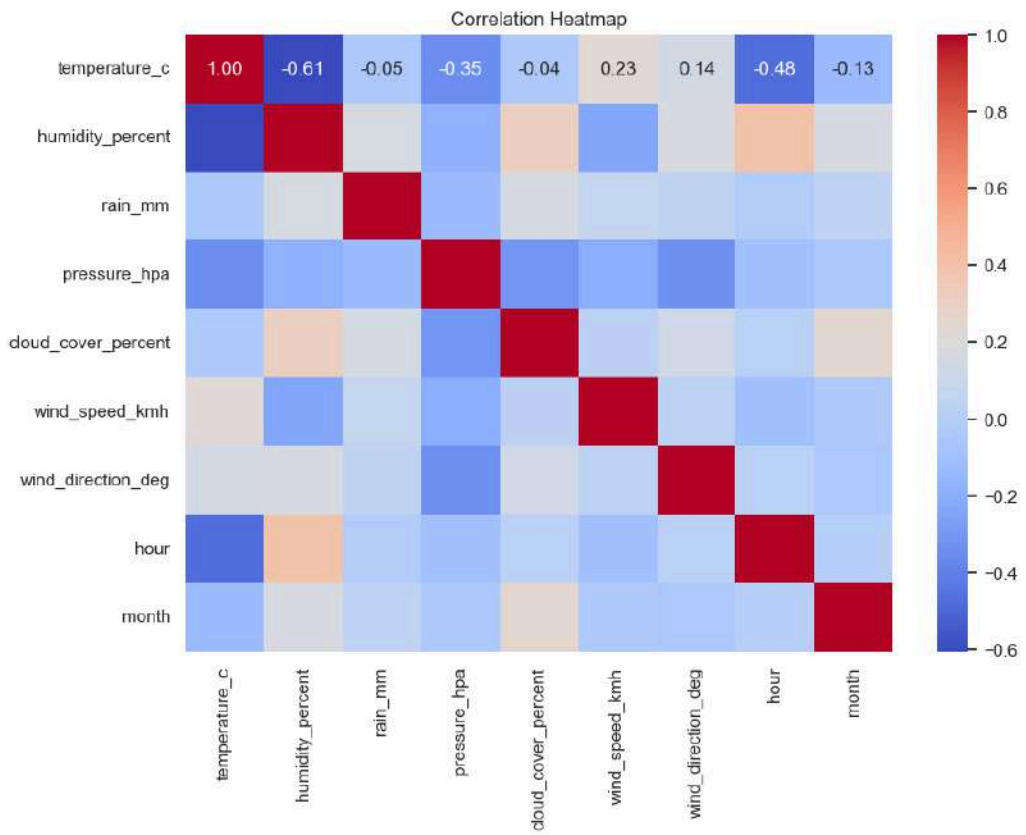


Figure C.4. Correlation Heatmap

The correlation heatmap provides an overview of pairwise relationships among the main variables. Temperature and humidity show a noticeable negative relationship, while rainfall has relatively weak direct linear correlation with individual predictors. This supports the idea that rainfall prediction is likely influenced by combined and non-linear interactions rather than by any single feature alone.

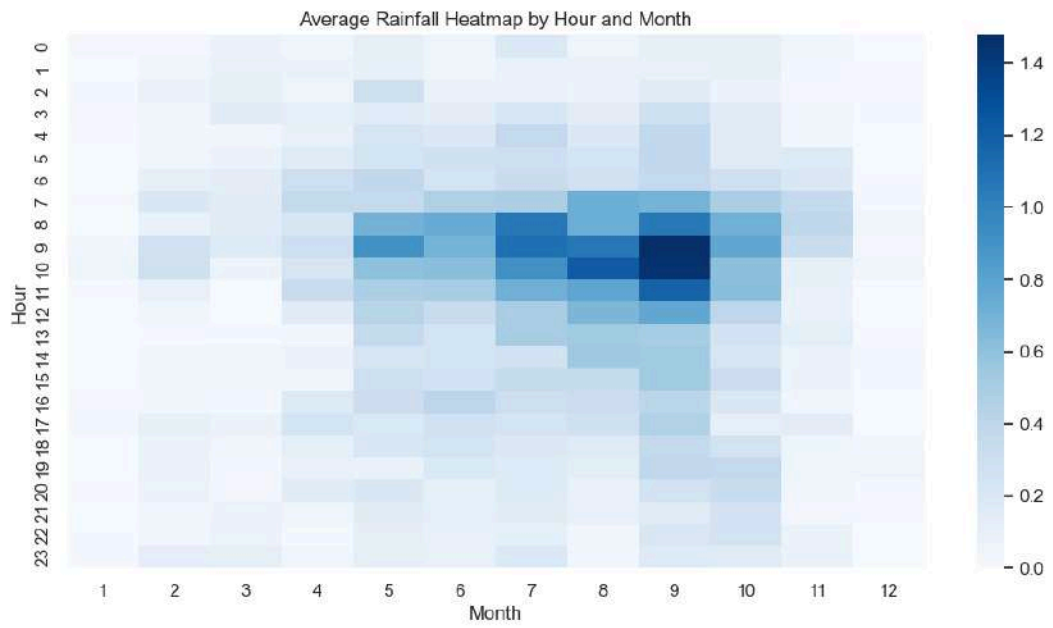


Figure C.5. Average Rainfall Heatmap by Hour and Month

This heatmap highlights the joint temporal pattern of rainfall by hour and month. Rainfall appears more pronounced during the mid-to-late wet-season months and is especially elevated during the morning to late-morning hours. The figure helps identify seasonal and intra-day rainfall concentration.

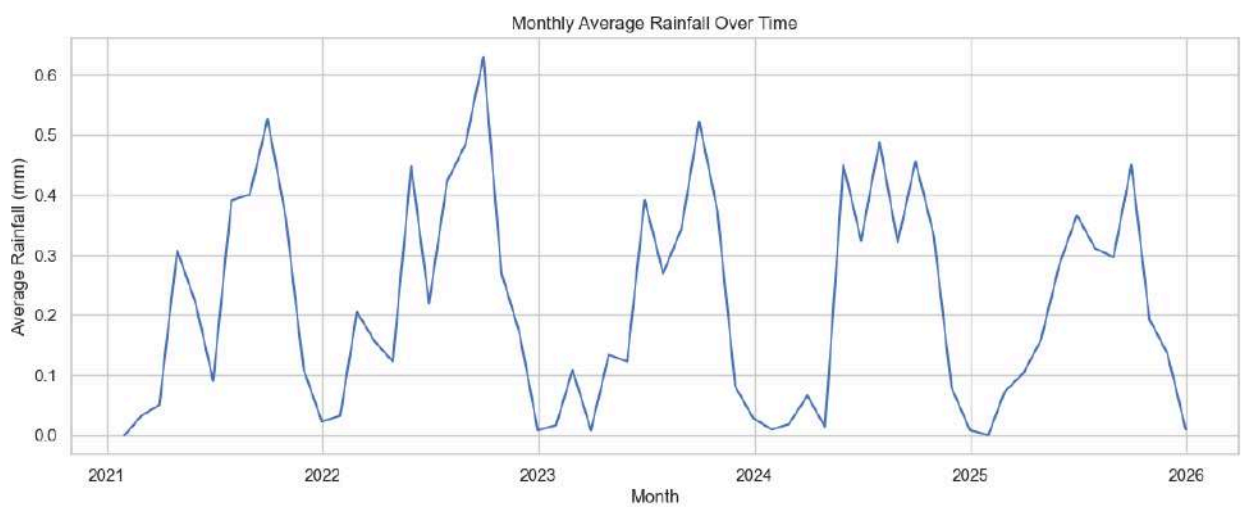


Figure C.6. Monthly Average Rainfall Over Time

The monthly average rainfall time series shows recurring seasonal variation across years. Rainfall tends to increase during the wet-season periods and decline during drier months, indicating a clear temporal structure in the dataset. This pattern supports the inclusion of time-based features such as month and rolling windows.

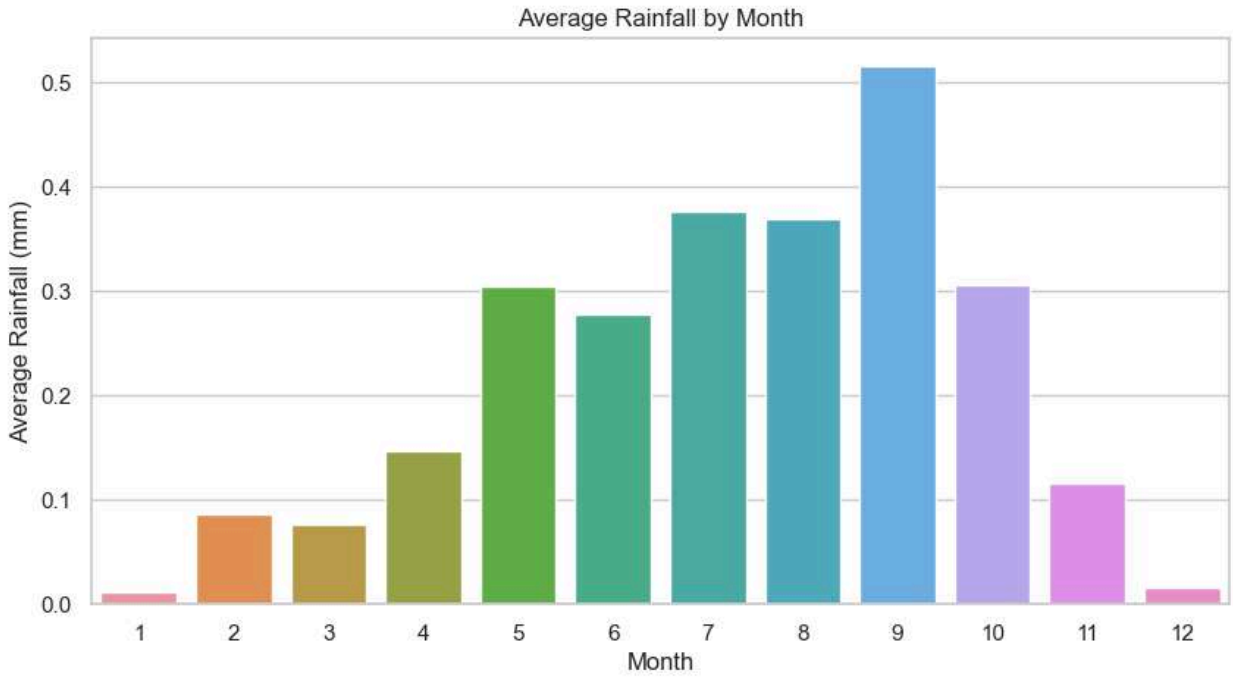


Figure C.7. Average Rainfall by Month

This bar chart summarizes the average rainfall for each month. Rainfall is lowest in the beginning and end of the year and rises notably during the middle months, with the highest average occurring around September. This confirms the presence of seasonal rainfall behavior in the study area.

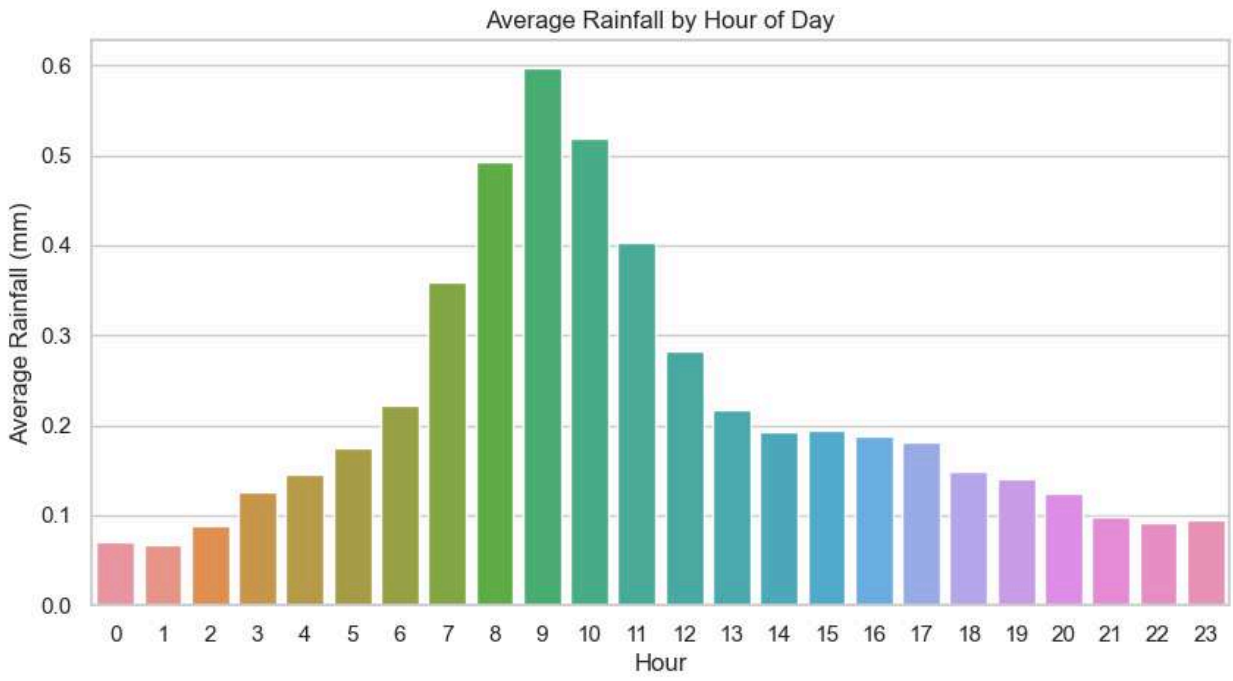


Figure C.8. Average Rainfall by Hour of Day

This figure shows that average rainfall varies across the day, with the highest rainfall observed during the morning period, especially around 08:00–10:00. After this peak, rainfall gradually declines throughout the afternoon and evening. This suggests that **hour of day** may be a useful temporal predictor in the forecasting model.

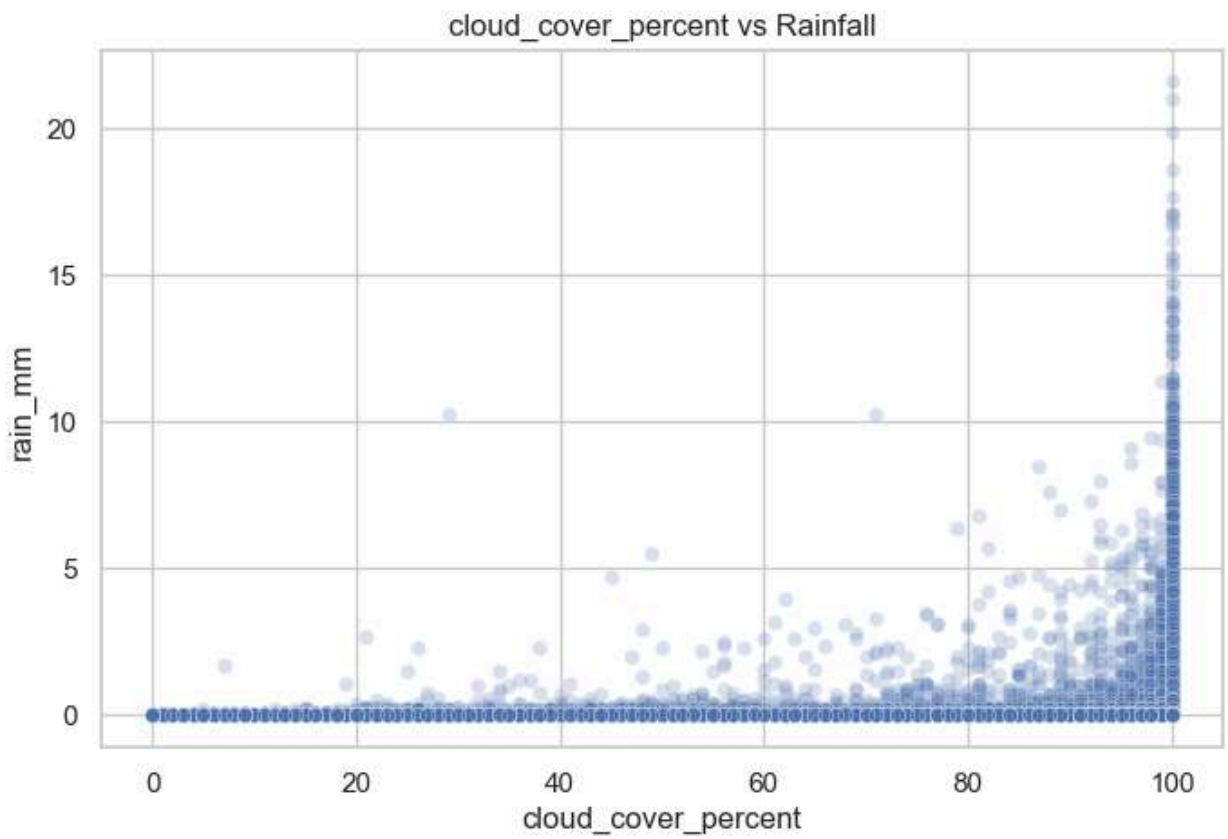


Figure C.9. Cloud Cover Percentage vs Rainfall

The scatter plot indicates that higher rainfall values are more likely to occur when cloud cover is high, especially close to 100%. However, the relationship is not strictly linear, since many observations with high cloud cover still correspond to little or no rainfall. This suggests that cloud cover is informative, but should be interpreted together with other variables.

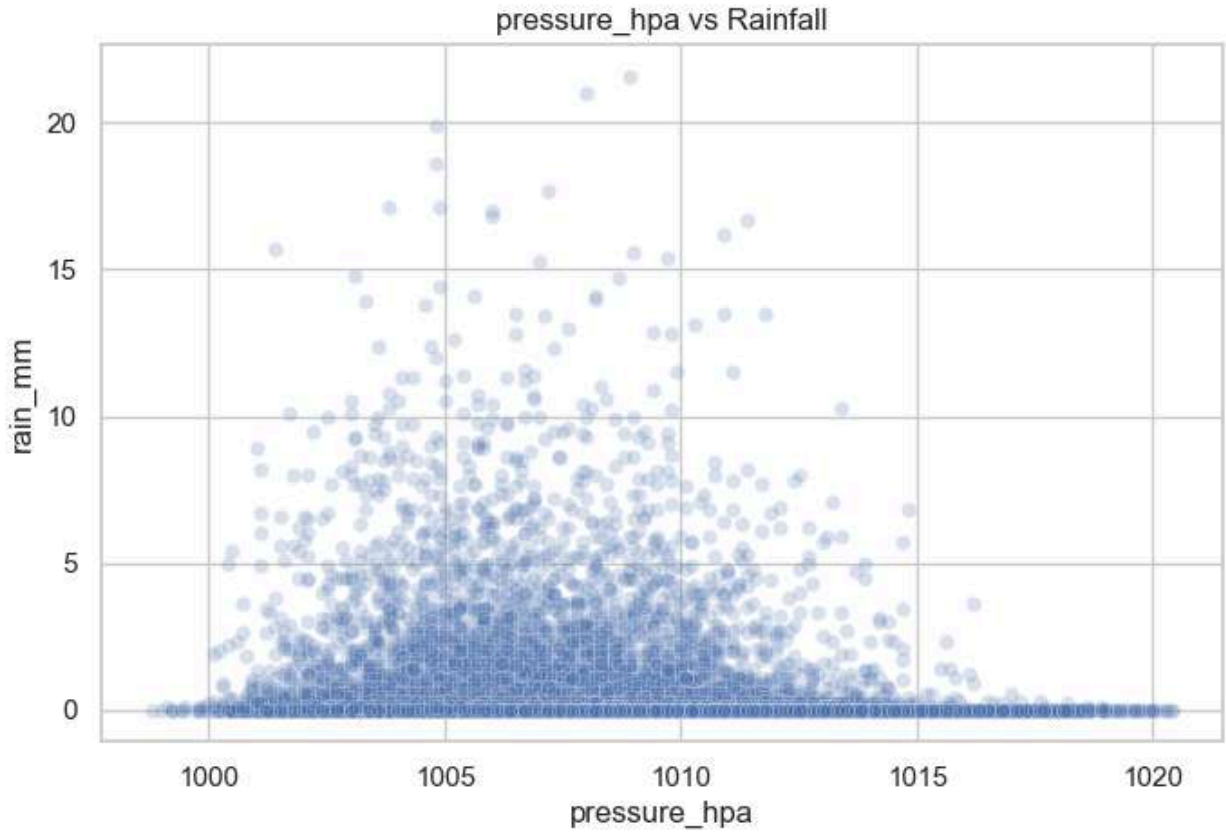


Figure C.10. Pressure (hPa) vs Rainfall

Rainfall appears more concentrated within a mid-range pressure band, while very low and very high pressure values are associated with fewer rainy observations. The pattern suggests that pressure may contribute to rainfall prediction, although its effect is not clearly linear on its own.

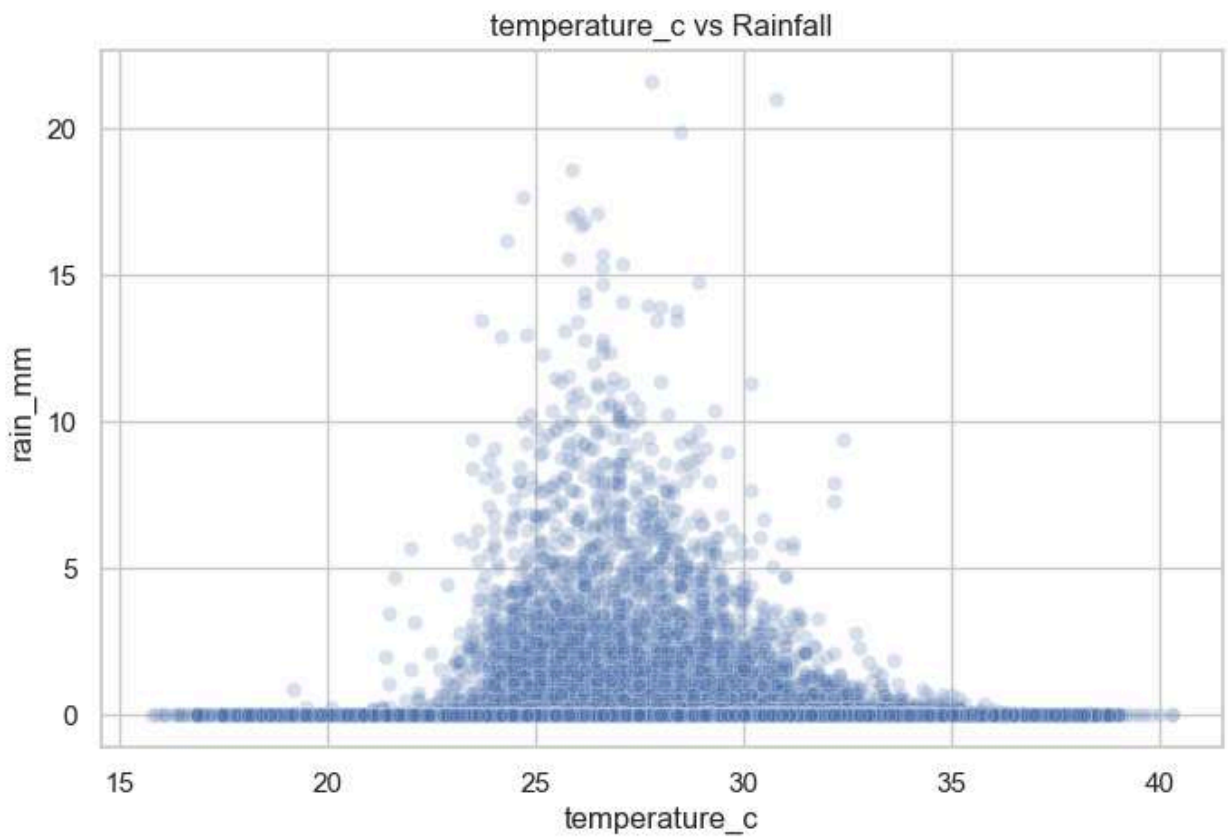


Figure C.11. Temperature (°C) vs Rainfall

The plot suggests that moderate-to-high rainfall is more commonly observed within a middle temperature range, roughly around 24–30°C. Extremely low or high temperatures are less frequently associated with heavy rainfall. This implies that temperature may provide supporting context for rainfall prediction, especially when combined with humidity and pressure

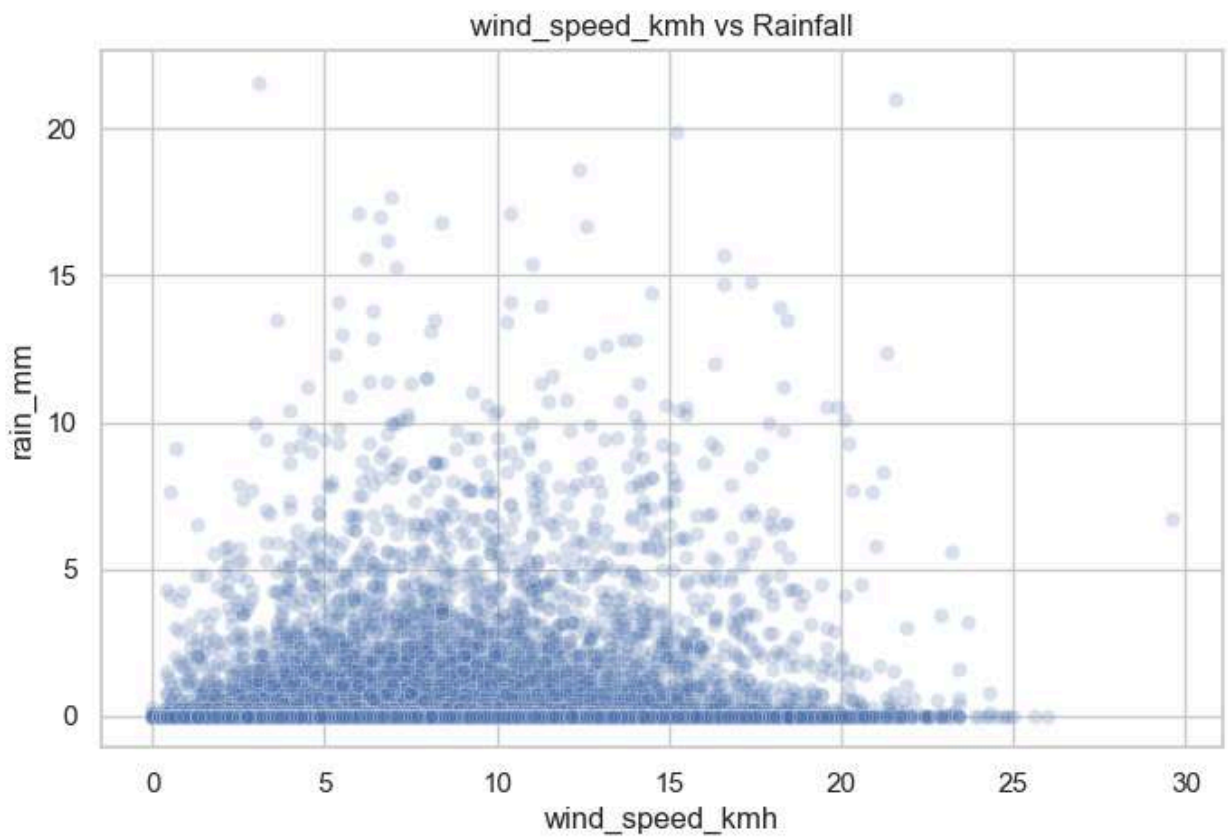


Figure C.12. Wind Speed (km/h) vs Rainfall

Rainfall occurs across a broad range of wind speeds, but heavier rainfall events are more frequently observed at moderate wind speeds rather than at very low or very high extremes. This indicates that wind speed may have predictive value, although its effect appears dispersed and should be modeled jointly with other weather variables.

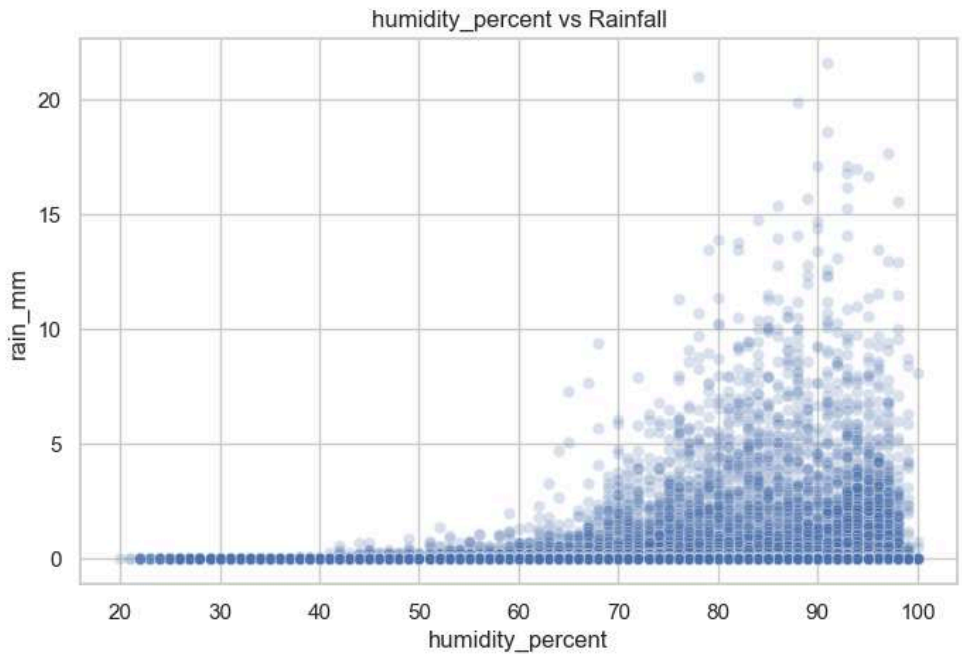


Figure C.13. Rainfall vs humidity

The scatter plot shows that higher rainfall values are more frequently observed when humidity is high, especially above roughly 70%. However, the relationship is not strictly linear, suggesting that humidity alone is not sufficient to explain rainfall intensity and should be interpreted together with other meteorological variables.

Summary

Overall, these figures suggest that rainfall in the dataset is influenced by both temporal patterns and combined meteorological conditions. Humidity and cloud cover show stronger visible association with rainfall, while pressure, temperature, and wind speed provide additional contextual information. The hourly rainfall pattern further supports the inclusion of time-based features, especially hour-of-day, in the final prediction framework.

APPENDIX D

Implementation Steps (Step-by-Step Guide)

The deployment of the proposed serverless architecture transitioned from an initial manual Proof-of-Concept (PoC) to an automated MLOps pipeline. Below are the sequential steps taken to establish the final operational system, corresponding to the visual evidence provided in this appendix:

- **Step 1: Cloud Resource Provisioning & IaC Readiness**
    - Manually provision the foundational storage layer (Amazon S3 buckets and Amazon DynamoDB table) via the AWS Management Console to maintain strict cost oversight within the Learner Lab environment (Figures D.4, D.7, D.9).
    - *Note:* An Infrastructure as Code (IaC) template (template.yaml) was also authored to ensure future reproducibility via AWS CloudFormation.
- **Step 2: Offline Model Training & Model Registry**
    - Execute the offline training pipeline using historical Open-Meteo data to train and evaluate the XGBoost model.
    - Serialize the finalized model into a .pkl artifact and securely upload it to the Amazon S3 Model Registry bucket (Figure D.6).
- **Step 3: Serverless Data Ingestion Configuration**
    - Deploy the Flood_Data_Ingestion AWS Lambda function directly via the AWS Console.
    - Configure Amazon EventBridge to trigger this ingestion function hourly, ensuring data is systematically fetched, transformed, and stored into S3 and DynamoDB (Figures D.8, D.10).
- **Step 4: Inference Containerization & CI/CD Automation**
    - To support the complex XGBoost library, containerize the inference environment using Docker and store the image in Amazon Elastic Container Registry (ECR) (Figure D.3).
    - Set up a private GitHub repository and configure GitHub Actions (deploy.yml) using GitHub Secrets for temporary AWS credentials. This CI/CD pipeline was

specifically implemented to automate the build and deployment process directly to the Flood_Prediction AWS Lambda function (Figures D.1, D.2).

- **Step 5: Real-time Prediction & Alert Routing Setup**
    - Deploy the Flood_Prediction AWS Lambda function using the automated ECR image (Figure D.11).
    - Integrate the LINE Messaging API within this function to execute the 4-Tier Adaptive Alert Logic and dispatch warnings to end-users (Figure D.15).
- **Step 6: Real-Time Observability & Monitoring Configuration**
    - Deploy the Flood_Monitoring AWS Lambda function to actively track system health and detect functional anomalies within the pipeline.
    - Configure Amazon EventBridge to systematically trigger this monitoring function, and integrate it with Amazon CloudWatch and Amazon SNS to automatically dispatch incident emails to the engineering team upon failures (Figure D.12).
- **Step 7: API Gateway & Frontend Dashboard Deployment**
    - Provision Amazon API Gateway to expose the DynamoDB/Prediction payload as a REST API, explicitly configuring CORS to allow secure client-side data fetching (Figure D.13).
    - Upload the HTML/JS frontend application files to the Amazon S3 Static Web Hosting bucket, creating the highly available public-facing Real-time Flood Alert Dashboard (Figures D.5, D.14).

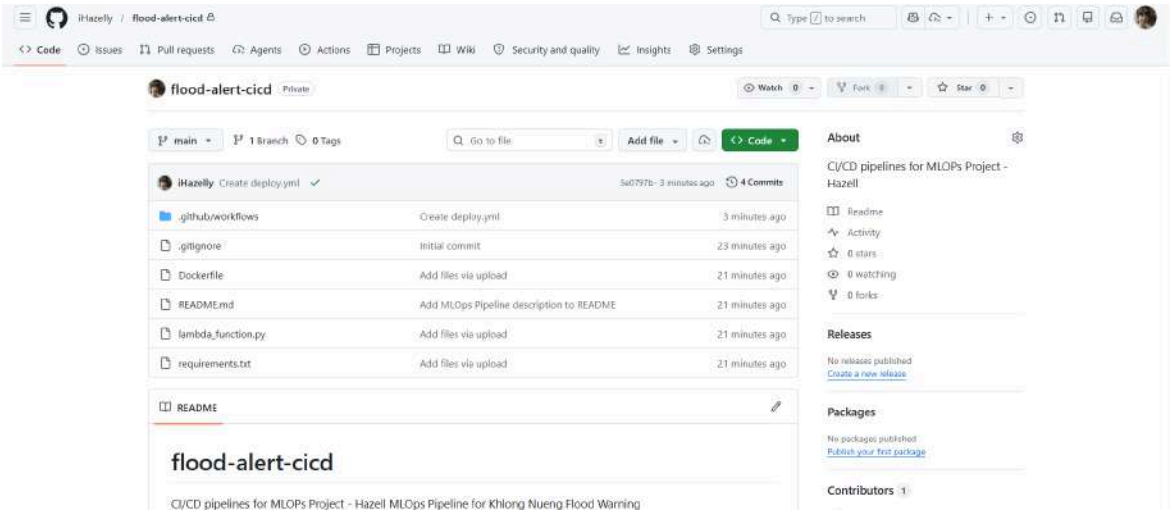


Figure D.1: GitHub repository structure containing the serverless MLOps codebase and CI/CD pipeline configurations.

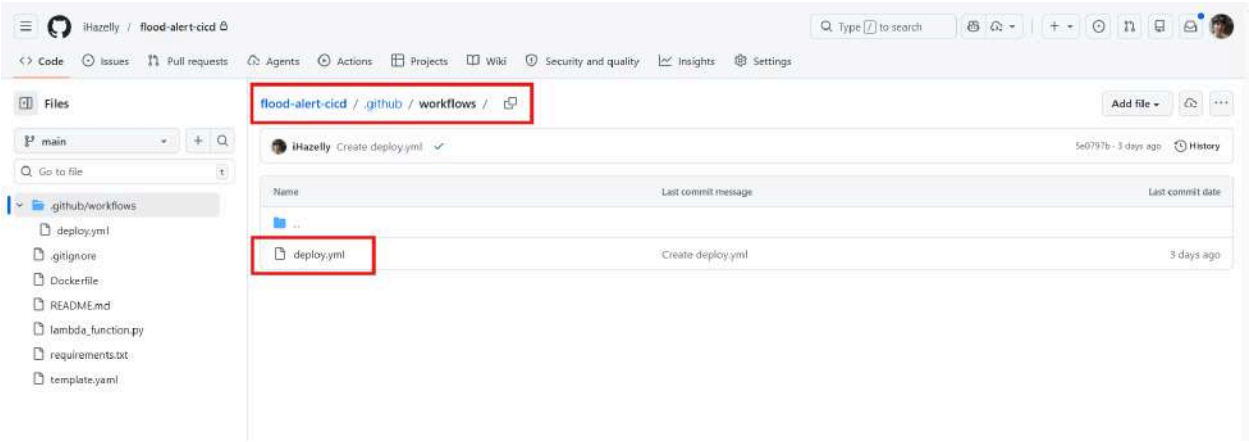


Figure D.2: GitHub Actions workflow (deploy.yml) enabling automated build and deployment of the Dockerized prediction model to AWS.

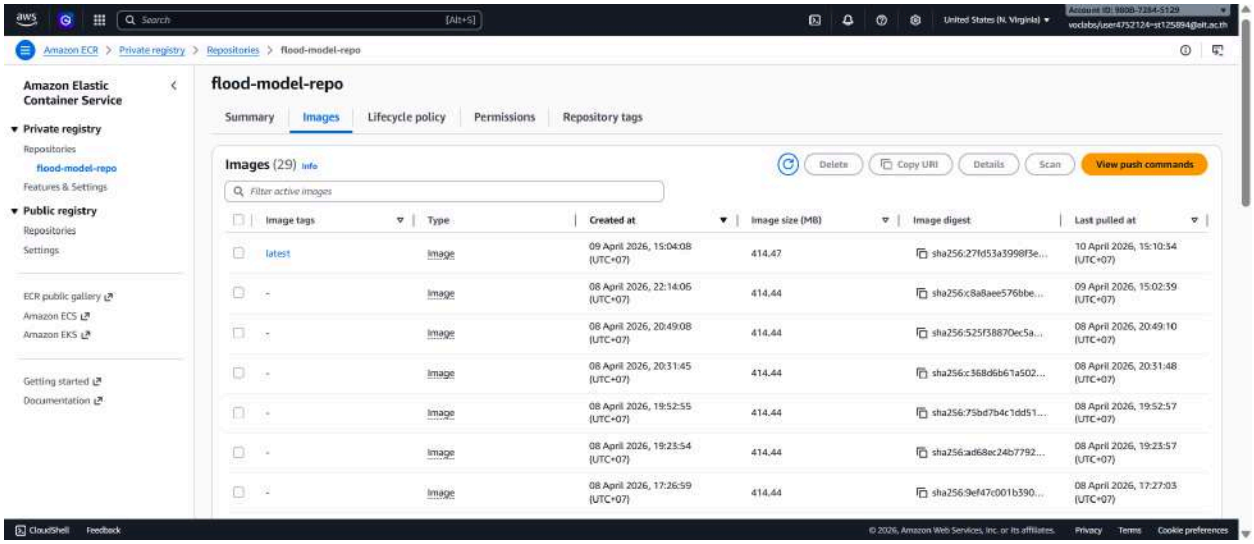


Figure D.3: Amazon Elastic Container Registry (ECR) housing the latest containerized XGBoost prediction environment.

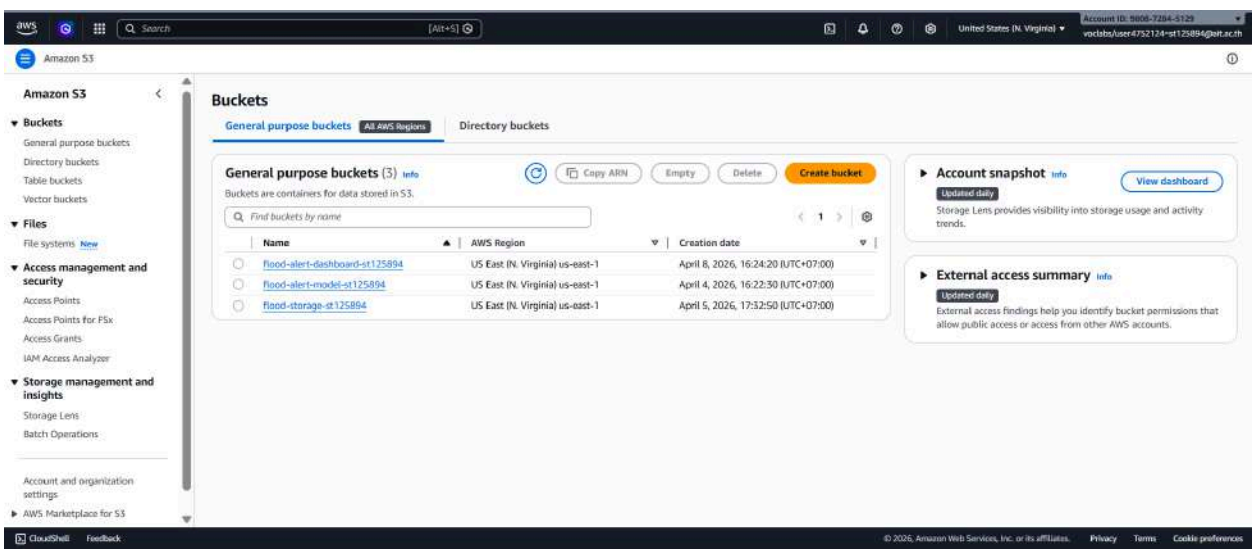
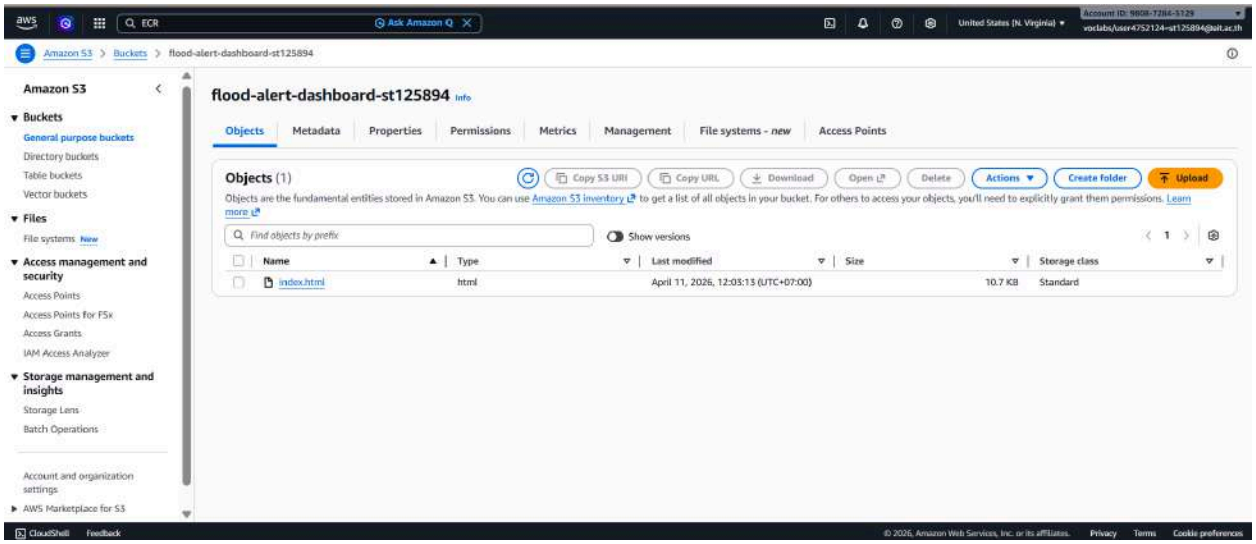


Figure D.4: Overview of Amazon S3 infrastructure provisioned for the Raw Data Lake, Model Registry, and Static Web Hosting.



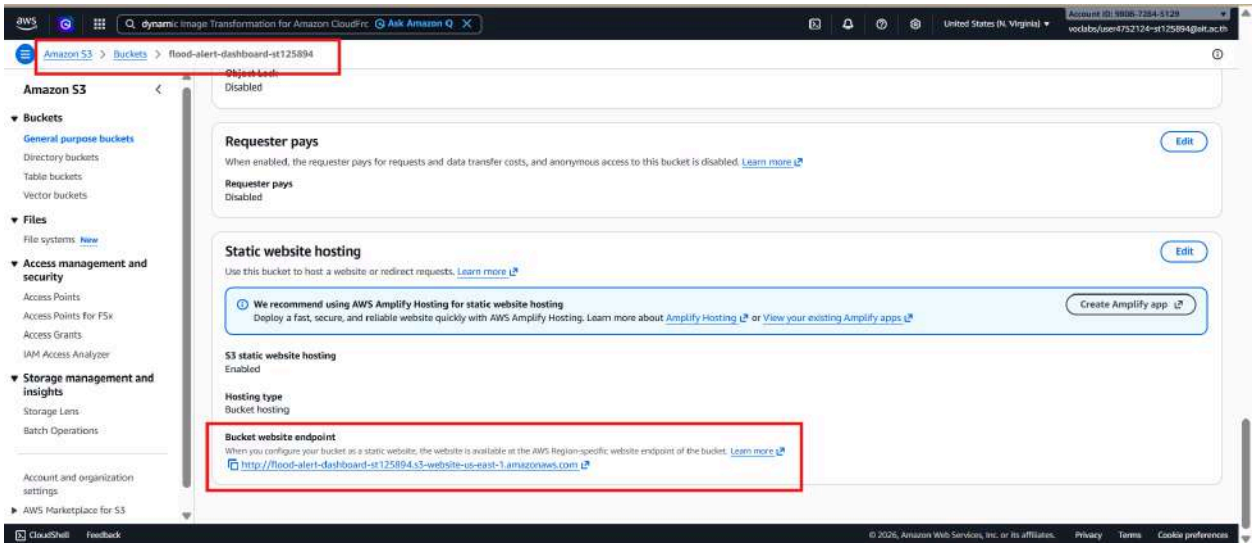


Figure D.5: Frontend web application deployed seamlessly via Amazon S3 Static Website Hosting

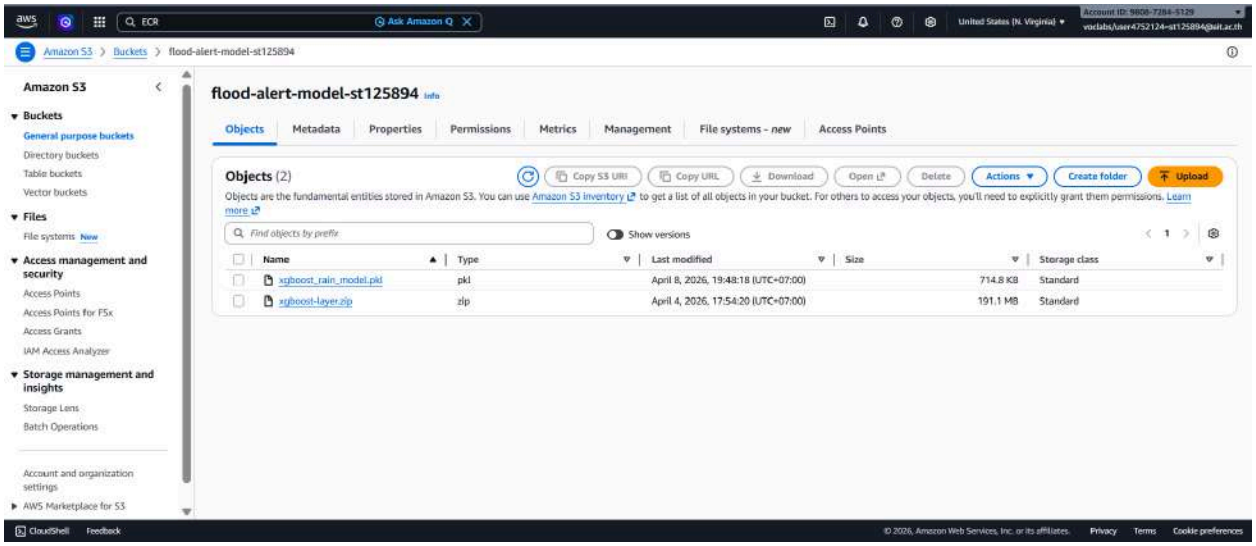


Figure D.6: Serialized XGBoost model artifact securely stored within the Amazon S3 Model Registry.

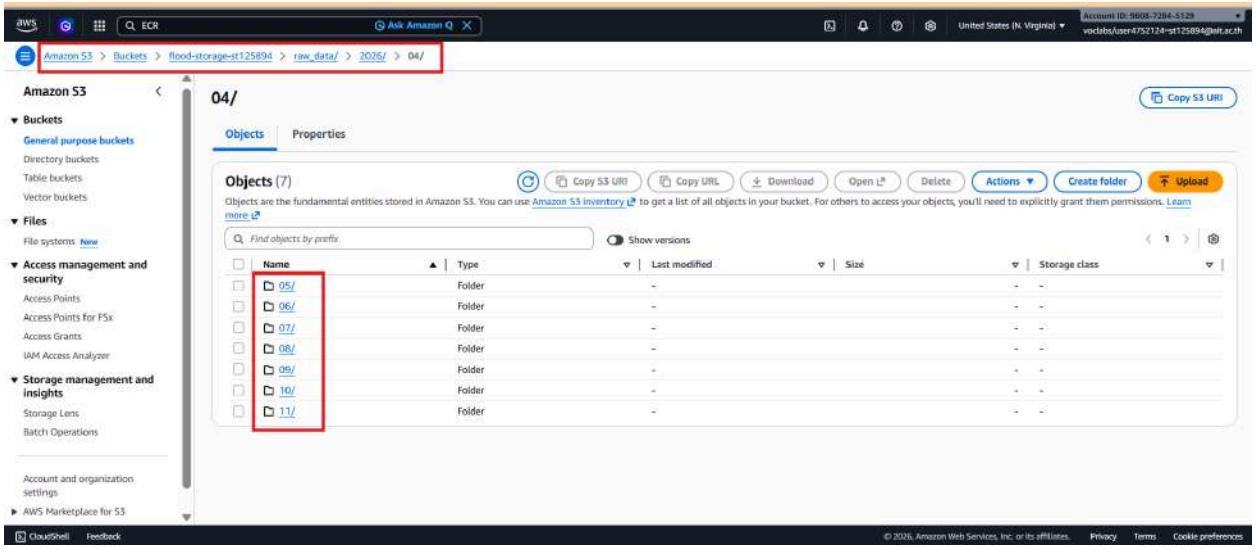


Figure D.7: Directory partitioning structure within the Amazon S3 Raw Data Lake, organized chronologically for efficient querying.

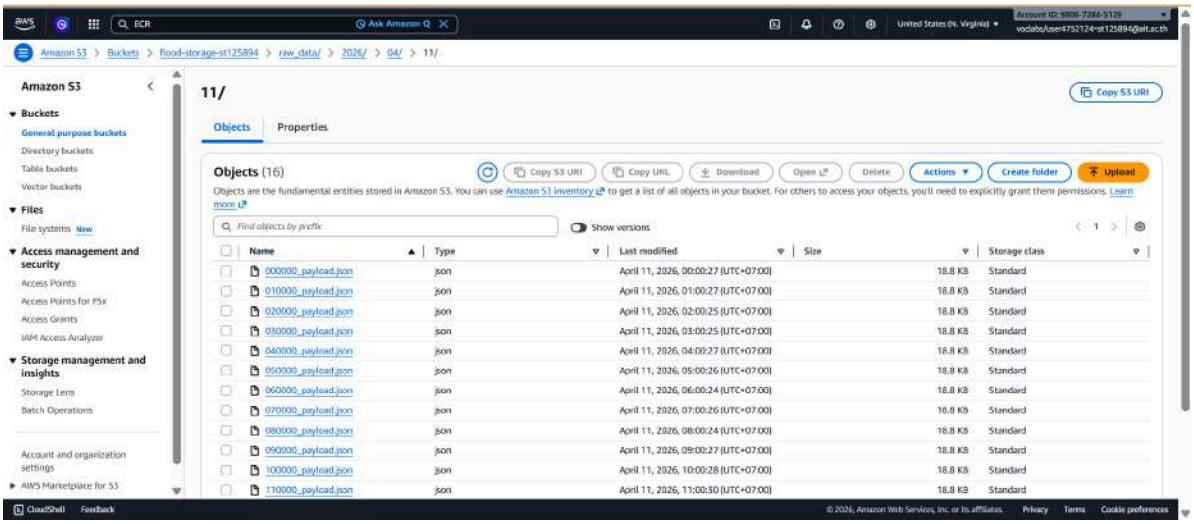
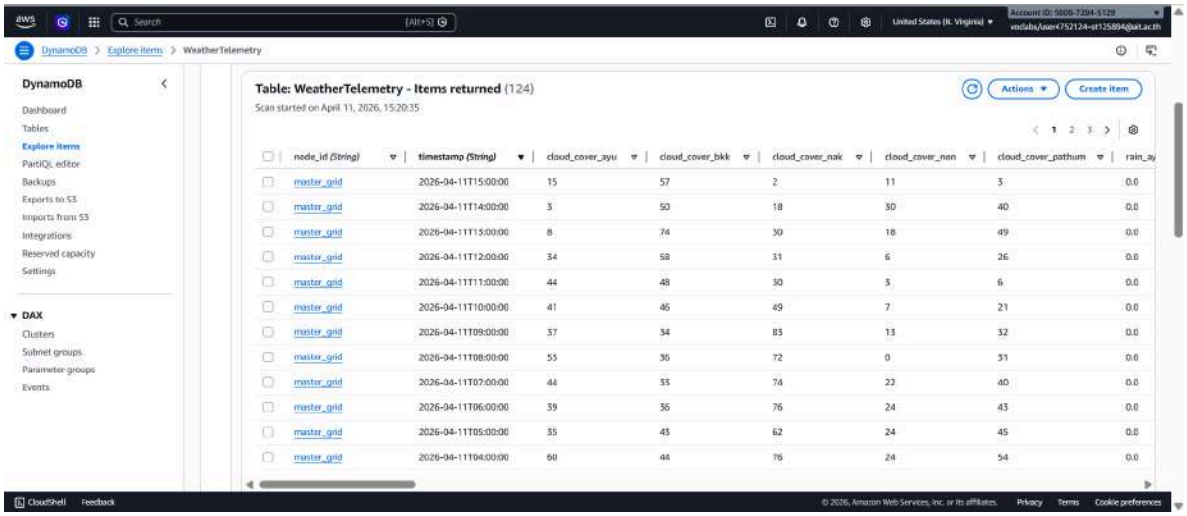


Figure D.8: Hourly meteorological JSON payloads successfully ingested from the API and archived in the Data Lake.



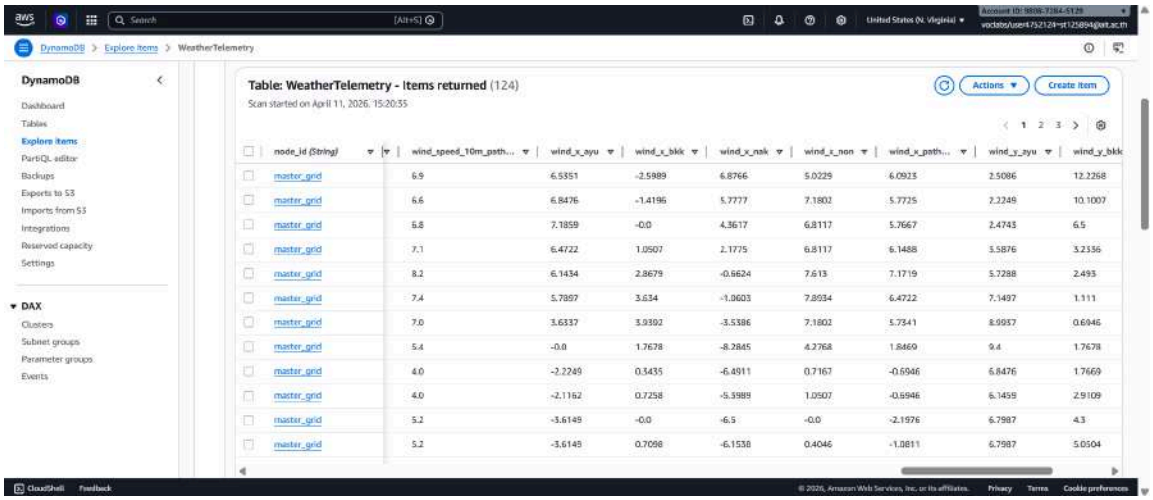


Figure D.9: Amazon DynamoDB operating as an ultra-low latency online feature store, aggregating and transformations.

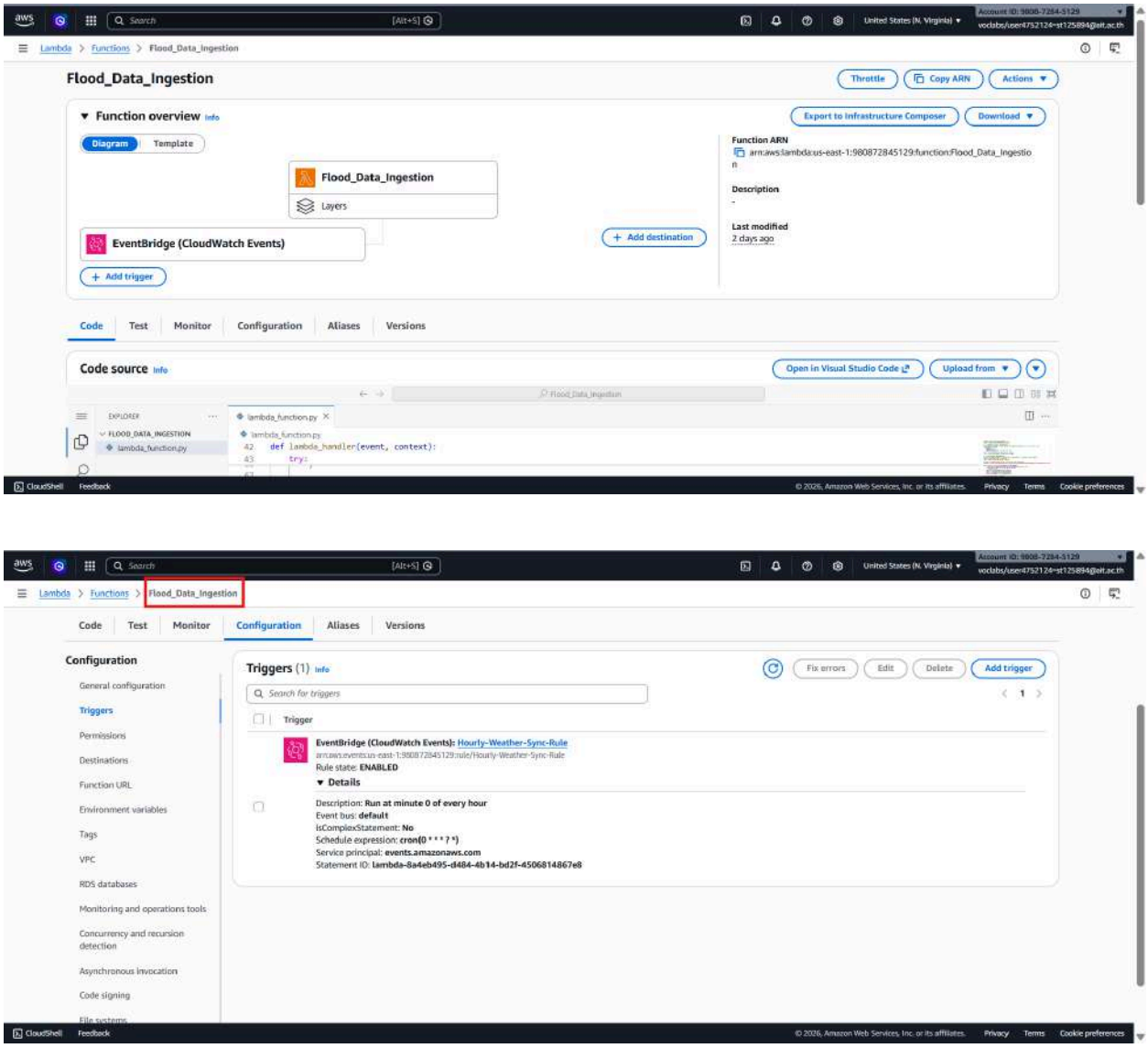


Figure D.10: Configuration of the Data Ingestion Lambda function and EventBridge

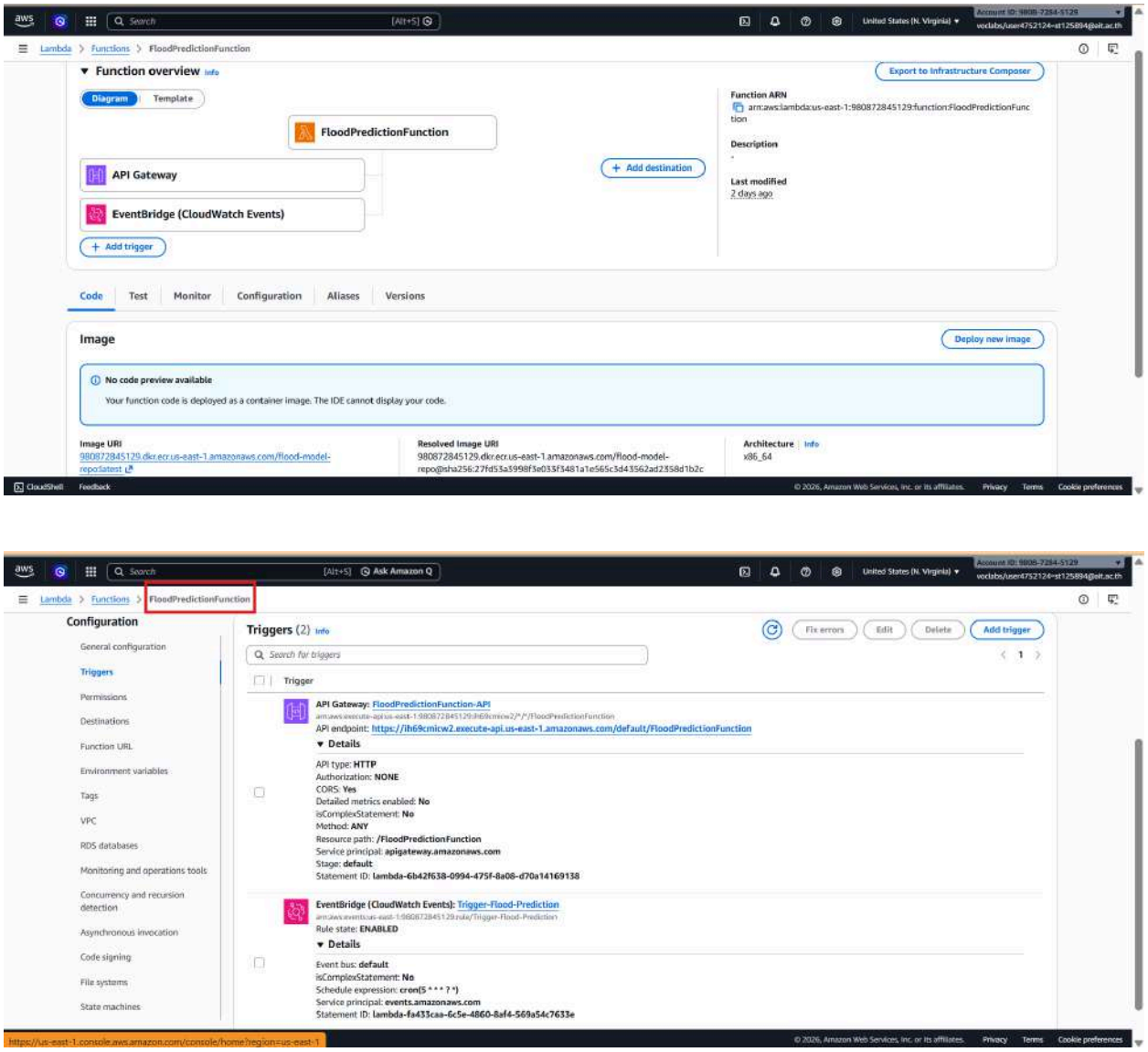
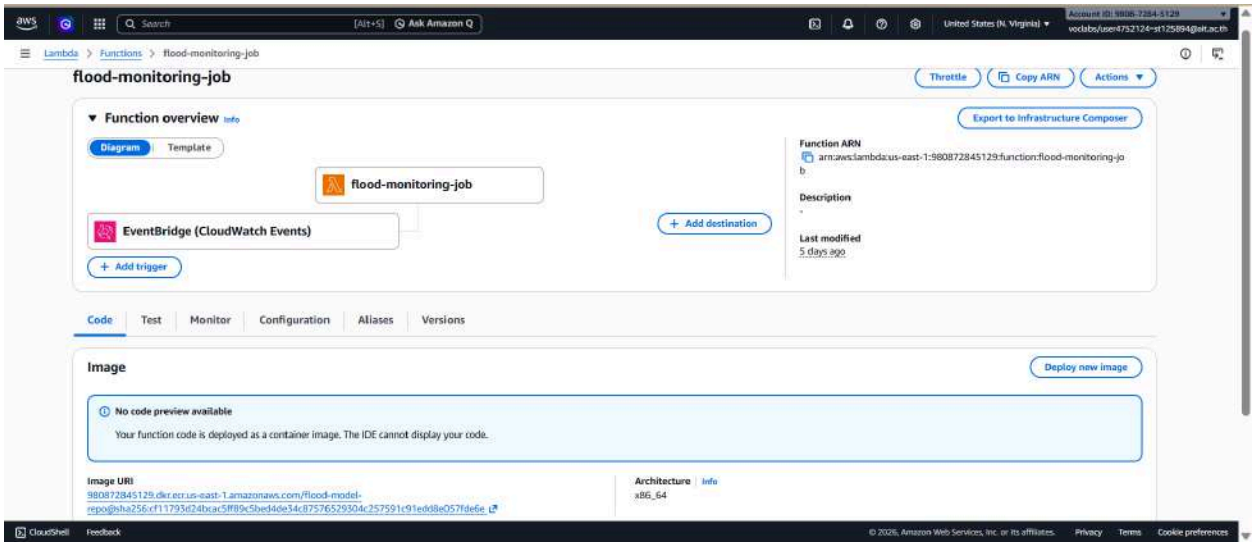


Figure D.11: Configuration of the flood prediction Lambda function deployed as a Docker container image, enabling the execution of complex machine learning dependencies.

- EventBridge cron expression configured to invoke the prediction pipeline systematically at 5 minutes past every hour.
- Amazon API Gateway provisioned as the secure HTTP entry point connecting the frontend web dashboard to the backend prediction function.



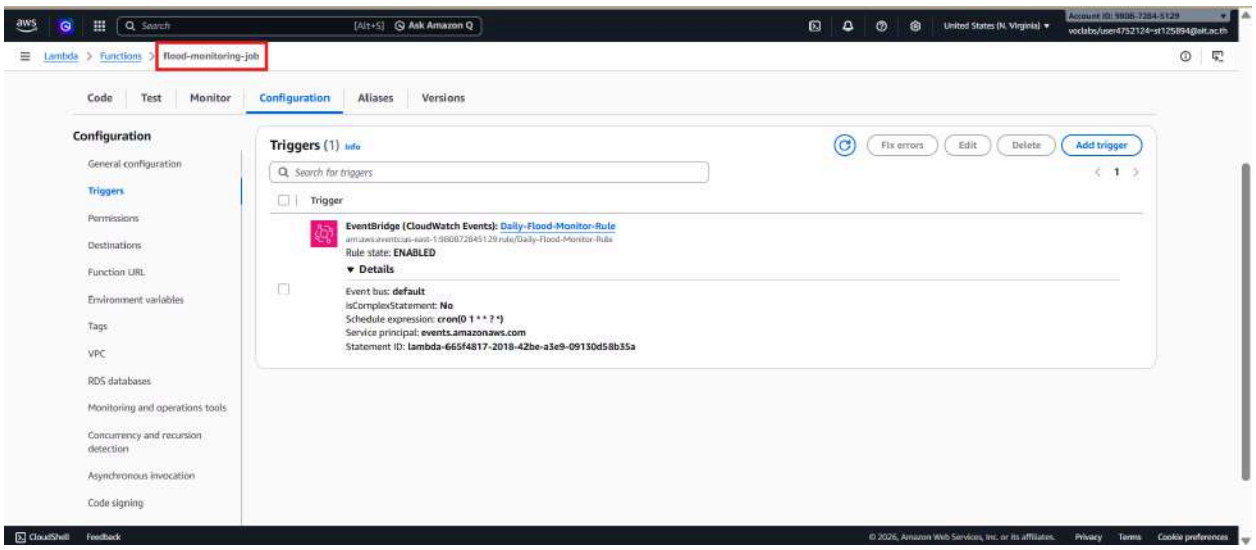


Figure D.12: Configuration of the Flood Monitoring Lambda function and EventBridge

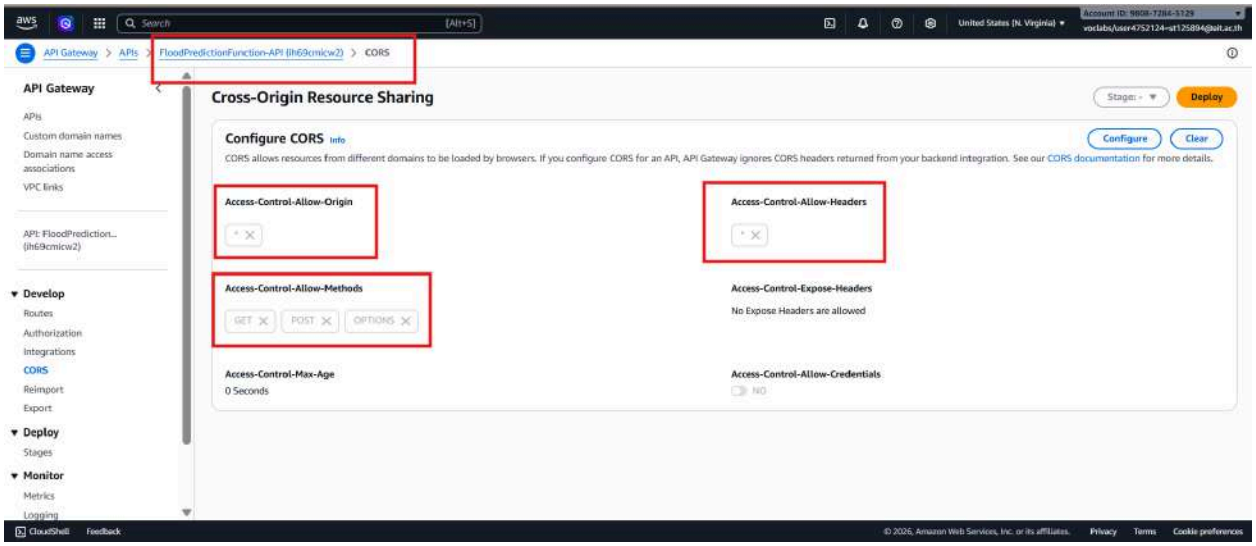


Figure D.13: API Gateway Cross-Origin Resource Sharing (CORS) configuration, ensuring secure data fetching from the client-side dashboard.

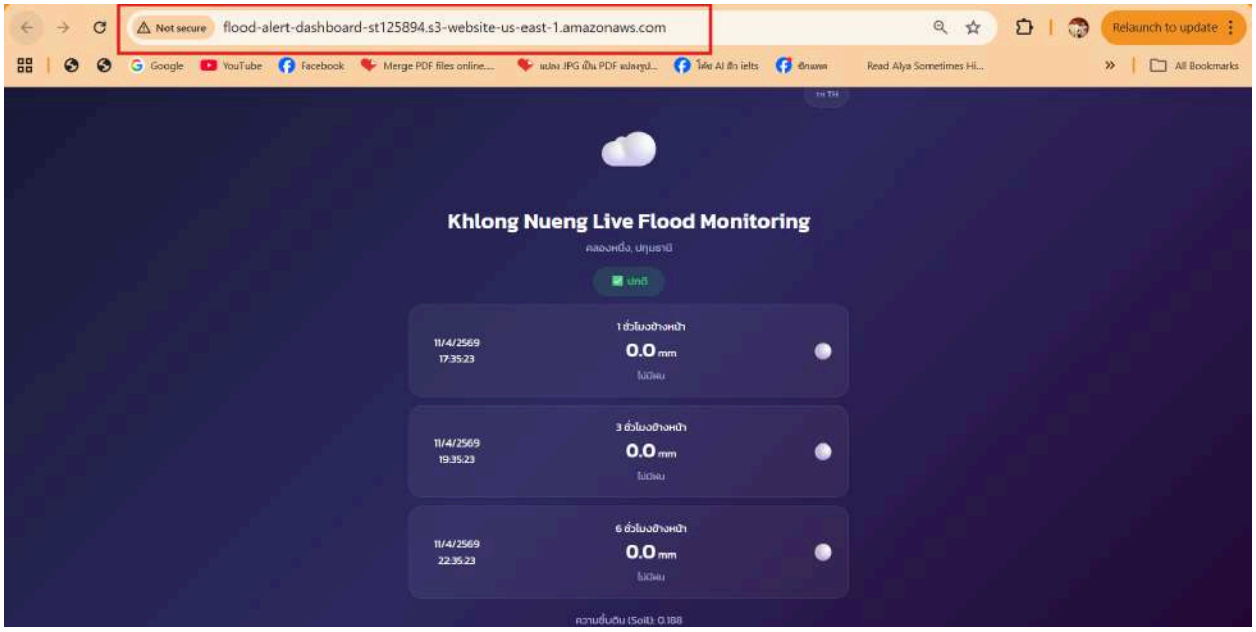


Figure D.14: Khlong Nueng Real-time Flood Alert Web Dashboard displaying hourly meteorological predictions and soil moisture status.

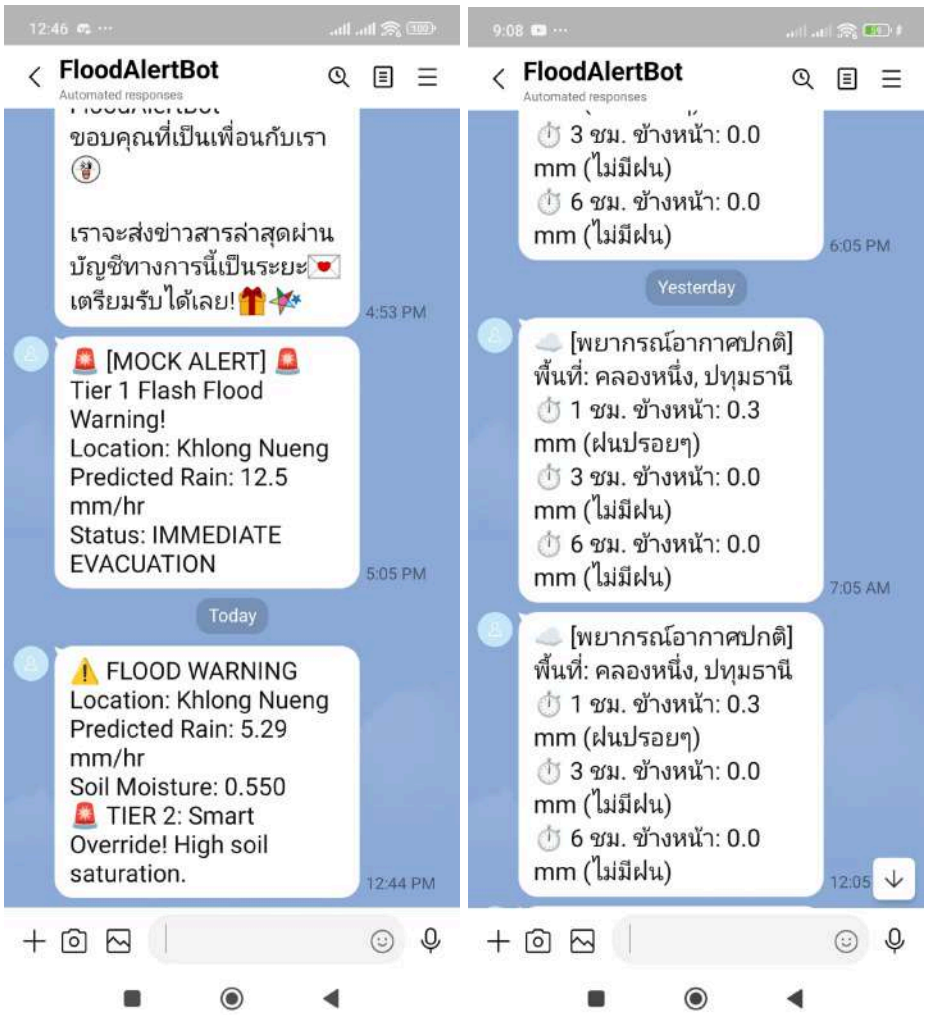


Figure D.15: End-to-end integration demonstrating automated routine meteorological forecasts dispatched to end-users via LINE Notify.

AI Use Declaration

During the development of this project, we used Artificial Intelligence tools to assist in several capacities. Specifically, Gemini was employed to translate text, enhance sentence cohesion, and refine the overall narrative flow. Furthermore, AI aided in conceptual brainstorming and suggested information to us for making decisions. Finally, logic, and conclusions were reviewed and verified by the authors.